كلية علوم وهندسة الحاسبات

FACULTY OF COMPUTER SCIENCE & ENGINEERING

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Faculty of Computer Science & Engineering
كلية علوم و هندسة الحاسبات

ALAMEIN INTERNATIONAL UNIVERSITY

Vision, Mission & Strategic Objectives

الرؤية والرسالة والأهداف الاستراتيجية

Vision

A leading educational and research faculty in the fields of computer engineering, computer science and artificial intelligence, offering multiple specialties to serve the community and improve the environment at the local, regional and global levels.

Mission

Providing a distinguished academic environment, to prepare distinguished graduate in computer engineering, computer sciences and artificial intelligence capable of meeting the needs of the local, regional and international market, developing innovative products of research value in the knowledge society, establishing sustainable partnerships with the industrial community and applying best educational, research and management practices to serve the environment. And society while preserving societal values and ethics.

الرؤيــة

كلية رائدة تعليمياً وبحثياً في مجالات هندسة وعلوم الحاسبات والذكاء الاصطناعي، تقدم تخصصات متعددة لخدمة المجتمع وتحسين البيئة على المستويات المحلية والإقليمية والعالمية.

الرسالة

توفير بيئة أكاديمية متميزة، تقوم بتخريج كوادر متميزة في هندسة وعلوم الحاسبات والذكاء الاصطناعي، قادرة على تلبية احتياجات السوق المحلي والإقليمي والعالمي وتطوير منتجات مبتكرة ذات قيمة بحثية في ظل مجتمع المعرفة، وإقامة شراكات مستدامة مع المجتمع الصناعي وتطبيق أفضل الممارسات التعليمية والبحثية والإدارية لخدمة البيئة والمجتمع مع الحفاظ على القيم والاخلاقيات المجتمعية.

Strategic Objectives

- ***Preparation of knowledgeable, skilful, and professional outstanding cadres on regional and international level through the following:***
  - Offering advanced educational programs and courses.
  - Developing students' scientific and technical skills.
  - Encourage and support students' participation in international competitions, and to prepare them for the future job market, at the regional and international levels.
  - Partnerships with international universities for offering joint scientific programs.
  - Attracting industrial companies to share in integrating practical courses and supervise graduation projects.
  - Granting the Faculty, the local and international accreditation.
  - Strengthening the infrastructure and interest for continuous development.

الأهداف الاستراتيجية

- **إعداد كوادر متميزة معرفياً ومهارياً ومهنياً، على المستوى الإقليمي والدولي من خلال:**
  - تقديم برامج ومقررات تعليمية متطورة
  - تنمية المهارات العلمية والتقنية لدى الطلاب
  - تحفيز ودعم اشتراك الطلاب في المسابقات الدولية. وتأهيلهم لسوق العمل المستقبلي، على المستوى الإقليمي والدولي.
  - عقد شراكات مع جامعات عالمية لتقديم برامج علمية مشتركة.
  - جذب الشركات العاملة بالصناعة، للمشاركة في دمج المقررات بالجانب العملي والإشراف على مشروعات التخرج.
  - حصول الكلية على الاعتماد المحلى والدولي.
  - تعزيز البنية التحتية والاهتمام بالتطوير المستمر.

- ***Configuring the appropriate environment for scientific research through the following:***
  - Developing the competitiveness of researchers.
  - Building and activating strategic partnerships with Artificial Intelligence companies and institutions.
  - Upgrading the research quality and level through continuous interaction with community issues.
  - Self-development of resources to enhance the Faculty research environment.
- ***Development of community services through the following:***
  - Strengthening the Faculty role in presenting specialized scientific services and consulting to private and governmental agencies
  - Strengthening ways of cooperation with companies and institutions that depend on artificial intelligence.
  - Establishment of a consulting and community service center.

- **تهيئة البيئة المناسبة للبحث العلمي. وذلك من خلال:**
  - تطوير القدرة التنافسية للباحثين.
  - بناء وتفعيل علاقات شراكة استراتيجية مع المؤسسات والشركات التي تعتمد على الذكاء الاصطناعي.
  - الارتقاء بنوعية ومستوى البحوث من خلال التفاعل المستمر مع قضايا المجتمع.
  - تنمية الموارد الذاتية لتحسين البيئة البحثية بالكلية.
- **تطوير الخدمات المجتمعية. وذلك من خلال:**
  - تعزيز دور الكلية في تقديم الخدمات والاستشارات العلمية المتخصصة للجهات الحكومية والخاصة.
  - تعزيز سبل التعاون مع المؤسسات الداعمة لدمج الذكاء الاصطناعي والشركات العاملة به.
  - إنشاء مركز للاستشارات وخدمة المجتمع.

Governing Values

- *In addition to the values adopted by the University, the Faculty adopts the following professional values:*
  - Creativity … Innovation … Integration … Excellence … Transparency … Quality … Social Responsibility … Applied Research.
- *The main areas of the continuous development plan of the Faculty to continue excellence:*
  - Making partnerships with international universities to present Dual or Double Degree and achieve continuous students' exchange.
  - Preparing distinguished post graduate programs to increase the scientific research level and guarantee building of research capabilities of the students and the teacher assistants.
  - Making partnerships with industry and community to offer innovative technological solutions that develop the Faculty surrounding environment.
  - Encourage the entrepreneurship culture for students and establish technological incubators and emerging companies; and establish a community service and technological solution marketing center.

القيم الحاكمة

- **بالإضافة إلي القيم التي تتبناها الجامعة، تتبنى الكلية القيم المهنية التالية:**
  - الابتكار ... الابداع ... التكامل ... التميز ... الشفافية ... الجودة ... المسئولية الاجتماعية ... البحث التطبيقي.
- **المحاور الرئيسية لمخطط التطوير المستمر للكلية لتستمر في التميز:**
  - عقد شراكات مع الجامعات العالمية لتقديم Double Degree أو Dual Degree ولتنفيذ التبادل الطلابي المستمر.
  - تجهيز برامج دراسات عليا مميزة، لرفع مستوى البحث العلمي وضمان بناء قدرات طلاب البحث والهيئة المعاونة.
  - عقد شراكات مع الصناعة والمجتمع لتقديم حلول تكنولوجية مبتكرة تطور المجتمع المحيط بالكلية.
  - تشجيع ثقافة ريادة الأعمال لدى الطلاب، وإنشاء حاضنات تكنولوجية وشركات ناشئة، وإنشاء مركز لخدمة المجتمع وتسويق الحلول التكنولوجية.

Scientific Departments

الأقسام العلمية

– Department of Computer Science & Engineering

– Programming – Algorithms – Operating Systems – Data and Information Processing – Computer Architecture – Data Security – Software Engineering – Graphics and Multimedia – Networks - System Building Technology

– Department of Artificial Intelligence Science & Engineering

– Artificial Intelligence (AI) – Machine Learning - Deep Learning - Logic-based AI - Natural Language Processing – Robotics.

– Department of Biomedical Informatics

– Medical Systems – Medical Image Processing – Medical informatics – Medical Statistics – Bioinformatics

– قسم علوم وهندسة الحاسب

– البرمجة – الخوارزميات – نظم التشغيل – معالجة البيانات والمعلومات – بناء الحاسب – أمن المعلومات – هندسة البرمجيات – الرسوم والوسائط المتعددة - الشبكات – تكنولوجيا بناء النظم

– قسم علوم وهندسة الذكاء الاصطناعي

– الذكاء الاصطناعي - تعلم الآلة - التعلم العميق - الذكاء الاصطناعي المبني على المنطق - معالجة اللغات الطبيعية - الروبوتات

– قسم المعلوماتية الطبية الحيوية

– الأنظمة الطبية – معالجة الصور الطبية – المعلوماتية الطبية – الإحصاء الطبي –المعلوماتية الحيوية

Laboratories

المعامل

- General Computer Lab
- معامل الحاسبات العامة

Computer Science & Engineering

علوم وهندسة الحاسب

- Logic Design and Computer Architecture Lab
- معمل التصميم المنطقي وعمارة الحاسبات
- Embedded Systems & Internet of Things Lab
- معمل النظم المدمجة وإنترنت الأشياء
- Microprocessor Lab
- معمل المعالجات الدقيقة
- Mobile Programming Lab
- معمل برمجة الأجهزة المحمولة
- Computer Networks Lab
- معمل الشبكات
- Cyber Security Lab
- معمل الأمن السيبراني
- Virtual & Augmented Reality Lab
- معمل الواقع الافتراضي والمعزز

Artificial Intelligence Engineering & Science

هندسة الذكاء الاصطناعي وعلومه

- Artificial Intelligence and Smart Robotics Lab
- معمل الذكاء الاصطناعي والروبوتات
- High Performance Computing Lab
- معمل الحوسبة عالية الأداء
- Big Data Lab
- معمل البيانات الضخمة

Biomedical Informatics

The Department uses basic labs that are affiliated to Faculty of Science and applied labs that are affiliated to computer science and engineering department and Artificial Intelligence Science and Engineering department.

المعلوماتية الطبية الحيوية

يستعين القسم بمعامل كلية العلوم لخدمة معامل العلوم الأساسية (الفيزياء – الكيمياء - الأحياء)، كما يستعين بكافة معامل الكلية لخدمة علوم الحاسبات التطبيقية.

كلية علوم وهندسة الحاسبات... قائمة المقررات المطروحة

Faculty of Computer Science & Engineering ... List of Courses

Department of Computer Science & Engineering

قسم علوم وهندسة الحاسب

Code	Course Name	اسم المقرر	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
CSE011	Computer Skills	مهارات الحاسب الآلي	1	3	0	0	4	0	60	0
CSE012	Scientific Applications of Computers	التطبيقات العلمية للحاسب الآلي	2	1	0	0	3	2	105	4
CSE013	Introduction to Information Systems & Technology	مقدمة في نظم وتكنولوجيا المعلومات	2	1	0	0	3	2	105	4
CSE014	Structured Programming	برمجة هيكلية	2	3	0	0	5	3	180	6
CSE015	Object Oriented Programming	برمجة شيئية	2	3	0	0	5	3	180	6
CSE081	Digital Branding	العلامة التجارية الرقمية	3	3	0	0	6	3	180	6
CSE111	Data Structures	هياكل البيانات	2	3	0	0	5	3	180	6
CSE112	Design & Analysis of Algorithms	تصميم و تحليل الخوارزميات	3	0	3	0	6	4	195	8
CSE113	Electric & Electronic Circuits	دوائر كهربية وإلكترونية	1	3	2	0	6	3	180	6
CSE131	Logic Design	التصميم المنطقي	2	2	2	0	6	3	195	6
CSE132	Computer Architecture & Organization	بناء وتنظيم الحاسب	2	2	1	0	5	3	165	6
CSE191	Field Training 1 in Computer Science	تدريب ميداني ١ في علوم الحاسب	2	0	0	0	2	2	60	4
CSE192	Field Training 1 in Information Technology	تدريب ميداني ١ في تكنولوجيا المعلومات	2	0	0	0	2	2	60	4
CSE211	Web Programming	برمجة الويب	2	2	1	0	5	3	165	6
CSE221	Database Systems	نظم قواعد البيانات	2	2	1	0	5	3	180	6
CSE232	Operating Systems Administration	إدارة نظم التشغيل	2	2	1	0	5	3	180	6
CSE233	Operating Systems	نظم التشغيل	2	2	1	0	5	3	165	6
CSE234	Embedded Systems	النظم المدمجة	2	2	1	0	5	3	180	6

Code	Course Name	اسم المقرر	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
CSE241	Security of Information Systems	أمن نظم المعلومات	2	2	1	0	5	3	150	6
CSE242	Cryptography	علم التشفير	2	1	2	0	5	3	180	6
CSE243	Secure Programming	البرمجة الآمنة	2	3	0	0	5	3	180	6
CSE251	Software Engineering	هندسة البرمجيات	2	2	1	0	5	3	180	6
CSE261	Computer Networks	شبكات الحاسب	2	2	1	0	5	3	165	6
CSE262	Introduction to Telecommunications, Networking and Security	مقدمة في الاتصالات والشبكات والأمن	2	2	1	0	5	3	180	6
CSE271	Introduction to Parallel Computing	مقدمة في الحوسبة الموازية	2	2	1	0	5	3	180	6
CSE272	Parallel & Distributed Systems	النظم المتوازية والموزعة	2	2	1	0	5	3	150	6
CSE281	Image Processing	معالجة الصور	2	2	1	0	5	3	180	6
CSE291	Field Training 1 in Computer Engineering	تدريب ميداني ١ في هندسة الحاسب	2	0	0	0	2	2	60	4
CSE292	Field Training 2 in Computer Science	تدريب ميداني ٢ في علوم الحاسب	2	0	0	0	2	2	60	4
CSE293	Field Training 2 in Information Technology	تدريب ميداني ٢ في تكنولوجيا المعلومات	2	0	0	0	2	2	60	4
CSE311	Design of Compilers	تصميم مترجمات البرامج	2	2	1	0	5	3	180	6
CSE312	Advanced Web Programming	برمجة الويب المتقدمة	2	2	1	0	5	3	165	6
CSE313	Mobile Programming	برمجة الأجهزة المحمولة	2	2	1	0	5	3	165	6
CSE314	Android and iOS Development	تطوير برامج الأندرويد وiOS	2	2	1	0	5	3	165	6
CSE315	Discrete Mathematics	الرياضيات المقطعة	2	0	2	0	4	3	150	6
CSE321	Societal and Ethical Issues in Information Technology	القضايا المجتمعية والأخلاقية في تكنولوجيا المعلومات	2	2	1	0	5	3	180	6
CSE322	Big Data Analytics 1	تحليل البيانات الكبيرة ١	2	2	1	0	5	3	180	6
CSE323	Advanced Database Systems	نظم قواعد بيانات متقدمة	2	2	1	0	5	3	180	6
CSE324	IT Management and Operations	إدارة وعمليات تكنولوجيا المعلومات	2	2	1	0	5	3	180	6
CSE325	Database Systems Administration	إدارة نظم قواعد البيانات	2	2	1	0	5	3	180	6

Code	Course Name	اسم المقرر	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
CSE335	Real Time & Embedded Systems Design	تصميم النظم المدمجة وذات الوقت الحقيقي	2	2	1	0	5	3	180	6
CSE344	Introduction to Cyber Security	مقدمة في الأمن السيبراني	2	2	1	0	5	3	180	6
CSE351	Software Project Management	إدارة مشروع البرمجيات	2	2	1	0	5	3	180	6
CSE352	Systems Analysis & Design	تحليل وتصميم النظم	2	2	1	0	5	3	165	6
CSE353	Interaction Design	التصميم التفاعلي	2	2	1	0	5	3	165	6
CSE361	Networks Virtualization and Administration	إدارة الشبكات الافتراضية	2	2	1	0	5	3	165	6
CSE362	Industrial Networks	الشبكات الصناعية	2	2	1	0	5	3	180	6
CSE363	Cloud Computing	الحوسبة السحابية	2	2	1	0	5	3	180	6
CSE373	Parallel Programming	البرمجة المتوازية	2	2	1	0	5	3	180	6
CSE382	Computer Graphics	الرسم بالحاسب الالي	2	2	1	0	5	3	180	6
CSE383	Computer Vision	الرؤية للحاسب	2	2	1	0	5	3	180	6
CSE392	Field Training 2 in Computer Engineering	تدريب ميداني ٢ في هندسة الحاسب	2	0	0	0	2	2	60	4
CSE411	Theory of Computation & Compilers	نظرية الحوسبة ومترجمات البرامج	2	2	1	0	5	3	180	6
CSE424	Data Warehousing	مستودع البيانات	2	2	1	0	5	3	180	6
CSE425	Big Data Analytics 2	تحليل البيانات الكبيرة ٢	2	2	1	0	5	3	180	6
CSE426	Selected Topics in Data Science	موضوعات مختارة في علم البيانات	2	2	1	0	5	3	180	6
CSE427	Selected Topics in Big Data	موضوعات مختارة في البيانات الضخمة	2	2	1	0	5	3	180	6
CSE436	Selected Topics in Embedded Systems	موضوعات مختارة في النظم المدمجة	2	2	1	0	5	3	180	6
CSE445	Selected Topics in Information Security	موضوعات مختارة في أمن المعلومات	2	2	1	0	5	3	180	6
CSE446	Information & Computer Networks Security	أمن المعلومات والشبكات الحاسوبية	2	2	1	0	5	3	180	6
CSE447	Selected Topics in Computer Security	موضوعات مختارة في أمن الحاسب	2	2	1	0	5	3	180	6
CSE448	Cyber Forensics	الطب الشرعي السيبراني	2	2	1	0	5	3	180	6
CSE453	Building Information Systems	بناء نظم المعلومات	2	2	1	0	5	3	180	6

Code	Course Name	اسم المقرر	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
CSE454	Advanced Software Engineering	هندسة برمجيات متقدمة	2	2	1	0	5	3	180	6
CSE455	Selected Topics in Software Engineering	موضوعات مختارة في هندسة البرمجيات	2	2	1	0	5	3	180	6
CSE463	Advanced Networking	الشبكات المتقدمة	2	2	1	0	5	3	180	6
CSE464	Internet of things	انترنت الأشياء	2	2	1	0	5	3	180	6
CSE465	Selected Topics in Cloud Computing	موضوعات مختارة في الحوسبة السحابية	2	2	1	0	5	3	180	6
CSE466	Selected Topics in IoT	موضوعات مختارة في انترنت الأشياء	2	2	1	0	5	3	180	6
CSE467	Client/Server Technologies & Applications	تقنيات وتطبيقات العميل / الخادم	2	2	1	0	5	3	180	6
CSE468	Web Server Administration	إدارة خادم الويب	2	2	1	0	5	3	180	6
CSE474	Distributed Information Systems	نظم المعلومات الموزعة	2	2	1	0	5	3	180	6
CSE475	High Performance Computing	الحوسبة عالية الأداء	2	2	1	0	5	3	180	6
CSE476	Selected Topics in High Performance Computing	موضوعات مختارة في الحوسبة عالية الأداء	2	2	1	0	5	3	180	6
CSE484	Interactive Multimedia	أنظمة وسائط متعددة تفاعليه	2	2	1	0	5	3	180	6
CSE485	Computer Games Theory	نظرية ألعاب الحاسب الآلي	2	2	1	0	5	3	180	6
CSE486	Selected Topics in Computer Vision	موضوعات مختارة في الرؤية بالحاسب	2	2	1	0	5	3	180	6
CSE487	Mixed & Augmented Reality	الواقع المختلط والمعزز	2	2	1	0	5	3	180	6
CSE488	Visualization & Animation	التصور والرسوم المتحركة	2	2	1	0	5	3	180	6
CSE493	Graduation Project 1	مشروع التخرج ١	2	3	0	0	5	3	180	6
CSE494	Graduation Project 2	مشروع التخرج ٢	2	3	0	0	5	3	180	6

Department of Artificial Intelligence Science & Engineering

قسم علوم وهندسة الذكاء الاصطناعي

Code	Course Name	اسم المقرر	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
AIE111	Artificial Intelligence	الذكاء الاصطناعي	2	2	1	0	5	3	180	6
AIE121	Machine Learning	تعلم الالة	2	2	1	0	5	3	180	6
AIE122	Basics of Machine Learning and Data Science	أساسيات التعلم الآلي وعلوم البيانات	2	2	1	0	5	3	180	6
AIE191	Field Training 1 in AI Science	تدريب ميداني ١ في علوم الذكاء الاصطناعي	2	0	0	0	2	2	60	4
AIE212	Knowledge-Based Systems	النظم القائمة على المعرفة	2	2	1	0	5	3	180	6
AIE213	Optimization Techniques	طرق إيجاد الحل الأمثل	3	0	3	0	6	3	180	6
AIE231	Neural Networks	الشبكات العصبية	2	2	1	0	5	3	180	6
AIE241	Natural Language Processing	معالجة اللغات الطبيعية	2	2	1	0	5	3	180	6
AIE291	Field Training 1 in Artificial Intelligence Engineering	تدريب ميداني ١ في هندسة الذكاء الاصطناعي	2	0	0	0	2	2	60	4
AIE292	Field Training 2 in AI Science	تدريب ميداني ٢ في علوم الذكاء الاصطناعي	2	0	0	0	2	2	60	4
AIE314	Ai-Based Programming	البرمجة المبنية علي الذكاء الاصطناعي	2	2	1	0	5	3	180	6
AIE315	Computational Logic	المنطق الحسابي	2	2	1	0	5	3	180	6
AIE316	Evolutionary Algorithms	الخوارزميات التطورية	2	2	1	0	5	3	180	6
AIE317	Artificial Intelligence in Medicine	الذكاء الاصطناعي في الطب	2	2	1	0	5	2	120	4
AIE322	Advanced Machine Learning	تعلم الآلة المتقدم	2	2	1	0	5	3	180	6
AIE323	Data Mining	التنقيب في البيانات	2	2	1	0	5	3	180	6
AIE332	Deep Learning	التعلم العميق	2	2	1	0	5	3	180	6
AIE342	Advanced Methods for Data Analysis	طرق متقدمة لتحليل البيانات	2	2	1	0	5	3	180	6

Code	Course Name	اسم المقرر	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
AIE343	Machine Learning for Text Mining	تعلم الآلة للتنقيب في النص	2	2	1	0	5	3	180	6
AIE351	Robotics Design	تصميم الروبوتات	2	2	1	0	5	3	180	6
AIE392	Field Training 2 in Artificial Intelligence Engineering	تدريب ميداني ٢ في هندسة الذكاء الاصطناعي	2	0	0	0	2	2	60	4
AIE417	Selected Topics in Artificial Intelligence 1	موضوعات مختارة في الذكاء الاصطناعي ١	2	2	1	0	5	3	180	6
AIE418	Selected Topics in Artificial Intelligence 2	موضوعات مختارة الذكاء الاصطناعي ٢	2	2	1	0	5	3	180	6
AIE419	Artificial Intelligence for Computer Games	الذكاء الاصطناعي لألعاب الكمبيوتر	2	2	1	0	5	3	180	6
AIE424	Intelligent Decision Support Systems	نظم دعم القرار الذكي	2	2	1	0	5	3	180	6
AIE425	Intelligent Recommender Systems	أنظمة التوصية الذكية	2	2	1	0	5	3	180	6
AIE426	Decision Making Under Uncertainty	صنع القرار مع مبدأ الريبة	2	2	1	0	5	3	180	6
AIE427	Statistical Pattern Recognition	التعرف على الأنماط	2	2	1	0	5	3	180	6
AIE444	Question Answering Systems	نظم الإجابة على الأسئلة	2	2	1	0	5	3	180	6
AIE452	Cognitive Robotics	الروبوتات المعرفية	2	2	1	0	5	3	180	6
AIE453	Planning Techniques for Robotics	تقنيات التخطيط للروبوتات	2	2	1	0	5	3	180	6
AIE454	Robot Kinematics & Dynamics	كينماتيكا وديناميكا الروبوت	2	2	1	0	5	3	180	6
AIE455	Robot Mapping & Localization	روبوتات الخرائط والتوطين	2	2	1	0	5	3	180	6
AIE456	Human Robot Interaction	تفاعل الانسان والروبوت	2	2	1	0	5	3	180	6
AIE457	Mobile Robot Development	خوارزميات الروبوت المحمول	2	2	1	0	5	3	180	6
AIE493	Graduation Project 1	مشروع التخرج ١	2	3	0	0	5	3	180	6
AIE494	Graduation Project 2	مشروع التخرج ٢	2	3	0	0	5	3	180	6

Department of Biomedical Informatics

قسم المعلوماتية الطبية الحيوية

Code	Course Name	اسم المقرر	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
BMD191	Field Training 1 in Biomedical Informatics	تدريب ميداني ١ في المعلوماتية الطبية الحيوية	2	0	0	0	2	2	60	4
BMD241	Human Physiology	علم وظائف الأعضاء البشرية	2	2	1	0	5	3	180	6
BMD292	Field Training 2 in Biomedical Informatics	تدريب ميداني ٢ في المعلوماتية الطبية الحيوية	2	0	0	0	2	2	60	4
BMD310	Introduction to Programming for Biostatisticians	مقدمة في البرمجة للإحصائيين الحيويين	1	2	0	0	3	2	135	4
BMD311	Introduction to Bioinformatics	مقدمة في المعلوماتية الحيوية	2	2	1	0	5	3	180	6
BMD312	Clinical Informatics	المعلوماتية العلاجية	2	2	1	0	5	3	180	6
BMD313	Dental Informatics	معلوماتية الأسنان	2	2	0	0	4	2	120	4
BMD320	Data Analysis Using R	تحليل البيانات باستخدام R	2	2	0	0	4	3	150	4
BMD351	Biomedical Data Acquisition	اكتساب البيانات الطبية الحيوية	2	2	1	0	5	3	180	6
BMD361	Biomedical Statistics	الاحصاءات الطبية الحيوية	2	2	1	0	5	3	180	6
BMD413	Structural Bioinformatics	المعلوماتية الهيكلية	2	2	1	0	5	3	180	6
BMD414	Selected Topics in Biomedical Informatics 1: Data Analysis & Visualization	موضوعات مختارة في المعلوماتية الطبية الحيوية ١: تحليل البيانات والتصور	2	2	1	0	5	3	180	6
BMD415	Selected Topics in Biomedical Informatics 2: Data Mining & Machine Learning	موضوعات مختارة في المعلوماتية الطبية الحيوية ٢ : التنقيب في البيانات وتعلم الآلة	3	2	2	0	7	3	195	6
BMD421	Biomedical Information Systems	نظم المعلومات الطبية الحيوية	2	2	1	0	5	3	180	6

Code	Course Name	اسم المقرر	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
BMD422	Systems Biology	علم أحياء الأنظمة	2	2	1	0	5	3	180	6
BMD431	Medical Image Informatics	الصور الطبية المعلوماتية	2	2	1	0	5	3	180	6
BMD452	Biomedical Text Processing	معالجة النصوص الطبية الحيوية	2	2	1	0	5	3	180	6
BMD462	Bio-Inspired Computing	الحوسبة المستوحاة من الحيوية	2	2	1	0	5	3	180	6
BMD493	Graduation Project 1	مشروع التخرج ١	2	3	0	0	5	3	180	6
BMD494	Graduation Project 2	مشروع التخرج ٢	2	3	0	0	5	3	180	6

كلية علوم وهندسة الحاسبات

FACULTY OF COMPUTER SCIENCE & ENGINEERING

Department of Computer Science & Engineering

CSE011 Computer Skills

0 Cr. Hrs. = (*1* LCT + *0* TUT + *2* LAB + *0* OTH) – SWL = *60* – ECTS = *0*

Prerequisite - - -

Types of computers – Computer hardware and software components – Data representation and number systems – Introduction to networking – Introduction to internet –Algorithm development – algorithm representation – flowcharts – stepwise refinement – problem solving methods and tools.

CSE012 Scientific Applications of Computers

2 Cr. Hrs. = (*2* LCT + *0* TUT + *0* LAB + *0* OTH) – SWL = *105* – ECTS = *4*

Prerequisite - - -

Windows-based operating environment, electronic mail, the World Wide Web, computerized library skills. Word processing and electronic spreadsheets. Desktop Publishing and Computer Graphics.

CSE013 Introduction to Information systems & technology

2 Cr. Hrs. = (*2* LCT + *0* TUT + *0* LAB + *0* OTH) – SWL = *105* – ECTS = *4*

Prerequisite - - -

An Introduction to Information Systems. Computer Hardware. Data Resource Management. Telecommunications and Networks. Electronic Business Systems. Enterprise Business Systems& Electronic Commerce Systems. Decision Support Systems.

CSE014 Structured Programming

3 Cr. Hrs. = (*2* LCT + *0* TUT + *3* LAB + *0* OTH) – SWL = *180* – ECTS = *6*

Prerequisite - - -

•. Primitive data types, control structures, loops and decisions. •. Functions and parameter passing, top-down design, arrays. •. Mechanics of compiling, running, testing, and debugging programs.

CSE015 Object Oriented Programming

3 Cr. Hrs. = (*2* LCT + *0* TUT + *3* LAB + *0* OTH) – SWL = *180* – ECTS = *6*

Prerequisite CSE014

Introduction to the Unified Modelling Language. Classes, subclasses, and inheritance. Concepts of encapsulation and information hiding. Polymorphism and Abstract classes. Implementation and reuse of application programming interfaces API. Operator overloading, delegates, and events. Searching and Sorting and Algorithms.

CSE081 Digital Branding

3 Cr. Hrs. = (*2* LCT + *0* TUT + *3* LAB + *0* OTH) – SWL = *180* – ECTS = *6*

Prerequisite - - -

Compare and contrast marketer control versus consumer control.

Examine consumer empowerment. Explore consumer participation and engagement. Examine content marketing and determine its value. Explore different strategies for content distribution. Identify some of the challenges associated with content marketing. Explore the concept of owned media and its importance to brands. Investigate the impact of owned media decision making. Examine a range of owned media assets and determine their value. Investigate brand engagement and why is it important. Examine and evaluate a range of engagement platforms. Identify different levels of engagement. Explore strategies for shaping earned media.

CSE111 Data Structures

3 Cr. Hrs. = (***2*** LCT + ***0*** TUT + ***3*** LAB + ***0*** OTH) – SWL = ***180*** – ECTS = ***6***

Prerequisite **CSE015**

Arrays, stacks, queues, lists, doubly linked lists. Trees, dynamic storage allocation, graphs. Advanced Searching and Sorting and Algorithms.

CSE112 Design & Analysis of Algorithms

4 Cr. Hrs. = (***3*** LCT + ***3*** TUT + ***0*** LAB + ***0*** OTH) – SWL = ***195*** – ECTS = ***8***

Prerequisite **CSE111**

Introduction. Fundamental techniques for designing and analyzing algorithms. Asymptotic analysis. Divide-and-conquer algorithms. Recurrences. Merge sort. Linear-time median. Greedy algorithms. Quick-sort algorithm. Dynamic programming. Graph algorithms. Graph search and Dijkstra's algorithm. Minimum Spanning Trees. Randomized algorithms. Hashing.

CSE113 Electric &Electronic Circuits

3 Cr. Hrs. = (***1*** LCT + ***2*** TUT + ***3*** LAB + ***0*** OTH) – SWL = ***180*** – ECTS = ***6***

Prerequisite **PHY212**

Circuits and Circuit Elements: Concept of a circuit, V-I relationships for R, L and C. Series and parallel combinations of elements. Voltage and current sources. Stored energy, power dissipation Circuit Laws and Theorems: Kirchoff's Laws, loop and nodal analysis. Time Varying Signals: The sinusoid, amplitude, phase, frequency. Response of L, C R to AC. AC Circuits: Phasor representation. Complex number notation. Analysis of ac circuits. Transient Response: First order R L and R C response. Stored energy. Time constants. Basic Diode Behavior. Diode Applications: Clipping, clamping, rectifiers, simple smoothing.

CSE131 Logic Design

3 Cr. Hrs. = (***2*** LCT + ***2*** TUT + ***2*** LAB + ***0*** OTH) – SWL = ***195*** – ECTS = ***6***

Prerequisite **- - -**

Number systems and coding. Boolean Algebra. Combinational circuits. Decoders and multiplexers. Synchronous sequential circuits. Counters, Registers and Memory. Advanced Arithmetic Circuits.

CSE132 Computer Architecture &Organization

3 Cr. Hrs. = (***2*** LCT + ***1*** TUT + ***2*** LAB + ***0*** OTH) – SWL = ***165*** – ECTS = ***6***

Prerequisite **CSE131 or ELE234**

Design of Basic Computer; Design concepts of Processors; Basic Assembly Language; Design of Channels and Controllers; Interconnections; Memory Structures and Design; Memory Management; Cache Memory Systems; firmware Design; Reliability; Testing and Fault Tolerance; CISC Computer; RISC Computers; Computer Interfacing; Computer Architecture Examples.

CSE191 Field Training 1 in Computer Science

2 Cr. Hrs. = (**2** LCT + **0** TUT + **0** LAB + **0** OTH) – SWL = **60** – ECTS = **4**

Prerequisite **- - -**

This course allows students to combine academic knowledge with practical experience by providing a range of internships in computer-related disciplines, such as Computer networks, software engineering, information systems, and information technology.

CSE192 Field Training 1 in Information Technology

2 Cr. Hrs. = (**2** LCT + **0** TUT + **0** LAB + **0** OTH) – SWL = **60** – ECTS = **4**

Prerequisite **- - -**

Hands on professional network administration, where students can install, support and manage the networks and computer systems. They implement and maintain network hardware and software, troubleshoot network problems, and ensure network security, availability & performance standards.

CSE211 Web Programming

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **165** – ECTS = **6**

Prerequisite **CSE015**

Web architecture and HTTP: History and architecture of the World Wide Web. •. Overview of the Hyper Text Transfer Protocol, other related protocols. •. Client-side scripting and Server-side scripting. •. Database Connectivity. •. Sending mail, cookies, and sessions.

CSE221 Database Systems

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **CSE111**

Evolution of database management systems. Relational Data Model and Relational Algebra. Structured Query Language. Entity Relationship Modelling and Design. Tables Normalization. Forms/ Reports/ Menus Implementation.

CSE232 Operating Systems Administration

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **CSE014**

Introduction to Operating systems such as Windows and LINUX/UNIX environments, administration of computer servers, usage of script languages in various operating systems, back up procedures and remote access, communication protocol used among different systems, managing information and system security.

CSE233 Operating Systems

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **165** – ECTS = **6**

Prerequisite **CSE111**

Principles of operating systems, sequential processes, concurrent processes, concurrency, functional mutual exclusion, processor cooperation and deadlocks, processor management. Control and scheduling of large information processing systems, Resource allocation, dispatching, processor access methods, job control languages, Memory management, memory addressing, paging and store multiplexing, Multiprocessing and time sharing, batch processing, Scheduling algorithms, file systems, protection and security, design and implementation methodology, performance evaluation and case studies.

CSE234 Embedded Systems

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **CSE132**

Introduction: the importance of microcontrollers, the roles and functions of microcontrollers. Acquaintance with microcontrollers and their simulators and debuggers. Understanding different addressing modes. Programming, debugging, and simulating assembly language programs. Developing a prototype for an embedded system. Interrupts and serial I/O. Memory Expansion. Microcontroller interfaces.

CSE241 Security of Information Systems

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **150** – ECTS = **6**

Prerequisite **- - -**

Common attacking techniques. Common security policies. Basic cryptographic tools. Authentication and authorization. Access control. Software security. Operating system security. Legal and ethical issues in information systems security.

CSE242 Cryptography

3 Cr. Hrs. = (**2** LCT + **2** TUT + **1** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **CSE112**

Computer Security Concepts: OSI security architecture, security attacks, security services, security mechanisms, network security model. Classical Encryption Techniques: symmetric cipher model, cryptanalysis, substitution techniques (Ceaser, Monoalphabetic, Play-fair, Hill cipher), transposition techniques, rotor machines, steganography. BlockCiphers and the Data Encryption Standard (DES): block cipher principles, Data Encryption Standard (DES), strength of DES, differential and linear cryptanalysis. Public-Key Cryptography and RSA: principles of public-key cryptosystems, RSA algorithm. Diffie-Hellman Key Exchange: Discrete logarithm, key exchange and generation algorithm, attacks on Diffie-Hellman protocol. Cryptographic Hash Functions: applications of cryptographic hash functions, requirements and security, hash functions based on Cipher Block Chaining (CBC), Secure Hash Algorithm (SHA). Digital Signatures: essential elements, limitations of symmetric key, Digital Signature Standard (DSS). Distribution of public keys and X.50. Network Security Protocols: Authentication, key exchange and key distribution protocols. Network Security Standards: IP security (IPsec), Secure Sockets Layer (SSL), Transport Layer Security (TLS), Hypertext Transfer Protocol Secure (HTTPS). Security analysis: Use of formal tools, e.g., Automated Validation of Internet Security Protocols and Applications (AVISPA).

CSE243 Secure Programming

3 Cr. Hrs. = (**2** LCT + **0** TUT + **3** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **CSE241 or CSE344**

Introduction. Secure code development principles. Best practices. Security strategies and controls. Malicious code and defensive techniques. Code review and testing. Security documentation and error messages. Secure coding techniques. Access control. Input validation. Threat identifications and modeling. Vulnerability analysis. Automated code analysis. Risk assessment. Secure code development life-cycle: development, maintenance, and refinement. Knowledge catalog: principles, guidelines, vulnerabilities, attack patterns, and historical risks. Coding errors. Breaking software. Web-applications threats and vulnerabilities.

CSE251 Software Engineering

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **CSE015**

Principles and techniques for the design and construction of reliable, maintainable, and useful software systems; Software life cycle, requirements specifications, and verification and validation issues; Implementation strategies (e.g. top-down, bottom-up, teams), support for reuse, and performance improvement; Concepts of software engineering: requirements definition, modularity; structured design; data specifications; functional specifications; verification, documentation; software maintenance; Software support tools;. Software project organization; quality assurance; management and communication skills.

CSE261 Computer Networks

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **165** – ECTS = **6**

Prerequisite **CSE014, CSE132**

Architecture and protocols of computer networks; Protocol layers; network topology; Data-communication principles, including circuit switching, packet switching and error control techniques; Sliding window protocols, protocol analysis and verification; Routing and flow control; Local and wide area networks; Network interconnection; Client-server interaction; Emerging networking trends and technologies; topics in security and privacy.

CSE262 Introduction to Telecommunications, Networking and Security

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **CSE014**

Introduction to telecommunications, networking and security concepts. Technologies and protocols for networked applications and services, Command line and graphical interfaces for network configuration, Implementation and configuration of networked applications and services, Securing wireless and wireline networks, network services, and protocols, Network analysis, Network virtualization, Integration and troubleshooting of networked apps, network security methods and services.

CSE271 Introduction to Parallel Computing

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **CSE112**

Introduction to Parallelism. Parallel Programming. Parallel Architectures. Parallel Algorithms. Parallel Applications. Other Parallel Models.

Interfacing techniques. Interfacing requirements. A typical microcontroller system is utilized in this course with typical software-based applications. Interfacing with USB, I2C, SPI, CAN, LIN.

CSE272 Parallel &distributed Systems

3 Cr. Hrs. = (**2** LCT + **1** TUT + **1** LAB + **0** OTH) – SWL = **150** – ECTS = **6**

Prerequisite **CSE132**

Motivations for parallel programming. Instruction Level Parallelism (ILP). Parallel Random Access Machines (PRAM). Cluster computing and grid computing. Message passing systems and applications.

Message Passing Interface (MPI) and configuration of MPI cluster. MPI programming algorithms and implementation of PRAM through MPI. Peer-to-Peer (P2P) systems, mobile agents. GPUs, Multi-Core, Distributed file systems. Distributed coordination systems. Replication and consistency. Fault tolerance. Grid computing paradigm. Cloud computing: properties and characteristics, service models, deployment models.

CSE281 Image Processing

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **CSE015, MAT212**

Digital image fundamentals; Image enhancement in the spatial domain: grey level transformation; Histogram processing; Spatial filters; Image enhancement in frequency domain: D Fourier transform; Other transforms; Smoothing filters; Sharpening filters; Image restoration; Noise model; Estimating the degradation function; filters; Geometric transformations; Image segmentation: detection of discontinuities; edge linking and boundary detection; Thresholding; Region based segmentation ; Morphological image processing : operation concepts ; some basic algorithms, Image Compression.

CSE291 Field Training 1 in Computer Engineering

2 Cr. Hrs. = (**2** LCT + **0** TUT + **0** LAB + **0** OTH) – SWL = **60** – ECTS = **4**

Prerequisite **- - -**

Hands-on introduction to computer engineering practice and research, including computer hardware, robotics, and embedded systems. Internship in software, cloud computing, mobile computing, embedded systems, and information network companies. Establish a student organization that works with communities to develop/apply technology to serve social community.

CSE292 Field Training 2 in Computer Science

2 Cr. Hrs. = (**2** LCT + **0** TUT + **0** LAB + **0** OTH) – SWL = **60** – ECTS = **4**

Prerequisite **CSE191**

This course allows students to combine academic knowledge with practical experience by providing a range of internships in computer-related disciplines, such as Computer networks, software engineering, information systems, information technology, and more advanced topics.

CSE293 Field Training 2 in Information Technology

2 Cr. Hrs. = (**2** LCT + **0** TUT + **0** LAB + **0** OTH) – SWL = **60** – ECTS = **4**

Prerequisite **CSE192**

Hand on professional computational infrastructures analysis (e.g., networks, servers, and DBs), where students can apply modern techniques for analyzing and modeling the structure and dynamics of complex systems, including major attacks. The main focus will be on applying statistical algorithms, methods, and visualization techniques.

CSE311 Design of Compilers

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **150** – ECTS = **6**

Prerequisite **CSE015**

Fundamental concepts in automata theory and formal languages including grammar, deterministic and nondeterministic finite automata, regular expression, formal language, pushdown automaton, Turing machines, the halting problem, diagonalization and reduction, decidability, Rice's theorem, P, NP, and NP-completeness. Systems software, compilers, interpreters. Byte-codes. Lexical analysis: interface with input, parser and symbol table, token, lexeme and patterns. Syntax

analysis: context-free grammars, ambiguity, precedence, top-down parsing, recursive descent parsing, transformation on the grammars, predictive parsing. Bottom up parsing, operator precedence grammars, LR parsers. Regular expressions and semantics. Error detection, type-checking and run-time environments. Code generation, code optimizations, code improvement techniques.

CSE312 Advanced Web Programming

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **165** – ECTS = **6**

Prerequisite **CSE211**

Introduction to HTML5 and JavaScript. HTML5 and Forms. HTML5; Intro to Flash, Canvas, Local storage, and Geolocation. HTML5 and JavaScript. HTML5, CSS Animation. Adobe Muse and Dreamweaver.

CSE313 Mobile Programming

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **165** – ECTS = **6**

Prerequisite **CSE015**

Mobile application development frameworks, Design techniques, Methodologies for mobile application development, Android development w/ Java, iOS development w/ Swift, Using Native React, RESTful and Non-RESTful apps, Creating Web/Cloud services, Mobile sensors, Security and trust management, Privacy and ethics, Usability.

CSE314 Android and iOS Development

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **165** – ECTS = **6**

Prerequisite **CSE313**

Introduction to Android and iOS. Android Activities and UI Design. Advanced UI Programming. iOS Architecture and SDK Framework. Case studies and projects.

CSE315 Discrete Mathematics

3 Cr. Hrs. = (**2** LCT + **2** TUT + **0** LAB + **0** OTH) – SWL = **150** – ECTS = **6**

Prerequisite **- - -**

Propositional Logic. Predicate Logic and Quantification. Methods of Proof. Sets and Functions. Arithmetic Algorithms. Growth of Functions. Computational Complexity of Algorithms. Integer properties and Matrices. Mathematical Induction. Recursion. Sequences and Summations. Program Correctness. Graphs and its Applications. Trees and its Applications. Languages and Grammars. Finite-State Machines. Automata and Language Recognition. Turing Machines.

CSE321 Societal and Ethical Issues in Information Technology

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **---**

Overview of philosophical and professional ethics in computing, Societal and ethical issues in computing including privacy, intellectual property, computer security, computer crime, product liability, and the societal, environmental and economic impacts of computers. Students develop skills in assuming and defending positions on societal and ethical issues through oral presentations and written reports.

CSE322 Big Data Analytics 1

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **AIE121**

This course provides a broad and practical introduction to big data: data analysis techniques including databases, data mining, and machine learning; data analysis tools including spreadsheets, relational databases and SQL, Python, and R; data visualization techniques and

tools; pitfalls in data collection and analysis. Tools and techniques are hands-on but at a cursory level, providing a basis for future exploration and application.

CSE323 Advanced database systems

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **CSE221**

Query processing and optimization. Database tuning. Transaction processing. Concurrency control. Database recovery. Object databases: standards, languages, and design. Object-relational databases. Database security. Distributed database systems: architecture, data fragmentation, distributed read/update transparency, access primitives, integrity constrains, distributed database design, queries, optimization, concurrency and reliability control. XML, semi-structured, federated, and Internet databases. Data warehousing. Introduction to data mining.

CSE324 IT Management and Operations

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **---**

This course gives an overview of IT operations and the management of IT operations, Roles of IT Operations, Recent information technologies (virtualization, cloud, etc), monitoring and tuning IT operations, Governance of IT operations.

CSE325 Database Systems Administration

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **CSE221**

Introduction to the knowledge, skills, and tools needed to successfully administer database systems. The conceptual and operational tools for analysis and resolution of problems such as performance, recovery, design, and technical issues are provided. Tools used to assist in the administration process, administer, backup, and recover Database instances, tuning and enhancing the performance of the Database. Also, introduces deployment and administration of a database on a server or the cloud.

CSE335 Real Time & Embedded Systems Design

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **CSE234**

Introduction to automotive embedded networking - Automotive CAN network simulation using CANoe (Vector Germany) evaluation version - Principles of CAPL script to simulate external events and network communications - Introduction to CAN bus protocol - TIVA C embedded development using CAN bus - MISRA static code checking guidelines - MISRA and Code Composer Texas Instruments tools - Real Time Operating System on TIVA C - OSEK network management standard - OSEK NM simulation using CANoe - OSEK state machine C development - Introduction to AutoSar Automotive embedded development standard - AutoSar Real Time Environment (RTE) - AutoSar Basic Software (BSW) - AutoSar Software Components (SWC).

CSE344 Introduction to Cyber Security

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **- - -**

This course provides an overview of Cyberspace, defines the scope of Cybersecurity, and addresses information classification and system compartmentalization. Course includes an appreciation of information

confidentiality, integrity, and availability, and covers Cybersecurity architecture, strategy, services, hardware, software, and cloud services. The course also examines national security issues, critical infrastructure, and the potential for cybercrime and cyber terrorism, as well as the need for corporations to align their security with business needs and consider the threat from malicious employees, contractors, and/or vendors.

CSE351 Software Project Management

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **- - -**

This course develops a foundation of concepts and solutions that supports the planning, scheduling, controlling, resource allocation, and performance measurement activities required for successful completion of an information system project.

CSE352 Systems Analysis &Design

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **165** – ECTS = **6**

Prerequisite **CSE251**

The Systems Development Environment. Rapid Application Development. Introduction to agile methodologies. Managing the Information Systems project. Automated Tools for Systems Development. Determining & Structuring System Requirements. Structuring System Data & Logic Requirements. Designing Distributed and Internet Systems.

CSE353 Interaction Design

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **---**

Study of and practice with processes, principles, tools, models, and techniques for designing interactions between humans and digital products and systems. Topics include physiological and psychological factors affecting interaction design, interaction design process, interaction models, styles and paradigms, design notations and representations, prototyping, and interaction design evaluation.

CSE361 Networks Virtualization and Administration

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **CSE261 or CSE262**

Introduction to Data-center networking, Overview on Virtualization technologies and techniques, Software Defined Networking, Network Function Virtualization concepts, impact of virtualization on the design, deployment and management of ICT services. Administration of virtualized networks.

CSE362 Industrial Networks

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **CSE261 or CSE262**

The course is presented through a layered top-down approach starting from the application layer down to the physical layer, focusing on basic networking concepts and typical application layer examples. Focusing on the Internet and the fundamentally important issues of networking, this course provides a foundation for students interested in computer science and electrical engineering, without requiring extensive knowledge of programming or mathematics. A typical outline of the course goes by the following sequence: Application layer (e.g. e-mail, the Web, PHP, wireless Web, MP3, and streaming audio). Transport layer essentials and requirements. Network layer functions and fundamentals of routing, congestion control, QoS, IPv4, and IPv. Data

link layer and MAC Sublayer with emphasis on gigabit Ethernet, 80. 11, broadband wireless, and switching. Physical layer (e.g. copper, fiber, wireless, satellites, and Internet over cable). The course dissects and depicts the principles associated with each layer and then focuses on Fieldbus networks, Control Area Networks (CAN, LIN, FLEXRAY) and SCADA systems.

CSE363 Cloud Computing

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **CSE261 or CSE262**

Trends of computing, introduction to distributed computing. Introduction to Cloud Computing: Cloud computing properties and characteristics, service models, deployment models; Attributes of Cloud computing: Multi-tenancy; a single instance of software or other computing resource serving several clients, massive scalability; ability to support hundreds of thousands of clients at the same time, elasticity. Infrastructure-as-a-Service (IaaS): Introduction to IaaS, resource (i.e., server, storage and network) virtualization, case studies; Platform-as-a-Service (PaaS): Introduction to PaaS. Cloud platform, management of computation and storage, case studies; Software-as-a-Service (SaaS): Introduction to SaaS, Web services, Web. 0, Web OS, case studies.

CSE373 Parallel programming

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **CSE015**

Introduction. Parallel versus distributed algorithms. Message passing and shared memory. Parallel algorithm design: parallel graph algorithms, parallel searching and sorting algorithms. Parallel computational algorithms. Basic distributed problems and protocols. Synchronous computation: communicators, pipeline, transformers, waiting, guessing, synchronous problems. Algorithms in systems with no failures. Election: election in trees, rings, mesh networks, cube networks, and complete networks, universal election protocols. Message routing: shortest path routing, coping with changes, routing in static systems. Distributed set operations: distributed selection, distributed sorting. Stable properties detection. Continuous computations. Computing in presence of faults: faults and failure, modeling faults, the crushing impact failure, localized entity and link failures, ubiquitous faults. Failure detectors. Parallel and distributed matrix algorithms. Optimization in parallel and distributed algorithms. Complexity analysis of distributed and parallel algorithms. Applications.

CSE382 Computer Graphics

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **CSE281**

Introduction to computer graphics hardware, algorithms, and software. Graphics Programming, OpenGL. Displaying images. 3D transformations. Light and shading. Ray tracing. Hidden surface removal. Color technology. Image morphing. Texture mapping. Line drawing. Local illumination models. Curves and Surfaces. Geometric Modelling. Animation.

CSE383 Computer Vision

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **CSE281**

Local Feature Extraction. Projective Geometry. Stereo Vision. Point Matching. 3D Reconstruction. Motion Detection. Object Recognition.

CSE392 Field Training 2 in Computer Engineering

2 Cr. Hrs. = (**2** LCT + **0** TUT + **0** LAB + **0** OTH) – SWL = **60** – ECTS = **4**

Prerequisite **CSE291**

Hands-on advanced topics in computer engineering practice and research, including computer hardware, robotics, and embedded systems. Internship in software, cloud computing, mobile computing, embedded systems, and information network companies. Establish a student organization that works with communities to develop/apply technology to serve social community.

CSE411 Theory of Computation &Compilers

3 Cr. Hrs. = (**2** LCT + **2** TUT + **1** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **CSE014**

Finite automata, Regular languages, converting DFA to NFA, Context-free languages, Pushdown automata, and Turing Machine. Lexical analysis; parsing theory; symbol tables; type systems; scope; semantic analysis; intermediate representations; runtime environments; code generation; and basic program analysis and optimization.

CSE424 Data Warehousing

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **CSE323**

Introduction: Comparison of operational and decisional systems; Metadata. Data warehousing architectures: Corporate Information Factory; DW. 0. Multidimensional modeling and OLAP tools: Structure; Integrity constraints; Operations; Advanced concepts. Database optimization: Basic concepts; Phases and goals. Database physical design for analytical queries: Star-join and join indexes; Bitmaps; Materialized views; Implementations (relational and NOSQL). Extraction, Transformation and Load: Data quality; Integration; ETL management.

CSE425 Big Data Analytics 2

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **CSE322**

Map Reduce; Clustering algorithms for high-dimensional data; predictive analytics; Dimensionality reduction; Application of machine learning algorithms for analyzing structure of large graphs like social network graphs Technologies for extracting important properties of large datasets.

CSE426 Selected Topics in Data Science

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **CSE322**

The course provides topics that are selected from different recent trends in Data Science that are not covered in the description of the courses listed in the curriculum.

CSE427 Selected Topics in Big Data

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **CSE322**

The course provides topics that are selected from different recent trends in Big Data that are not covered in the description of the courses listed in the curriculum.

CSE436 Selected Topics in Embedded Systems

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **CSE234**

Topics which are not included in the curriculum and seems to be needed should be suggested as an elective course by the Department Faculty Council.

CSE445 Selected Topics in Information Security

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **CSE241 or CSE344**

Topics are selected from different areas in Information Security that are not covered in the description of the courses listed in the curriculum. This course will cover recent trends and issues in the field of Information Security and will be chosen at the discretion of the Program Administration Council and the Faculty Council.

CSE446 Information &Computer Networks Security

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **CSE261 or CSE262, (CSE241 or CSE344)**

Attacks and threats, symmetric key cryptography, public key cryptography, authentication protocols, digital signature, viruses, worms, Trojan horses, malicious programs, computer crimes, web-security, firewalls, intrusion detection, TLS, IPSec, SET, digital homeland security, offensive and defensive tools, security issues in wireless technologies and mobile computing, ethics and hacking in laws.

CSE447 Selected Topics in Computer Security

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **CSE241 or CSE344**

Topics are selected from different areas in Information Security that are not covered in the description of the courses listed in the curriculum. This course will cover recent trends and issues in the field of Information Security and will be chosen at the discretion of the Program Administration Council and the Faculty Council.

CSE448 Cyber Forensics

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **- - -**

Computer and Cyber Forensic Basics- Introduction to Computers, Computer History, Software, Hardware, Classification, Computer Input-Output Devices, Basic Computer Terminology, Internet, Networking, Computer Storage, Cell Phone / Mobile Forensics, Computer Ethics and Application Programs, Cyber Forensic Basics-Introduction to Cyber Forensics, Storage Fundamentals, File System Concepts, Data Recovery, Operating System Software and Basic Terminology.

CSE453 Building Information Systems

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **CSE 251**

Define the information systems basics that support different managerial levels using modern environments. -List deep understanding of database design and optimization in local/remote and centralized /distributed deployment models. -Propose and evaluate the information system components and infrastructure. -Operate modern computing, programming, information technology, and database design skills to

build information system. -Implement comprehensive computing knowledge of information systems to solve practical information problems. -Work in groups and manage team, time and organizational skills to build information systems. -Retrieve the modern information efficiently in building information systems. -Use the modern computing facilities in building information systems.

CSE454 Advanced Software Engineering

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **CSE251**

Differences between structured and object-oriented paradigms. The Unified Modeling Language (UML). Use-case modeling. Class modeling: noun extraction, Class-Responsibility-Collaboration (CRC) cards. Dynamic modeling. State diagrams. Testing during the object-oriented analysis phase. CASE tools for object-oriented analysis and design. Object-oriented design: interaction diagram, detailed class diagram, clients of objects, detailed design and program description languages.

CSE455 Selected Topics in Software Engineering

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **CSE251**

Topics which are not included in the curriculum and seems to be needed should be suggested as an elective course by the Program Administration Council and the Faculty Council.

CSE463 Advanced Networking

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **CSE261 or CSE262**

Network architectures/technologies, Principles of efficient and secure networks for providing voice, data, and video and mobility applications. Hands-on knowledge of network components (host, router, switch, hub, bridges, etc.), network protocols (HTTP, IP, TCP, SQL, etc.), network interfaces (Ethernet, Serial, USB, Wi-Fi, BT, Zigbee, Cellular, etc.), network facilities (wired, wireless), services (DHCP, DNS, Proxy, etc.), network analysis and troubleshooting, network management and policies (maintenance, operations and provisioning), network security (hardening, firewalls, IDS, IPS, etc.) and networked applications (remote access, email server, web server, multicasting/streaming, etc.).

CSE464 Internet of Things

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **CSE261 or CSE262**

Introduction to IoT. Concepts and architecture. Connected devices. Managing IoT resources in the cloud. Fog computing. Programming frameworks. Virtualization on Embedded boards. Collecting and managing data. Reliability, privacy, and security. IoT applications.

CSE465 Selected Topics in cloud computing

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **CSE363**

Topics which are not included in the curriculum and seems to be needed should be suggested as an elective course by the Department Faculty Council.

CSE466 Selected Topics in IoT

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **CSE464**

Topics which are not included in the curriculum and seems to be needed should be suggested as an elective course by the Department Faculty Council.

CSE467 Client/Server Technologies &Applications

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **- - -**

Design of Information Systems and Fundamentals of the development of application software systems. The presentation of tools and methodology development are in line to obtain the skills to identify and solve specific information problems. Selected teaching strategy is to create skills for students to explore literature and to explore the problem originated information and finding the right tools to solve it. Overall, the course aims to develop additional skills to build complex three-layer systems based on client / server technologies. Requirements: students to have training in programming, using programming environments and knowledge of basic algorithms and data structures.

CSE468 Web Server Administration

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **165** – ECTS = **6**

Prerequisite **CSE262**

Introduction to basic Linux command, Basic Windows Server commands; Administration of Apache Web Server and Internet Information Server, Domain Name Service (DNS) Configuration, Web server Management, Web server environment analysis and performance optimization.

CSE474 Distributed Information Systems

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **CSE221, CSE261 or CSE262**

An introduction to computer systems. Introduction to Internetworking, protocol stacks and data transport. Design and operation of distributed systems and applications. Central concepts in distributed systems, including transparency, scalability, middleware, synchronization, failure handling, consistency, and parallelism. Operating systems, scheduling, processes, memory systems and cloud abstractions. Security considerations, basic cryptography and network security. Basic designs and constraints of Internet-of-Things, including energy, scalability, privacy, and semantical interoperability. Analysis and presentation of a network based distributed system. Examining running networks with observing tools.

CSE475 High Performance Computing

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **CSE271**

Overview of existing HPC software and hardware. Basic software design patterns for high performance parallel computing. CUDA for parallel computing on the Graphics Processing Unit (GPU). Message Passing Interface (MPI) parallel programming. OpenMP and POSIX threads solution to enable parallelism across multiple CPU cores. Standard algorithms utilizing parallelism. Matrix and vector operations. Collective communications. The use of Graphics Processing Units (GPUs) for general purpose computations (GPGPU). Multi-GPU and Multi-CPU solutions. Optimizing HPC-based programs. Designing GPU-based systems. Applications.

CSE476 Selected Topics in high performance computing

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **CSE475**

Topics which are not included in the curriculum and seems to be needed should be suggested as an elective course by the Department Faculty Council.

CSE484 Interactive Multimedia

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **CSE382**

Introduction to Multimedia Studies. Data Representation. Basic Compression Techniques. Video and Audio Data Compression Techniques. Multimedia Networks and QoS Support. Multimedia Wireless Networks, Heterogeneous Networks, and advanced QoS Support. Multimedia Applications. Topics in Multimedia Technologies.

CSE485 Computer Games Theory

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **165** – ECTS = **6**

Prerequisite **CSE382**

Preferences, Utility, and Goals. Strategic Form Non-Cooperative Games. Iterated Games. Extensive Form Non-Cooperative Games. Cooperative Games. Social Choice.

CSE486 Selected Topics in Computer Vision

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **165** – ECTS = **6**

Prerequisite **CSE383**

The course provides topics that are selected from different advanced and recent trends in Computer Vision that are not covered in the description of the courses listed in the curriculum.

CSE487 Mixed &Augmented Reality

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **165** – ECTS = **6**

Prerequisite **CSE382**

3D Vision. Approaches to Augmented Reality. Alternative Interface Paradigms. Spatial Augmented Reality. Lighting and Illumination Issues in Augmented Reality.

CSE488 Visualization &Animation

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **CSE382**

Perception and its applications. Graphical perception. Visual encoding principles. Interaction principles. Single-view methods. Multiple-view methods. Item reduction methods. Attribute reduction methods. Tabular data. Visualization toolkits. Graphs and trees. Flow visualization. Geo-spatial visualization. Volume visualization. Vector visualization. High-dimensional Visualization. Visualizing relational data. Design and evaluation. Visualizing structure. Visualizing time. Scaling. Key-framing. Storyboarding. Animation software. Spacing and timing. Digital animation techniques. 2D and 3D animatic, special effects design, 3D paint techniques and integration. Sequence planning, non-photorealistic rendering. Kinematics, physically based dynamics modeling. Motion capture. Scene composition, lighting, and sound track generation. Visual effects process. Texture-mapping, rendering and camera tracking techniques. Live action films.

CSE493 Graduation Project 1

3 Cr. Hrs. = (**2** LCT + **0** TUT + **3** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **SENIOR STANDING**

During the first of the two semesters, students will begin their work on the project and are expected to complete at least half the project by the end of the semester. Students will develop and work on their projects under faculty supervision.

CSE494 Graduation Project 2

3 Cr. Hrs. = (***2*** LCT + ***0*** TUT + ***3*** LAB + ***0*** OTH) – SWL = ***180*** – ECTS = ***6***

Prerequisite **CSE493**

During the second of the two semesters, students will continue their work on the project and are expected to complete the project by the end of the semester. Students will develop and work on their projects under faculty supervision.

Department of Artificial Intelligence Science & Engineering

AIE111 Artificial Intelligence

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **CSE015**

Knowledge representation and organization, Search strategies & inference methods, and. AI problem solving tools and techniques. Agent Architecture, Multi-Agent Systems, Reasoning with uncertain or incomplete knowledge;

AIE121 Machine Learning

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **AIE 111**

Linear Regression and regularization. Instance-based Learning and Decision Tree Induction. Maximum Likelihood (Linear and Logistic regression). Probabilistic (Bayesian) Inference (Linear regression, Logistic regression with the Laplace approximation, Intro to Sampling). Support Vector Machines. Artificial Neural Networks: perceptron, MLPs, back propagation, intro to Deep Learning. Ensemble learning, bagging, boosting, stacking, random forests. Clustering algorithms, k-means, Expectation-Maximization, Hierarchical Clustering, Dimensionality reduction techniques, SVD/PCA, Multi-dimensional scaling.

AIE122 Basics of Machine Learning and Data Science

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **MAT131**

Introduction to data science and machine learning landscapes, Data Collection and Manipulation, Developing and implementing programs using ready tools that facilitate model development and deployment to a variety of computing platforms.

AIE191 Field Training 1 in AI Science

2 Cr. Hrs. = (**2** LCT + **0** TUT + **0** LAB + **0** OTH) – SWL = **60** – ECTS = **4**

Prerequisite **- - -**

Granting students practical employability skills required in the AI Science job market by providing training in companies that working in natural language processing and smart home components manufacturers to eventually be able to develop intelligent software and systems.

AIE212 Knowledge-based Systems

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **AIE111**

Propositional and predicate logic, non-classical logic, computational intelligence (recap). Knowledge based systems (KBS): information

management, objectives, components, and types of KBS, expert systems. KBS Architecture: source of knowledge, types of knowledge, skills components, structure KBS, knowledge base, reasoning based on rules of inference mechanisms, forward and backward chaining, the reasoning in the presence of uncertainty, KBS based on fuzzy logic, application KBS, Semantic Web. Knowledge representation: representational models, predicate logic, rules, frames and objects, descriptive logic, semantic networks, ontologies, formal concepts, conceptual graphs. KBS development: development methodology, mechanisms of recovery and recycling of knowledge and tools to develop KBS: C Language Integrated Production System (CLIPS), Java Expert System Shell (JESS), Protégé, and Web Ontology Language (OWL).

AIE213 Optimization techniques

3 Cr. Hrs. = (**3** LCT + **3** TUT + **0** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **MAT312**

Linear Algebra and Matrices. Probability Theory Review. Linear Programming. One-dimensional Search Techniques. Gradient-based Techniques. Quasi-Newton Methods. Constrained Optimization. Non-linear Constrained Optimization.

AIE231 Neural Networks

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **AIE121**

Simple perceptron for classification, BackProp and Multilayer Perceptron for deep learning, Statistical Classification by deep networks, Regularization and Tricks of the Trade in deep learning, Error landscape and optimization methods for deep networks, Convolutional networks, Sequence prediction and recurrent networks, Bellman equation and SARSA, Variants of SARSA, Q-learning, n-step-TD learning, Policy gradient, Deep reinforcement learning: applications, Reinforcement learning and the brain.

AIE241 Natural Language Processing

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **AIE121**

Overview of NLP. Statistical Machine Translation. Word Alignment Models for Statistical MT. Machine Translation: Word Alignment, Parallel Corpora, Decoding, Evaluation. Modern MT Systems (Phrase-based, Syntactic). N-Grams, Final Project Discussion. Syntax and parsing. Competitive Grammar Writing. Dependency Parsing. Coreference Resolution. Computational Semantics.

AIE291 Field Training 1 in AI Engineering

2 Cr. Hrs. = (**2** LCT + **0** TUT + **0** LAB + **0** OTH) – SWL = **60** – ECTS = **4**

Prerequisite **- - -**

This is an advanced course that grants students practical employability skills required in the AI Science job market by providing training in companies that work in natural language processing and smart home components manufacturers to eventually be able to develop intelligent software and systems.

AIE292 Field Training 2 in AI Science

2 Cr. Hrs. = (**2** LCT + **0** TUT + **0** LAB + **0** OTH) – SWL = **60** – ECTS = **4**

Prerequisite **AIE191**

This is an advanced course that grants students practical employability skills required in the AI Science job market by providing training in companies that work in natural language processing and smart home components manufacturers to eventually be able to develop intelligent

software and systems.

AIE314 AI-based programming

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **AIE212**

Introduction to Artificial Intelligence, Classification and Regression Using Supervised Learning, Predictive Analytics with Ensemble Learning, Detecting Patterns with Unsupervised Learning, Building Recommender Systems, Logic Programming, Heuristic Search Techniques, Natural Language Processing, Probabilistic Reasoning for Sequential Data, Building A Speech Recognizer, Object Detection and Tracking, Artificial Neural Networks, Reinforcement Learning, Deep Learning with Convolutional Neural Networks, Genetic Algorithms, Building Games With Artificial Intelligence.

AIE315 Computational Logic

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **AIE111**

Topics include the syntax and semantics of Propositional Logic, Relational Logic, and Herbrand Logic, validity, contingency, un-satisfiability, logical equivalence, entailment, consistency, natural deduction (Fitch), mathematical induction, resolution, compactness, soundness, and completeness.

AIE316 Evolutionary Algorithms

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **AIE213**

Introduction: Simple example of Evolution, Comparison and Motivation. The Model of Biological Evolution. a. Genotypes and Phenotypes. b. Basics of the Neo-Darwinian Paradigm. Optimization. Evolutionary Algorithms – General Overview. Genetic Algorithms. a. Basic Algorithm. b. Schema Processing Interpretation of Genetic Algorithms. c. Schema Theorem. d. Convergence Velocity Perspective. e. Practical Applications: From Airline Crew Scheduling to Car Crash Optimization. Evolution Strategies. a. Basic Algorithms. b. Convergence Velocity Perspective. c. Practical Applications. d. Advanced Techniques (mixed-integer, multi objective). Genetic Programming. a. Basic Algorithm. b. Practical Applications. Advanced Topics. a. Mixed-Integer Representations. b. Self-Adaptation in Genetic Algorithms. c. Optimizing Evolutionary Algorithms.

AIE317 Artificial intelligence in medicine

2 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **120** – ECTS = **4**

Prerequisite **- - -**

an intensive introduction to artificial intelligence and its applications to problems of medical diagnosis, therapy selection, and monitoring and learning from databases.

AIE322 Advanced Machine Learning

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **AIE121**

This course emphasizes practical skills, and focuses on giving you skills to make these algorithms work. You will learn in depth the commonly used learning techniques including supervised learning algorithms (logistic regression, linear regression, SVM, neural networks), unsupervised learning algorithms (k-means, Gaussian Mixture Models, spectral clustering), reinforcement learning, as well as learn about specific applications such as anomaly detection and building recommender systems.

AIE323 Data Mining

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **AIE121**

Knowledge discovery in databases. Data mining process and Models. Data cleaning and preparation. Mining association rules, Classification, Prediction, and Clustering. Web mining and Text Mining. Data Warehouse and OLAP Technology for Data Mining. Model Evaluation and Cross Validation. Applications of data mining.

AIE332 Deep Learning

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **AIE 231**

Deep Learning: A revolution in Artificial Intelligence, Limitations of Machine Learning, what is Deep Learning? Advantage of Deep Learning over Machine learning, 3 Reasons to go for Deep Learning, Real-Life use cases of Deep Learning, Review of Machine Learning: Regression, Classification, Clustering, Reinforcement Learning, Under-fitting and Overfitting, Optimization.

AIE342 Advanced Methods for Data Analysis

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **AIE 323**

Introduction and regression, the truth about linear regression, Error and validation, Kernel regression, the bootstrap, Degrees of freedom, smoothing splines, Additive models, Inference with linear smoothers, Logistic regression, Generalized linear models, Principal components analysis, Other dimension reduction techniques, Clustering, High-dimensional regression, Time series.

AIE343 Machine Learning for Text Mining

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **AIE323, AIE241**

Basic algorithms: the classical algorithms for text analytics, such as pre-processing, similarity computation, topic modelling, matrix factorization, clustering, classification, regression, and ensemble analysis. Domain-sensitive learning: learning models in heterogeneous settings such as a combination of text with multimedia or Web links. The problem of information retrieval and Web search is also discussed in the context of its relationship with ranking and machine learning methods. Sequence-centric mining: various sequence-centric and natural language applications, such as feature engineering, neural language models, deep learning, text summarization, information extraction, opinion mining, text segmentation, and event detection.

AIE351 Robotics Design

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **AIE111**

This course presents an overview of robotics in practice and research with topics including vision, motion planning, mobile mechanisms, kinematics, inverse kinematics, and sensors. In course projects, students construct robots which are driven by a microcontroller, with each project reinforcing the basic principles developed in lectures. Students usually work in teams of three: an electrical engineer, a mechanical engineer, and a computer scientist. Groups are typically self-formed except for the first lab. This course will also expose students to some of the contemporary happenings in robotics, including current robotics research, applications, robot contests and robot web surfing.

AIE392 Field Training 2 in AI Engineering

2 Cr. Hrs. = (**2** LCT + **0** TUT + **0** LAB + **0** OTH) – SWL = **60** – ECTS = **4**

Prerequisite **AIE291**

Hands-on advanced topics in modern AI as well as some of the representative applications of AI including how to develop intelligent algorithms capable of learning, analyzing, and predicting future events. Internship in machine learning, data science, data analysis, deep learning, computer vision, and natural language processing companies. Establish a student organization that works with communities to develop/apply technology to serve social community.

AIE417 Selected Topics in Artificial Intelligence 1

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **AIE111**

The course provides topics that are selected from different advanced and recent trends in Artificial Intelligence that are not covered in the description of the courses listed in the curriculum.

AIE418 Selected Topics in Artificial Intelligence 2

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **AIE111**

The course provides topics that are selected from different advanced and recent trends in advanced topics Artificial Intelligence that are not covered in the description of the courses listed in the curriculum.

AIE419 Artificial intelligence for computer games

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **- - -**

Knowledge representation and organization, Search strategies & inference methods, and. AI problem solving tools and techniques. Agent Architecture, Multi-Agent Systems, Reasoning with uncertain or incomplete knowledge;

AIE424 Intelligent Decision Support Systems

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **AIE323**

Artificial intelligence in multi-criteria decision analysis (sorting, ranking, classification, programming in fuzzy environments and rule induction systems); - Intelligent systems in knowledge-based systems (knowledge discovery and representation, approximate reasoning and management of uncertainty); - GIS-based multi-criteria decision analysis (spatial data mining and visual analytics). The objective of this course is achieving a profound understanding of Intelligent Decision Support Systems in terms of its tools, current practices and impacts. The students should acquire knowledge on how to design IDSS for different decision making problems.

AIE425 Intelligent Recommender Systems

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **AIE323**

Neighborhood-based Collaborative Filtering. Model-based Collaborative Filtering. Content-based Recommender Systems. Knowledge-based Recommender Systems. Evaluation Recommender Systems.

AIE426 Decision Making under Uncertainty

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **AIE323**

The course will cover computational methods for solving decision problems with stochastic dynamics, model uncertainty, and imperfect state information. Topics include Bayesian networks, influence diagrams, dynamic programming, reinforcement learning, and partially observable Markov decision processes. Applications cover: air traffic control, aviation surveillance systems, autonomous vehicles, and robotic planetary exploration.

AIE427 Statistical Pattern Recognition

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **AIE322**

Statistical Pattern Recognition: A Review. Statistical Decision Theory. Notes on Neyman-Pearson decision rule. Notes on error rate of a linear discriminant function. Parameter Estimation. Bayes Estimator for multivariate Gaussian density with unknown covariance matrices. Component analysis and Discriminants. Readings on Isomap and LLE. Principle Component Analysis (PCA). PCA for face Recognition. Non-parametric Technique. Curse of Dimensionality. A Problem of Dimensionality: A Simple Example. Feature Selection: Evaluation, Application, and Small Sample Performance. Decision Trees. Support Vector Machine. Error Rate Estimation, Bagging, Boosting, Classifier Combination. Logistic Regression. Unsupervised and semi-supervised learning. Clustering and Multidimensional Scaling.

AIE444 Question Answering Systems

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **AIE241**

Introduction to QA systems, Answer Validation, Sentence Annotation: Named Entity Annotation, Dependency Parsing, Semantic Role Labeling. Question Analysis, Question Classification, Query Construction, Sentence Retrieval: Sentences vs. Documents, Word Relationship, Answer Extraction, Opinion and Polarity Classification.

AIE452 Cognitive Robotics

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **AIE351**

Robots: Beyond the computer metaphor in cognitive science. Varieties of embodiment. Evolution of robots. Developmental robotics. Learning intrinsic environment representations from sensory-motor interactions. Designing sociable robots. Eliza effect and its role in cognitive robotics: Robots and Autistic children. Theory of mind for robots. Internal value system in cognitive robotics architectures. Interaction theory in cognitive robotics.

AIE453 Planning Techniques for Robotics

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **AIE351**

Introduction; What is Planning? search algorithms: Uninformed A*, heuristics, weighted A*, Backward A*. interleaving planning and execution. planning representations: lattice-based graphs, explicit vs. implicit graphs. planning representations: PRM for continuous spaces. planning representations/search algorithms: RRT, RRT-Connect. search algorithms: IDA*, Beam Search, Multi-goal A*. search

algorithms: Markov Property, dependent vs. independent variables, Dominant Relationship. planning representations: state-space vs. symbolic representation for task planning. search algorithms: symbolic task planning algorithms. planning under uncertainty.

AIE454 Robot Kinematics &Dynamics

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **AIE351**

Transformations. forward kinematics. inverse kinematics. differential kinematics (Jacobians). manipulability. basic equations of motion.

AIE455 Robot Mapping &Localization

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **AIE351**

Introduction. SLAM entities: map, robot, sensor, landmarks, observations, estimator. Motion and observation models. EKF-SLAM. Operations of EKF-SLAM. Geometry (Rotation matrix - Reference frames - Motion of a body in the plane - Polar coordinates - Useful combinations). Probability. Generalities - Gaussian variables - Graphical representation.

AIE456 Human Robot Interaction

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **AIE351**

The field of human-robot interaction (HRI) is fast becoming a significant area of research in robotics. The basic objective is to create natural and effective interactions between people and robots. HRI is highly interdisciplinary, bringing together methodologies and techniques from robotics, artificial intelligence, human-computer interaction, psychology, education, and other fields. This course is primarily lecture-based, with in-class participatory mini-projects, homework assignments, a group term project that will enable students to put theory to practice, and a final. The topics covered will include technologies that enable human-robot interactions, the psychology of interaction between people and robots, how to design and conduct HRI studies, and real-world applications such as assistive robots. This course has no prerequisites, but some basic familiarity with robots is recommended (programming knowledge is not necessary, but is useful for the term project).

AIE457 Mobile Robot Development

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **AIE351**

This course investigates robot mobility, energetics, sensing, computing, software, payload, interface, and operating environment. The context is robotic pursuit of the Moon. Scope incorporates mechanism, electronics, software, locomotion, navigation, communication, sensing, power and thermal considerations. Additionally, space systems address challenges of low mass, energetics, space environment, and reliability of design. Media is incorporated to chronicle and represent the accomplishments. The course is appropriate for a broad range of student disciplines and interests. Course Learning Objectives include formulation, problem solving, robotics and developing space systems. Students work cooperatively in teams with guidance to produce mission-relevant results and practice technical communications through written and oral presentations. Teams generate term papers detailing the design, development, testing and lessons learned.

AIE493 Graduation Project 1

3 Cr. Hrs. = (***2*** LCT + ***0*** TUT + ***3*** LAB + ***0*** OTH) – SWL = ***180*** – ECTS = ***6***

Prerequisite **SENIOR STANDING**

During the first of the two semesters, students will begin their work on the project and are expected to complete at least half the project by the end of the semester. Students will develop and work on their projects under faculty supervision.

AIE494 Graduation Project 2

3 Cr. Hrs. = (***2*** LCT + ***0*** TUT + ***3*** LAB + ***0*** OTH) – SWL = ***180*** – ECTS = ***6***

Prerequisite **AIE493**

During the second of the two semesters, students will continue their work on the project and are expected to complete the project by the end of the semester. Students will develop and work on their projects under faculty supervision.

Department of Biomedical Informatics

BMD191 Field Training 1 In Biomedical Informatics

2 Cr. Hrs. = (*2* LCT + *0* TUT + *0* LAB + *0* OTH) – SWL = *60* – ECTS = *4*

Prerequisite - - -

Hands-on introduction to the Biomedical informatics field that combines ideas from computer science and quantitative disciplines (statistics, data science, and decision science) to solve challenging problems in biology and medicine. Internships in health informatics is available through hospitals and health sector institutions.

BMD241 Human physiology

3 Cr. Hrs. = (*2* LCT + *1* TUT + *2* LAB + *0* OTH) – SWL = *180* – ECTS = *6*

Prerequisite - - -

Basic physiological functions including circulatory system, digestive system, execration system, nervous system, reproduction system, and immune system. Physiochemical process for each system. Anatomy of the human body. Modelling of human physiological and anatomical systems.

BMD292 Field Training 2 In Biomedical Informatics

2 Cr. Hrs. = (*2* LCT + *0* TUT + *0* LAB + *0* OTH) – SWL = *60* – ECTS = *4*

Prerequisite **BMD191**

Hands-on advanced topics in the Biomedical informatics field that combines ideas from computer science and quantitative disciplines (statistics, data science, and decision science) to solve challenging problems in biology and medicine. Internships in health informatics is available through hospitals and health sector institutions.

BMD310 Introduction to Programming for Biostatisticians

2 Cr. Hrs. = (*1* LCT + *0* TUT + *3* LAB + *0* OTH) – SWL = *135* – ECTS = *4*

Prerequisite - - -

This course will focus on skills required for advanced computing applications in biostatistics. Students will learn statistical programming and methods such as loops, functions, macros as well as data visualization techniques in SAS and R. Furthermore, the course will provide and introduction to Linux and basic statistical programming in Python.

BMD311 Introduction to Bioinformatics

3 Cr. Hrs. = (*2* LCT + *1* TUT + *2* LAB + *0* OTH) – SWL = *180* – ECTS = *6*

Prerequisite - - -

Introduction to Bioinformatics and Biological Databases. NCBI Tools. Sequence manipulation and analysis. Sequence alignment theory and applications. Sequence alignment and matching. Multiple sequence alignment methods and algorithms. Evolution and Phylogenetic analysis. PCR primer Design. RNA Bioinformatics: secondary structure prediction. Comparative structure modelling.

BMD312 Clinical Informatics

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **BMD311**

Overview of clinical informatics. Medical departments and terminologies. Classification of diseases and biomedical ontologies. Structure and organization of medical data and information, including physiochemical measurement, imaging, genetics, bioanalysis. Standards and regulations in biomedical informatics for data representation, safety, and privacy. Clinical information systems: design, implementation, and operation of clinical information system serving patients of a healthcare entity. Electronic medical records: architecture and design, interoperability, implementation, and operation.

BMD313 Dental informatics

2 Cr. Hrs. = (**2** LCT + **0** TUT + **2** LAB + **0** OTH) – SWL = **120** – ECTS = **4**

Prerequisite **- - -**

Informatics. Dental Informatics. Data Management. Information Extraction. Dental Imaging and digital data.

BMD320 Data Analysis Using R

3 Cr. Hrs. = (**2** LCT + **0** TUT + **2** LAB + **0** OTH) – SWL = **150** – ECTS = **4**

Prerequisite **BMD310, PHS231, PHS236, MAT316**

As R statistical software is becoming a leading tool in data analysis, and as R software requires programming knowledge, this course aims to teach basics of R software programming and trains students to conduct statistical analysis on R for different dataset types and different statistical methods.

BMD351 Biomedical Data Acquisition

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **CSE111, BMD241**

Overview of biomedical equipment and types. Physiological (physiochemical) measurements (ECG, EEG, EMG, ECG, etc.), and related devices. Medical imaging, including X-ray, ultrasound, MRI. Life support equipment, including incubators, ventilators, dialysis, heart-lung machines, etc. Laboratory equipment, including bio-analyzers, PCR, and DNA sequencers.

BMD361 Biomedical Statistics

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **BMD311**

Review probability theory and statistical methods. Research methodology in biomedical sciences. Cohorts and groups establishment and power analysis. Hypothesis testing and its applications to group comparisons. Application of classification and clustering techniques in biomedical informatics. Biostatistics efficient association tests.

BMD413 Structural bioinformatics

3 Cr. Hrs. = (**2** LCT + **1** TUT + **2** LAB + **0** OTH) – SWL = **180** – ECTS = **6**

Prerequisite **BMD311**

Modelling of Protein and Nucleic Acid Structures. Protein Structure Classification and Databases. Prediction of Protein Structure: Homology Modelling. Prediction of Protein Motion: Molecular Dynamics Simulation. Modelling of Small Molecule Ligands and aspects in Ligand-Protein Interactions. Molecular Docking and Prediction of Protein Binding Site. Recent Approaches in Structural Bioinformatics and Drug Discovery.

BMD414 Selected Topics in Biomedical Informatics 1: Data Analysis &Visualization

3 Cr. Hrs. = (*2* LCT + *1* TUT + *2* LAB + *0* OTH) – SWL = *180* – ECTS = *6*

Prerequisite **BMD311**

Introduction to Data Analysis. The NumPy Library. Introduction to Pandas Library. Pandas Reading and Writing data. Pandas In-depth. Machine Learning with scikit-learn. Combining and importing data. Data exploration. Visualizing trends. Plotting 2D arrays. Statistical plots with Seaborn.

BMD415 Selected Topics in Biomedical Informatics 2: Data Mining & Machine Learning

3 Cr. Hrs. = (*2* LCT + *1* TUT + *2* LAB + *0* OTH) – SWL = *180* – ECTS = *6*

Prerequisite **AIE121, AIE323**

Linear Classification. Non-linear Classification. Clustering Techniques. Dimensionality Reduction and Feature Extraction. Mining unstructured data: text as an example. Finding Similar Items.

BMD421 Biomedical Information systems

3 Cr. Hrs. = (*2* LCT + *1* TUT + *2* LAB + *0* OTH) – SWL = *180* – ECTS = *6*

Prerequisite **BMD311**

Hospital information systems: design, implementation, and operation of clinical information system including medical and non-medical components. Structure of complex organizations. Concepts of business processes. Design and implementation of business workflows in hospital information systems. Healthcare information systems on the population scale. Sharing biomedical data among different healthcare entities. Use of geographical information systems. Role of large-scale information systems in public healthcare.

BMD422 Systems Biology

3 Cr. Hrs. = (*2* LCT + *1* TUT + *2* LAB + *0* OTH) – SWL = *180* – ECTS = *6*

Prerequisite **BMD311**

Introduction to Complex Systems and Systems biology. Mathematical representation of Biological Systems. Topologies of Biological Networks. Types of Biological Networks. Mathematical Modeling of Biological Networks. Differential Equations for Modeling of Biological Networks. Recent topics in Systems Biology: Synthetic biology. Recent topics in Systems Biology.

BMD431 Medical Image Informatics

3 Cr. Hrs. = (*2* LCT + *1* TUT + *2* LAB + *0* OTH) – SWL = *180* – ECTS = *6*

Prerequisite **CSE281**

Introduces the characteristics of medical images and basic techniques for analyzing medical images, including ultrasound images, X-ray images, and MRI images. The course also tackles extra challenges of medical imaging, including noise patterns, movement, tissue structure, and elasticity. Applications of medical image processing in computer aided diagnosis. Management of technical images, including DICOM standard and PACS system.

BMD452 Biomedical text processing

3 Cr. Hrs. = (*2* LCT + *1* TUT + *2* LAB + *0* OTH) – SWL = *180* – ECTS = *6*

Prerequisite **BMD311**

Basics of text processing: text parsing, text retrieval, tagging, natural language processing, and information extraction. Medical texts and public databases. Medical Languages systems. Mining biomedical literature: gene name extraction, disease name identification, relation identification.

BMD462 Bio-inspired Computing

3 Cr. Hrs. = (***2*** LCT + ***1*** TUT + ***2*** LAB + ***0*** OTH) – SWL = ***180*** – ECTS = ***6***

Prerequisite **BMD311**

An introduction to self-adapting methods also called artificial intelligence or machine learning. Schemes for classification, search and optimization based on bio-inspired mechanisms are introduced. This includes evolutionary computation, artificial neural networks and more specialized approaches like e.g. swarm intelligence and artificial immune systems. Further, an overview of alternative traditional methods will also be included. Bio-inspired hardware and computers. Applications in robotics, problem solving, and optimization problems.

BMD493 Graduation Project 1

3 Cr. Hrs. = (***2*** LCT + ***0*** TUT + ***3*** LAB + ***0*** OTH) – SWL = ***180*** – ECTS = ***6***

Prerequisite **SENIORSTANDING**

All students undertake a major project as part of the program. The aim of the project is to provide the students - in groups - with an opportunity to implement the appropriate concepts and techniques to a particular design. Students are required to choose and research the expected project to be designed and implemented in course project-. The student is expected to give an oral presentation to be approved.

BMD494 Graduation Project 2

3 Cr. Hrs. = (***2*** LCT + ***0*** TUT + ***3*** LAB + ***0*** OTH) – SWL = ***180*** – ECTS = ***6***

Prerequisite **BMD493**

All students undertake a major project as part of the program. The aim of the project is to provide the students - in groups - with an opportunity to implement the appropriate concepts and techniques to a particular design. Students are required to design and implement the project initiated in in course project-. The student is expected to give an oral presentation to be approved.

مجال هندسة الحاسبات

FIELD OF COMPUTER ENGINEERING

البرامج الدراسية

Study Programs

Degree Requirements متطلبات الحصول على الدرجة

The Program	Cr. Hrs.	University Requirement	Mathematic & Basic Science	Computer Science & Engineering	AI Science & Engineering	Advanced Specially Requirements	BiO-Medical Informatics	Project & Training	البرنامج
Computer Engineering	157	20	36	49	6	36	0	10	هندسة حاسب
AI Engineering	157	20	36	49	30	12	0	10	هندسة ذكاء اصطناعي

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برنامج هندسة الحاسب

Computer Engineering Program

Program Description

- The program concerns of qualifying an Engineering graduate specialized in Computer Engineering, capable of developing software and ensuring its quality. He can deal with cloud computing, mobile computing, embedded systems, and information networks and their security. The student can choose one of the following tracks:

Embedded Systems

- The program is concerned in preparing graduates specialized in designing embedded and distributed systems that are connected to the internet, in addition to analyzing the big data extracted from these systems.

Cloud Computing

- The track is concerned in preparing graduates capable of designing cloud-based systems and how to deal with distribution of big data associated with these systems.

وصف البرنامج

- يهتم البرنامج بتخريج مهندس متخصص في هندسة الحاسبات، قادر على تطوير البرمجيات وضمان جودتها، والتعامل مع الحوسبة السحابية، والحوسبة المتنقلة، والأنظمة المدمجة، وشبكات المعلومات وأمنها. ويمكن للطالب اختيار أحد المسارات التالية:

النظم المدمجة

- يهتم المسار بإعداد كوادر متخصصة في تصميم النظم المدمجة وكذلك النظم المتسم بتوزيع مكوناتها واتصالها بالإنترنت، وتحليل البيانات الضخمة المستخرجة من تلك النظم.

الحوسبة السحابية

- يهتم المسار بإعداد كوادر متخصصة في تصميم النظم المبنية علي الحوسبة السحابية وكيفية التعامل مع توزيع البيانات الضخمة المصاحبة لها.

High Performance Computing

– The track is concerned in preparing graduates capable of designing distributed systems and how to deal with distributed big data and analyzing them.

الحوسبة عالية الأداء

– يهتم المسار بإعداد كوادر متخصصة في برمجة الحاسبات المتوازية تصميم النظم الموزعة وكيفية التعامل مع البيانات الضخمة الموزعة وتحليلها.

Cyber Security

– The track is concerned in preparing graduates capable of securing computer systems and networks and distinguish between different models of information technology security and the risks complemented the transfer of information and its storage. They will be capable of designing and developing secure systems, and discovering security vulnerability and how to fix them.

الأمن السيبراني

– يهتم المسار بإعداد كوادر قادرة على تامين أنظمة الحاسب والشبكات والتفريق بين مختلف نماذج أمن تقنية المعلومات والمخاطر المرتبطة بنقل المعلومات وتخزينها. وقادرة على استخدام أحدث الأساليب في تصميم وتطوير أنظمة آمنة واكتشاف الثغرات الأمنية وطرق علاجها .

Graduate attributes

مواصفات الخريج

The main outcome of the proposed program is expert graduates in computer science specializations where the graduates achieve standards that exceeded NARS these graduate should be able to:

– Demonstrating inductive reasoning abilities, figuring general rules and conclusions about seemingly unrelated events

– Using current advanced techniques, skills, and tools necessary for computing practices to specify, design, and implement computer-based systems.

المخرج الرئيسي للبرامج المقترحة هو جيل من الخبراء في هندسة الحاسب بحيث يكون الخريج محققاً لمعايير تزيد عن المعايير القياسية الوطنية قادراً علي:

– الاستدلال الاستقرائي وتحديد القواعد العامة والاستنتاجات حول الأحداث التي تبدو غير ذات صلة.

– استخدام التقنيات المتقدمة الحالية والمهارات والأدوات اللازمة لممارسات الحوسبة لتحديد وتصميم وتنفيذ الأنظمة المعتمدة على الكمبيوتر.

- Recognizing the information requirements of various business activities on both operational and decision making levels.
- Tackling business problems using system analysis tools and techniques.
- Managing projects related to computer systems in diverse fields of applications.
- Implementing phases of the computer system development life cycle, procurement and installation of hardware, software design, data manipulation and system operations.

Specific graduate attributes embedded systems track:

- The ability to design embedded systems.
- The ability to design distributed systems connected to the internet.
- The ability to analyze big data from systems connected to the internet

Specific graduate attributes Cloud Computing track:

- The ability to design cloud-based systems.
- The ability to distribute the big data associated with cloud-based systems

Specific graduate attributes High Performance Computing track:

- The ability to design distributed systems.
- The ability to perform parallel programming.

- تعرف على متطلبات المعلومات الخاصة بأنشطة الأعمال المختلفة على المستويين التشغيلي ومستويات صنع القرار.
- معالجة مشاكل الأعمال باستخدام أدوات وتقنيات تحليل النظام.
- إدارة المشاريع المتعلقة بأنظمة الكمبيوتر في مختلف مجالات الاستخدام.
- تنفيذ مراحل دورة حياة تطوير نظام الكمبيوتر، وشراء وتركيب الأجهزة، وتصميم البرمجيات، ومعالجة البيانات وعمليات النظام.

المواصفات المميزة لخريج مسار النظم المدمجة:

- القدرة على تصميم النظم المدمجة
- القدرة على تصميم النظم الموزعة المتصلة بالإنترنت
- القدرة على تحليل البيانات الضخمة المستخرجة من النظم المتصلة بالإنترنت

المواصفات المميزة لخريج مسار الحوسبة السحابية:

- القدرة على تصميم النظم المبنية على الحوسبة
- القدرة على توزيع البيانات الضخمة المتصلة بالنظم المبنية علي الحوسبة

المواصفات المميزة لخريج مسار الحوسبة عالية الاداء:

- القدرة على تصميم النظم الموزعة
- القدرة على برمجة الحاسبات المتوازية

– The ability to distribute the big data associated with distributed systems.

– القدرة علي تحليل البيانات الضخمة المصاحبة للنظم الموزعة.

Specific graduate attributes of Cyber Security track:

المواصفات المميزة لخريج مسار الأمن السيبراني:

– The ability to distinguish between different models of information technology security and the risks complemented the transfer of information and its storage.

– القدرة على التفريق بين مختلف نماذج أمن تقنية المعلومات والمخاطر المرتبطة بنقل المعلومات وتخزينها.

– The ability to use the recent techniques in designing and developing secure systems.

– القدرة على استخدام أحدث الأساليب في تصميم وتطوير أنظمة آمنة.

– The ability to develop solutions to protect computer networks and information from threats and hacking on internal and external bands.

– القدرة على تطوير حلول لحماية شبكات الحاسب والمعلومات من التهديدات والاختراقات على النطاقين الداخلي والخارجي .

Job Opportunities

مجالات عمل الخريج

– International companies working in software – Computer components manufacturing companies – digital transformation companies' startups.

– الشركات الدولية العاملة في البرمجيات – شركات تصنيع أجزاء الحاسب – شركات التحول الرقمي الناشئة

Benchmark

- The program has adopted the general knowledge and skills issued from the National authority for quality assurance and accreditation of education (NARS) for computer engineering graduate which can be found through the following link.
  https://admin.naqaae.eg/api/v1/archive/download/34733
- The bylaws of computer engineering faculties have been studied and referenced in the following universities: Stanford University- Carnegie Mellon University Ain Shams University .
  https://www.cs.cmu.edu/bs-in-artificial-intelligence/curriculum

المرجعية

- وقد تبنى البرنامج المعارف والمهارات الصادرة من الهيئة القومية لضمان جودة التعليم والاعتماد (NARS) لخريجى الهندسة والتى يمكن الاطلاع عليها من خلال الرابط التالى:
  https://admin.naqaae.eg/api/v1/archive/download/34733
- تم الاطلاع والرجوع للوائح كليات الهندسة فى تخصصات الحاسبات فى جامعة ستانفورد – كارنيجى ميلون

Computer Engineering Program — برنامج هندسة الحاسب

I. Embedded Systems — I. النظم المدمجة

Study Plan — الخطة الدراسية

Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
1	MAT111	Mathematics I	3	0	1	0	4	3	150	6
	MAT123	Mechanics	2	2	0	0	4	3	150	6
	MEC011	Engineering Drawing (1)	2	0	3	0	5	3	135	4
	PHY212	Introduction to Engineering Physics	2	2	0	0	4	3	150	6
	CSE014	Structured Programming	2	3	0	0	5	3	180	6
	UC1	University Requirement (1)	2	0	0	0	2	2	90	4
							24	17	855	32
2	MAT112	Mathematics II	3	0	1	0	4	3	150	6
	MAT131	Probability & Statistics I	2	1	2	0	5	3	195	6
	CSE015	Object Oriented Programming	2	3	0	0	5	3	180	6
	CSE315	Discrete Mathematics	2	0	2	0	4	3	150	6
	UC2	University Requirement (2)	2	0	0	0	2	2	90	4
	UC3	University Requirement (3)	2	0	0	0	2	2	90	4
							22	16	855	32
	MAT212	Linear Algebra	2	0	2	0	4	3	150	6

Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
3	CSE111	Data Structures	2	3	0	0	5	3	180	6
	CSE113	Electric & Electronic Circuits	1	3	2	0	6	3	180	6
	CSE131	Logic Design	2	2	2	0	6	3	195	6
	UC4	University Requirement (4)	2	0	0	0	2	2	90	4
	UC5	University Requirement (5)	2	0	0	0	2	2	90	4
							25	16	885	32
4	MAT231	Probability & Statistics II	2	2	1	0	5	3	195	6
	MAT312	Differential Equations	2	0	2	0	4	3	150	6
	CSE132	Computer Architecture & Organization	2	2	1	0	5	3	165	6
	AIE111	Artificial Intelligence	2	2	1	0	5	3	180	6
	E1	Elective Course (1)	2	2	1	0	5	3	180	6
							24	15	870	30
5	E2	Elective Course (2)	2	2	1	0	5	3	180	6
	CSE221	Database Systems	2	2	1	0	5	3	180	6
	CSE233	Operating Systems	2	2	1	0	5	3	165	6
	AIE121	Machine Learning	2	2	1	0	5	3	180	6
	CSE291	Field Training 1 in Computer Engineering	2	0	0	0	2	2	60	4
	UE1	Elective University (1)	2	0	0	0	2	2	90	4
							24	16	855	32
6	CSE112	Design & Analysis of Algorithms	3	0	3	0	6	4	195	8
	CSE211	Web Programming	2	2	1	0	5	3	165	6
	CSE251	Software Engineering	2	2	1	0	5	3	180	6
	CSE261	Computer Networks	2	2	1	0	5	3	165	6

Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
	CSE234	Embedded Systems	2	2	1	0	5	3	180	6
							26	**16**	**885**	**32**
7	ELE432	Digital Signal Processing	2	1	1	0	4	3	150	6
	CSE242	Cryptography	2	1	2	0	5	3	180	6
	CSE281	Image Processing	2	2	1	0	5	3	180	6
	CSE344	Introduction to Cyber Security	2	2	1	0	5	3	180	6
	CSE363	Cloud Computing	2	2	1	0	5	3	180	6
	UE2	Elective University (2)	2	0	0	0	2	2	90	4
							26	**17**	**960**	**34**
8	CSE272	Parallel & Distributed Systems	2	2	1	0	5	3	150	6
	CSE335	Real Time & Embedded Systems Design	2	2	1	0	5	3	180	6
	CSE446	Information & Computer Networks Security	2	2	1	0	5	3	180	6
	CSE464	Internet of things	2	2	1	0	5	3	180	6
	CSE392	Field Training 2 in Computer Engineering	2	0	0	0	2	2	60	4
	UC6	University Requirement (6)	2	0	0	0	2	2	90	4
							24	**16**	**840**	**32**
9	E3	Elective Course (3)	2	2	1	0	5	3	180	6
	E4	Elective Course (4)	2	2	1	0	5	3	180	6
	CSE436	Selected Topics in Embedded Systems	2	2	1	0	5	3	180	6
	CSE493	Graduation Project 1	2	3	0	0	5	3	180	6
	UC7	University Requirement (7)	2	0	0	0	2	2	90	4

Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
							22	14	810	28
10	E5	Elective Course (5)	2	2	1	0	5	3	180	6
	E6	Elective Course (6)	2	2	1	0	5	3	180	6
	CSE466	Selected Topics in IoT	2	2	1	0	5	3	180	6
	CSE494	Graduation Project 2	2	3	0	0	5	3	180	6
	UE3	Elective University (3)	2	0	0	0	2	2	90	4
							22	14	810	28
							239	157	8625	312

Elective (E1)

CHE142	Engineering Chemistry
MAT121	Dynamics
MAT122	Statics
MAT315	Numerical Analysis
PHY211	Physics II

Elective (E2)

ELE115	Electrical Circuits & Measurements
ELE215	Electronics & Digital Circuits
ELE232	Fundamentals of Communication Systems
ELE233	Automatic Control Systems
ELE338	Wireless Communication Networks

Elective (E3)

CSE241	Security of Information Systems
CSE311	Design of Compilers
CSE271	Introduction to Parallel Computing
AIE231	Neural Networks

Elective (E4)

CSE322	Big Data Analytics 1
CSE323	Advanced Database Systems
CSE373	Parallel Programming
CSE383	Computer Vision

Elective (E5)

AIE322	Advanced Machine Learning
AIE323	Data Mining
AIE332	Deep Learning
AIE351	Robotics Design

Elective (E6)

CSE425	Big Data Analytics 2
CSE426	Selected Topics in Data Science
CSE465	Selected Topics in Cloud Computing
CSE474	Distributed Information Systems
CSE475	High Performance Computing

Computer Engineering Program — برنامج هندسة الحاسب

I. Cloud Computing — II. الحوسبة السحابية

Study Plan — الخطة الدراسية

Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
1	MAT111	Mathematics I	3	0	1	0	4	3	150	6
	MAT123	Mechanics	2	2	0	0	4	3	150	6
	MEC011	Engineering Drawing (1)	2	0	3	0	5	3	135	4
	PHY212	Introduction to Engineering Physics	2	2	0	0	4	3	150	6
	CSE014	Structured Programming	2	3	0	0	5	3	180	6
	UC1	University Requirement (1)	2	0	0	0	2	2	90	4
							24	17	855	32
2	MAT112	Mathematics II	3	0	1	0	4	3	150	6
	MAT131	Probability & Statistics I	2	1	2	0	5	3	195	6
	CSE015	Object Oriented Programming	2	3	0	0	5	3	180	6
	CSE315	Discrete Mathematics	2	0	2	0	4	3	150	6
	UC2	University Requirement (2)	2	0	0	0	2	2	90	4
	UC3	University Requirement (3)	2	0	0	0	2	2	90	4
							22	16	855	32
	MAT212	Linear Algebra	2	0	2	0	4	3	150	6

Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
3	CSE111	Data Structures	2	3	0	0	5	3	180	6
	CSE113	Electric & Electronic Circuits	1	3	2	0	6	3	180	6
	CSE131	Logic Design	2	2	2	0	6	3	195	6
	UC4	University Requirement (4)	2	0	0	0	2	2	90	4
	UC5	University Requirement (5)	2	0	0	0	2	2	90	4
							25	**16**	**885**	**32**
4	MAT231	Probability & Statistics II	2	2	1	0	5	3	195	6
	MAT312	Differential Equations	2	0	2	0	4	3	150	6
	CSE132	Computer Architecture & Organization	2	2	1	0	5	3	165	6
	AIE111	Artificial Intelligence	2	2	1	0	5	3	180	6
	E1	Elective Course (1)	2	2	1	0	5	3	180	6
							24	**15**	**870**	**30**
5	E2	Elective Course (2)	2	2	1	0	5	3	180	6
	CSE221	Database Systems	2	2	1	0	5	3	180	6
	CSE233	Operating Systems	2	2	1	0	5	3	165	6
	AIE121	Machine Learning	2	2	1	0	5	3	180	6
	CSE291	Field Training 1 in Computer Engineering	2	0	0	0	2	2	60	4
	UE1	Elective University (1)	2	0	0	0	2	2	90	4
							24	**16**	**855**	**32**
6	CSE112	Design & Analysis of Algorithms	3	0	3	0	6	4	195	8
	CSE211	Web Programming	2	2	1	0	5	3	165	6
	CSE251	Software Engineering	2	2	1	0	5	3	180	6

Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
	CSE261	Computer Networks	2	2	1	0	5	3	165	6
	CSE234	Embedded Systems	2	2	1	0	5	3	180	6
							26	**16**	**885**	**32**
7	ELE432	Digital Signal Processing	2	1	1	0	4	3	150	6
	CSE242	Cryptography	2	1	2	0	5	3	180	6
	CSE271	Introduction to Parallel Computing	2	2	1	0	5	3	180	6
	CSE344	Introduction to Cyber Security	2	2	1	0	5	3	180	6
	CSE363	Cloud Computing	2	2	1	0	5	3	180	6
	UE2	Elective University (2)	2	0	0	0	2	2	90	4
							26	**17**	**960**	**34**
8	CSE272	Parallel & Distributed Systems	2	2	1	0	5	3	150	6
	CSE322	Big Data Analytics 1	2	2	1	0	5	3	180	6
	CSE446	Information & Computer Networks Security	2	2	1	0	5	3	180	6
	CSE464	Internet of things	2	2	1	0	5	3	180	6
	CSE392	Field Training 2 in Computer Engineering	2	0	0	0	2	2	60	4
	UC6	University Requirement (6)	2	0	0	0	2	2	90	4
							24	**16**	**840**	**32**
9	E3	Elective Course (3)	2	2	1	0	5	3	180	6
	E4	Elective Course (4)	2	2	1	0	5	3	180	6
	CSE465	Selected Topics in Cloud Computing	2	2	1	0	5	3	180	6
	CSE493	Graduation Project 1	2	3	0	0	5	3	180	6

Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
	UC7	University Requirement (7)	2	0	0	0	2	2	90	4
							22	14	810	28
10	E5	Elective Course (5)	2	2	1	0	5	3	180	6
	E6	Elective Course (6)	2	2	1	0	5	3	180	6
	CSE427	Selected Topics in Big Data	2	2	1	0	5	3	180	6
	CSE494	Graduation Project 2	2	3	0	0	5	3	180	6
	UE3	Elective University (3)	2	0	0	0	2	2	90	4
							22	14	810	28
							239	157	8625	312

Elective (E1)

CHE142 Engineering Chemistry
MAT121 Dynamics
MAT122 Statics
MAT315 Numerical Analysis
PHY211 Physics II

Elective (E2)

ELE115 Electrical Circuits & Measurements
ELE215 Electronics & Digital Circuits
ELE232 Fundamentals of Communication Systems
ELE233 Automatic Control Systems
ELE338 Wireless Communication Networks

Elective (E3)

CSE241	Security of Information Systems
CSE281	Image Processing
AIE231	Neural Networks
BMD311	Introduction to Bioinformatics

Elective (E4)

CSE311	Design of Compilers
CSE373	Parallel Programming
CSE335	Real Time & Embedded Systems Design
CSE362	Industrial Networks
CSE383	Computer Vision

Elective (E5)

AIE322	Advanced Machine Learning
AIE323	Data Mining
AIE332	Deep Learning
AIE351	Robotics Design

Elective (E6)

CSE425	Big Data Analytics 2
CSE426	Selected Topics in Data Science
CSE447	Selected Topics in Computer Security
CSE466	Selected Topics in IoT
CSE474	Distributed Information Systems
CSE475	High Performance Computing

Computer Engineering Program — برنامج هندسة الحاسب

II. High Performance Computing — III. الحوسبة عالية الأداء

Study Plan — الخطة الدراسية

Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
1	MAT111	Mathematics I	3	0	1	0	4	3	150	6
	MAT123	Mechanics	2	2	0	0	4	3	150	6
	MEC011	Engineering Drawing (1)	2	0	3	0	5	3	135	4
	PHY212	Introduction to Engineering Physics	2	2	0	0	4	3	150	6
	CSE014	Structured Programming	2	3	0	0	5	3	180	6
	UC1	University Requirement (1)	2	0	0	0	2	2	90	4
							24	17	855	32
2	MAT112	Mathematics II	3	0	1	0	4	3	150	6
	MAT131	Probability & Statistics I	2	1	2	0	5	3	195	6
	CSE015	Object Oriented Programming	2	3	0	0	5	3	180	6
	CSE315	Discrete Mathematics	2	0	2	0	4	3	150	6
	UC2	University Requirement (2)	2	0	0	0	2	2	90	4
	UC3	University Requirement (3)	2	0	0	0	2	2	90	4
							22	16	855	32
	MAT212	Linear Algebra	2	0	2	0	4	3	150	6

Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
3	CSE111	Data Structures	2	3	0	0	5	3	180	6
	CSE113	Electric & Electronic Circuits	1	3	2	0	6	3	180	6
	CSE131	Logic Design	2	2	2	0	6	3	195	6
	UC4	University Requirement (4)	2	0	0	0	2	2	90	4
	UC5	University Requirement (5)	2	0	0	0	2	2	90	4
							25	**16**	**885**	**32**
4	MAT231	Probability & Statistics II	2	2	1	0	5	3	195	6
	MAT312	Differential Equations	2	0	2	0	4	3	150	6
	CSE132	Computer Architecture & Organization	2	2	1	0	5	3	165	6
	AIE111	Artificial Intelligence	2	2	1	0	5	3	180	6
	E1	Elective Course (1)	2	2	1	0	5	3	180	6
							24	**15**	**870**	**30**
5	E2	Elective Course (2)	2	2	1	0	5	3	180	6
	CSE221	Database Systems	2	2	1	0	5	3	180	6
	CSE233	Operating Systems	2	2	1	0	5	3	165	6
	AIE121	Machine Learning	2	2	1	0	5	3	180	6
	CSE291	Field Training 1 in Computer Engineering	2	0	0	0	2	2	60	4
	UE1	Elective University (1)	2	0	0	0	2	2	90	4
							24	**16**	**855**	**32**
6	CSE112	Design & Analysis of Algorithms	3	0	3	0	6	4	195	8
	CSE211	Web Programming	2	2	1	0	5	3	165	6
	CSE251	Software Engineering	2	2	1	0	5	3	180	6
	CSE261	Computer Networks	2	2	1	0	5	3	165	6

Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
	CSE234	Embedded Systems	2	2	1	0	5	3	180	6
							26	**16**	**885**	**32**
7	ELE432	Digital Signal Processing	2	1	1	0	4	3	150	6
	CSE271	Introduction to Parallel Computing	2	2	1	0	5	3	180	6
	CSE272	Parallel & Distributed Systems	2	2	1	0	5	3	150	6
	CSE363	Cloud Computing	2	2	1	0	5	3	180	6
	CSE344	Introduction to Cyber Security	2	2	1	0	5	3	180	6
	UE2	Elective University (2)	2	0	0	0	2	2	90	4
							26	**17**	**930**	**34**
8	CSE373	Parallel Programming	2	2	1	0	5	3	180	6
	CSE446	Information & Computer Networks Security	2	2	1	0	5	3	180	6
	CSE464	Internet of things	2	2	1	0	5	3	180	6
	CSE475	High Performance Computing	2	2	1	0	5	3	180	6
	CSE392	Field Training 2 in Computer Engineering	2	0	0	0	2	2	60	4
	UC6	University Requirement (6)	2	0	0	0	2	2	90	4
							24	**16**	**870**	**32**
9	E3	Elective Course (3)	2	2	1	0	5	3	180	6
	E4	Elective Course (4)	2	2	1	0	5	3	180	6
	CSE411	Theory of Computation & Compilers	2	2	1	0	5	3	180	6
	CSE493	Graduation Project 1	2	3	0	0	5	3	180	6
	UC7	University Requirement (7)	2	0	0	0	2	2	90	4
							22	**14**	**810**	**28**

Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
10	E5	Elective Course (5)	2	2	1	0	5	3	180	6
	E6	Elective Course (6)	2	2	1	0	5	3	180	6
	CSE476	Selected Topics in High Performance Computing	2	2	1	0	5	3	180	6
	CSE494	Graduation Project 2	2	3	0	0	5	3	180	6
	UE3	Elective University (3)	2	0	0	0	2	2	90	4
							22	14	810	28
							239	157	8625	312

Elective (E1)

CHE142 Engineering Chemistry
MAT121 Dynamics
MAT122 Statics
MAT315 Numerical Analysis
PHY211 Physics II

Elective (E2)

ELE115 Electrical Circuits & Measurements
ELE215 Electronics & Digital Circuits
ELE232 Fundamentals of Communication Systems
ELE233 Automatic Control Systems
ELE338 Wireless Communication Networks

Elective (E3)

CSE241 Security of Information Systems

CSE281	Image Processing
AIE231	Neural Networks
CSE311	Design of Compilers

Elective (E4)

CSE322	Big Data Analytics 1
CSE335	Real Time & Embedded Systems Design
CSE383	Computer Vision
BMD311	Introduction to Bioinformatics

Elective (E5)

AIE316	Evolutionary Algorithms
AIE322	Advanced Machine Learning
AIE323	Data Mining
AIE332	Deep Learning
AIE351	Robotics Design

Elective (E6)

CSE425	Big Data Analytics 2
CSE427	Selected Topics in Big Data
CSE436	Selected Topics in Embedded Systems
CSE465	Selected Topics in Cloud Computing
CSE466	Selected Topics in IoT

Computer Engineering Program — برنامج هندسة الحاسب

III. Cyber Security — IV. الأمن السيبراني

Study Plan — الخطة الدراسية

Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
1	MAT111	Mathematics I	3	0	1	0	4	3	150	6
	MAT123	Mechanics	2	2	0	0	4	3	150	6
	MEC011	Engineering Drawing (1)	2	0	3	0	5	3	135	4
	PHY212	Introduction to Engineering Physics	2	2	0	0	4	3	150	6
	CSE014	Structured Programming	2	3	0	0	5	3	180	6
	UC1	University Requirement (1)	2	0	0	0	2	2	90	4
							24	17	855	32
2	MAT112	Mathematics II	3	0	1	0	4	3	150	6
	MAT131	Probability & Statistics I	2	1	2	0	5	3	195	6
	CSE015	Object Oriented Programming	2	3	0	0	5	3	180	6
	CSE315	Discrete Mathematics	2	0	2	0	4	3	150	6
	UC2	University Requirement (2)	2	0	0	0	2	2	90	4
	UC3	University Requirement (3)	2	0	0	0	2	2	90	4
							22	16	855	32
	MAT212	Linear Algebra	2	0	2	0	4	3	150	6

Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
3	CSE111	Data Structures	2	3	0	0	5	3	180	6
	CSE113	Electric & Electronic Circuits	1	3	2	0	6	3	180	6
	CSE131	Logic Design	2	2	2	0	6	3	195	6
	UC4	University Requirement (4)	2	0	0	0	2	2	90	4
	UC5	University Requirement (5)	2	0	0	0	2	2	90	4
							25	**16**	**885**	**32**
4	MAT231	Probability & Statistics II	2	2	1	0	5	3	195	6
	MAT312	Differential Equations	2	0	2	0	4	3	150	6
	CSE132	Computer Architecture & Organization	2	2	1	0	5	3	165	6
	AIE111	Artificial Intelligence	2	2	1	0	5	3	180	6
	E1	Elective Course (1)	2	2	1	0	5	3	180	6
							24	**15**	**870**	**30**
5	E2	Elective Course (2)	2	2	1	0	5	3	180	6
	CSE221	Database Systems	2	2	1	0	5	3	180	6
	CSE233	Operating Systems	2	2	1	0	5	3	165	6
	AIE121	Machine Learning	2	2	1	0	5	3	180	6
	CSE291	Field Training 1 in Computer Engineering	2	0	0	0	2	2	60	4
	UE1	Elective University (1)	2	0	0	0	2	2	90	4
							24	**16**	**855**	**32**
6	CSE112	Design & Analysis of Algorithms	3	0	3	0	6	4	195	8
	CSE211	Web Programming	2	2	1	0	5	3	165	6
	CSE251	Software Engineering	2	2	1	0	5	3	180	6
	CSE261	Computer Networks	2	2	1	0	5	3	165	6

Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
	CSE234	Embedded Systems	2	2	1	0	5	3	180	6
							26	**16**	**885**	**32**
7	ELE432	Digital Signal Processing	2	1	1	0	4	3	150	6
	CSE241	Security of Information Systems	2	2	1	0	5	3	150	6
	CSE242	Cryptography	2	1	2	0	5	3	180	6
	CSE363	Cloud Computing	2	2	1	0	5	3	180	6
	CSE344	Introduction to Cyber Security	2	2	1	0	5	3	180	6
	UE2	Elective University (2)	2	0	0	0	2	2	90	4
							26	**17**	**930**	**34**
8	CSE243	Secure Programming	2	3	0	0	5	3	180	6
	CSE373	Parallel Programming	2	2	1	0	5	3	180	6
	CSE446	Information & Computer Networks Security	2	2	1	0	5	3	180	6
	CSE464	Internet of things	2	2	1	0	5	3	180	6
	CSE392	Field Training 2 in Computer Engineering	2	0	0	0	2	2	60	4
	UC6	University Requirement (6)	2	0	0	0	2	2	90	4
							24	**16**	**870**	**32**
9	E3	Elective Course (3)	2	2	1	0	5	3	180	6
	E4	Elective Course (4)	2	2	1	0	5	3	180	6
	CSE445	Selected Topics in Information Security	2	2	1	0	5	3	180	6
	CSE493	Graduation Project 1	2	3	0	0	5	3	180	6
	UC7	University Requirement (7)	2	0	0	0	2	2	90	4
							22	**14**	**810**	**28**

Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
10	E5	Elective Course (5)	2	2	1	0	5	3	180	6
	E6	Elective Course (6)	2	2	1	0	5	3	180	6
	CSE447	Selected Topics in Computer Security	2	2	1	0	5	3	180	6
	CSE494	Graduation Project 2	2	3	0	0	5	3	180	6
	UE3	Elective University (3)	2	0	0	0	2	2	90	4
							22	14	810	28
							239	157	8625	312

Elective (E1)

CHE142 Engineering Chemistry
MAT121 Dynamics
MAT122 Statics
MAT315 Numerical Analysis
PHY211 Physics II

Elective (E2)

ELE115 Electrical Circuits & Measurements
ELE215 Electronics & Digital Circuits
ELE232 Fundamentals of Communication Systems
ELE233 Automatic Control Systems
ELE338 Wireless Communication Networks

Elective (E3)

CSE271	Introduction to Parallel Computing
CSE281	Image Processing
CSE382	Computer Graphics
AIE231	Neural Networks

Elective (E4)

CSE311	Design of Compilers
CSE322	Big Data Analytics 1
CSE335	Real Time & Embedded Systems Design
CSE383	Computer Vision

Elective (E5)

AIE322	Advanced Machine Learning
AIE323	Data Mining
AIE332	Deep Learning
AIE351	Robotics Design

Elective (E6)

CSE272	Parallel & Distributed Systems
CSE425	Big Data Analytics 2
CSE448	Cyber Forensics
BMD311	Introduction to Bioinformatics

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برنامج هندسة الذكاء الاصطناعي

Artificial Intelligence Engineering Program

Program Description

- The program considers developing smart software that depends on machine learning and deep learning that depends on artificial neural networks and how to apply this software in Engineering systems.

وصف البرنامج

- يهتم البرنامج بتطوير البرمجيات الذكية المعتمدة على تعلم الآلة والتعلم العميق المعتمد على الشبكات العصبية الاصطناعية، وتطبيقات تلك البرمجيات في النظم الهندسية.

Graduate attributes

In addition, the graduate of the Artificial Intelligence Engineering program has all the modules of graduates of computer Engineering programs. The graduates have the following characteristics intensively

- The ability to develop smart software that uses machine learning and deep learning
- The ability to apply smart software on engineering applications.

مواصفات الخريج

بالإضافة الى تمتع خريج برنامج هندسة الذكاء الاصطناعي بكافة مقومات خريجي برامج هندسة الحاسب، يتسم الخريجين بالسمات التالية بشكل مكثف

- القدرة على تصميم برمجيات ذكية تستخدم طرق تعلم الآلة والتعلم العميق
- القدرة على تطبيق البرمجيات الذكية على التطبيقات الهندسية

Job Opportunities

مجالات عمل الخريج

- Self-driving car manufacturers (or manufacturing companies) – Factories and companies operating in smart systems– robot's manufacturers – Smart home components manufacturers – Ministries and government agencies related to smart cities – Manufacturers of smart personal health devices

- شركات تصنيع السيارات ذاتية القيادة - المصانع والشركات العاملة في النظم الذكية – شركات تصنيع الروبوتات – شركات تصنيع أجزاء المنازل الذكية – الوزارات والجهات الحكومية ذات العلاقة بالمدن الذكية – شركات تصنيع أجهزة الصحة الشخصية الذكية

Benchmark

المرجعية

- The program has adopted the general knowledge and skills issued from the National authority for quality assurance and accreditation of education (NARS) for computer engineering graduate which can be found through the following link. https://admin.naqaae.eg/api/v1/archive/download/34733
- The bylaws of computer science faculties have been studied and referenced in the following universities: Stanford University Carnegie Mellon University Ain Shams University. https://www.cs.cmu.edu/bs-in-artificial-intelligence/curriculum

- وقد تبنى البرنامج المعارف والمهارات الصادرة من الهيئة القومية لضمان جودة التعليم والاعتماد (NARS) لخريجى الهندسة والتى يمكن الاطلاع عليها من خلال الرابط التالى:
https://admin.naqaae.eg/api/v1/archive/download/34733
- تم الاطلاع والرجوع للوائح كليات الحاسبات فى جامعة ستانفورد - كارنيجى ميلون – جامعة عين شمس.

Artificial Intelligence Engineering Program

برنامج هندسة الذكاء الاصطناعي

Study Plan

الخطة الدراسية

Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
1	MAT111	Mathematics I	3	0	1	0	4	3	150	6
	MAT123	Mechanics	2	2	0	0	4	3	150	6
	MEC011	Engineering Drawing (1)	2	0	3	0	5	3	135	4
	PHY212	Introduction to Engineering Physics	2	2	0	0	4	3	150	6
	CSE014	Structured Programming	2	3	0	0	5	3	180	6
	UC1	University Requirement (1)	2	0	0	0	2	2	90	4
							24	17	855	32
2	MAT112	Mathematics II	3	0	1	0	4	3	150	6
	MAT131	Probability & Statistics I	2	1	2	0	5	3	195	6
	CSE015	Object Oriented Programming	2	3	0	0	5	3	180	6
	CSE315	Discrete Mathematics	2	0	2	0	4	3	150	6
	UC2	University Requirement (2)	2	0	0	0	2	2	90	4
	UC3	University Requirement (3)	2	0	0	0	2	2	90	4
							22	16	855	32
	MAT212	Linear Algebra	2	0	2	0	4	3	150	6

Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
3	CSE111	Data Structures	2	3	0	0	5	3	180	6
	CSE113	Electric & Electronic Circuits	1	3	2	0	6	3	180	6
	CSE131	Logic Design	2	2	2	0	6	3	195	6
	UC4	University Requirement (4)	2	0	0	0	2	2	90	4
	UC5	University Requirement (5)	2	0	0	0	2	2	90	4
							25	**16**	**885**	**32**
4	MAT231	Probability & Statistics II	2	2	1	0	5	3	195	6
	MAT312	Differential Equations	2	0	2	0	4	3	150	6
	CSE132	Computer Architecture & Organization	2	2	1	0	5	3	165	6
	AIE111	Artificial Intelligence	2	2	1	0	5	3	180	6
	E1	Elective Course (1)	2	2	1	0	5	3	180	6
							24	**15**	**870**	**30**
5	E2	Elective Course (2)	2	2	1	0	5	3	180	6
	CSE221	Database Systems	2	2	1	0	5	3	180	6
	CSE233	Operating Systems	2	2	1	0	5	3	165	6
	AIE121	Machine Learning	2	2	1	0	5	3	180	6
	AIE291	Field Training 1 in Artificial Intelligence Engineering	2	0	0	0	2	2	60	4
	UE1	Elective University (1)	2	0	0	0	2	2	90	4
							24	**16**	**855**	**32**
6	CSE112	Design & Analysis of Algorithms	3	0	3	0	6	4	195	8
	CSE251	Software Engineering	2	2	1	0	5	3	180	6
	CSE261	Computer Networks	2	2	1	0	5	3	165	6
	CSE281	Image Processing	2	2	1	0	5	3	180	6

Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
	CSE234	Embedded Systems	2	2	1	0	5	3	180	6
							26	**16**	**900**	**32**
7	ELE432	Digital Signal Processing	2	1	1	0	4	3	150	6
	CSE363	Cloud Computing	2	2	1	0	5	3	180	6
	CSE344	Introduction to Cyber Security	2	2	1	0	5	3	180	6
	AIE241	Natural Language Processing	2	2	1	0	5	3	180	6
	AIE323	Data Mining	2	2	1	0	5	3	180	6
	UE2	Elective University (2)	2	0	0	0	2	2	90	4
							26	**17**	**960**	**34**
8	CSE383	Computer Vision	2	2	1	0	5	3	180	6
	AIE212	Knowledge-Based Systems	2	2	1	0	5	3	180	6
	AIE213	Optimization Techniques	3	0	3	0	6	3	180	6
	AIE231	Neural Networks	2	2	1	0	5	3	180	6
	AIE392	Field Training 2 in Artificial Intelligence Engineering	2	0	0	0	2	2	60	4
	UC6	University Requirement (6)	2	0	0	0	2	2	90	4
							25	**16**	**870**	**32**
9	E3	Elective Course (3)	2	2	1	0	5	3	180	6
	E4	Elective Course (4)	2	2	1	0	5	3	180	6
	AIE351	Robotics Design	2	2	1	0	5	3	180	6
	AIE493	Graduation Project 1	2	3	0	0	5	3	180	6
	UC7	University Requirement (7)	2	0	0	0	2	2	90	4
							22	**14**	**810**	**28**
	E5	Elective Course (5)	2	2	1	0	5	3	180	6

Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
10	E6	Elective Course (6)	2	2	1	0	5	3	180	6
	AIE332	Deep Learning	2	2	1	0	5	3	180	6
	AIE494	Graduation Project 2	2	3	0	0	5	3	180	6
	UE3	Elective University (3)	2	0	0	0	2	2	90	4
							22	14	810	28
							240	157	8670	312

Elective (E1)

CHE142 Engineering Chemistry
MAT121 Dynamics
MAT122 Statics
MAT315 Numerical Analysis
PHY211 Physics II

Elective (E2)

ELE115 Electrical Circuits & Measurements
ELE215 Electronics & Digital Circuits
ELE232 Fundamentals of Communication Systems
ELE233 Automatic Control Systems
ELE338 Wireless Communication Networks

Elective (E3)

CSE211 Web Programming

CSE271	Introduction to Parallel Computing
CSE272	Parallel & Distributed Systems
CSE322	Big Data Analytics 1
CSE382	Computer Graphics

Elective (E4)

CSE335	Real Time & Embedded Systems Design
CSE446	Information & Computer Networks Security
CSE464	Internet of things
CSE475	High Performance Computing
CSE485	Computer Games Theory
CSE487	Mixed & Augmented Reality
CSE488	Visualization & Animation
AIE314	AI-Based Programming

Elective (E5)

AIE315	Computational Logic
AIE316	Evolutionary Algorithms
AIE322	Advanced Machine Learning
AIE342	Advanced Methods for Data Analysis
AIE343	Machine Learning for Text Mining
AIE417	Selected Topics in Artificial Intelligence 1
AIE424	Intelligent Decision Support Systems
AIE425	Intelligent Recommender Systems
AIE444	Question Answering Systems

Elective (E6)

AIE418	Selected Topics in Artificial Intelligence 2

AIE426	Decision Making Under Uncertainty
AIE427	Statistical Pattern Recognition
AIE452	Cognitive Robotics
AIE453	Planning Techniques for Robotics
AIE454	Robot Kinematics & Dynamics
AIE455	Robot Mapping & Localization
AIE456	Human Robot Interaction
AIE457	Mobile Robot Development

مجال علوم الحاسبات
FIELD OF COMPUTER SCIENCE

البرامج الدراسية
Study Programs

Degree Requirements متطلبات الحصول على الدرجة

The Program	Cr. Hrs.	University Requirement	Mathematic & Basic Science	Computer Science & Engineering	AI Science & Engineering	Advanced Specially Requirements	BiO-Medical Informatics	Project & Training	البرنامج
Computer Science	133	20	24	31	6	42	0	10	علوم حاسب
AI Science	133	20	24	31	6	42	0	10	علوم ذكاء اصطناعي
Bio-Informatics	133	20	27	31	3	0	42	10	المعلومات الحيوية
Information Technology	132	20	24	36	0	42	0	10	تكنولوجيا المعلومات

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برنامج علوم الحاسب

Computer Science Program

Program Description

The program concerns of preparing specialized graduates in Computers and Informatics fields capable of developing software and building information systems. They can deal with decision support systems, data science and computer graphics. The student can choose one of the following tracks.

Big Data Analytics

– The program is concerned in preparing graduates capable of preparing Big Data and extract knowledge and Information from it and distributing Big Data processing on various servers and collect results from them.

Computer Vision

– The program is concerned in preparing graduates capable of processing digital images, increase their quality, building computer vision software and systems and maintaining them, and building pattern recognition systems, and embed them with

وصف البرنامج

يهتم البرنامج بإعداد خريجين متخصصين في مجال الحاسبات والمعلومات قادرين على تطوير البرمجيات وبناء نظم المعلومات والتعامل مع نظم دعم القرار، وعلم البيانات، والرسم بالحاسب. يمكن للطالب اختيار أحد المسارات التالية.

تحليل البيانات الضخمة

– يهتم البرنامج بإعداد كوادر قادرة على تجهيز البيانات الضخمة واستخراج المعلومات والمعارف منها وتوزيع معالجة البيانات الضخمة على خوادم متعددة وتجميع النتائج منها.

الرؤية بالحاسب

– يهتم المسار بإعداد كوادر قادرة على معالجة الصور الرقمية ورفع جودتها وإنشاء برمجيات وأنظمه الرؤية بالحاسب وصيانتها وإنشاء نظم التعرف

interactive multimedia systems along with mixed and augmented reality.

على الأنماط ودمجها بأنظمة الوسائط المتعددة التفاعلية والواقع المختلط والمعزز.

Software Engineering

– The program is concerned in preparing graduates capable of building information systems and decision support systems, and designing embedded and distributed systems.

هندسة البرمجيات

– يهتم البرنامج بتخريج كوادر قادرة على إنشاء أنظمة المعلومات وأنظمة دعم اتخاذ القرار وتطوير البرمجيات المستخدمة في النظم المدمجة والموزعة.

Graduate attributes

مواصفات الخريج

The main outcome of the proposed program is expert graduates in computer science specializations where the graduates achieve standards that exceeded NARS these graduate should be able to:

– Familiarity with the fundamental areas of computer science such as: databases, operating systems, artificial intelligence, software engineering, digital image processing, computer vision systems, interactive multimedia systems, mixed and augmented reality, and others.

– Work efficiently and integrally in a multidisciplinary team whether they are in the same specialization other specialization (medical, engineering, agricultural, and others).

– Ability to demonstrate the knowledge of mathematical, scientific, and technical fundamentals to solve problems in computer Engineering and science fields.

المخرج الرئيسي للبرامج المقترحة هو جيل من الخبراء في تخصصات علوم الحاسب، بحيث يكون الخريج محققا لمعايير تزيد عن المعايير القياسية الوطنية قادراً علي:

– الإلمام بالتطبيقات المختلفة في مجال علوم الحاسب مثل نظم قواعد البيانات وأنظمة التشغيل والذكاء الاصطناعي وهندسة البرمجيات ومعالجة الصور الرقمية وانظمه الرؤية بالكمبيوتر وأنظمة الوسائط المتعددة التفاعلية والواقع المختلط والمعزز وغيرها.

– العمل بكفاءة وتكامل في فريق متعدد التخصصات سواءً كانوا في نفس تخصصه أو في تخصصات أخرى (طبية، هندسية، زراعية، وغيرها).

– التمكن من الأساس الرياضي والعلمي والتقني لحل مشاكل مجال هندسة وعلوم الحاسب.

- Use the acquired sciences and adapt it to develop and design software programs and systems that are suitable for the market and environment needs.
- Communicate professionally orally and by writing with his/her coworkers.
- Learning, continuous development, and gaining new skills.
- Awareness of the surrounding social aspects that sometime rule the graduate in designing and developing products and services appropriate for his/her surroundings and the society in computer Engineering and science fields.
- Ability to gain higher scientific degrees after finishing the bachelor degree (for outstanding excellent students).

Specific attributes of Big Data Analytics track:

- The ability to prepare Big Data and extract knowledge and Information from it.
- The ability to distribute Big Data processing on various servers and collect results from them.

Specific attributes Computer Vision track:

- The ability to process digital images and increase their quality.
- The ability to build computer vision software and systems and maintain them.
- The ability to build pattern recognition systems and embed them with interactive multimedia systems along with mixed and augmented reality.

- استخدام العلوم المكتسبة وتطويعها لتطوير وتصميم البرامج والنظم في مجال علوم الحاسب بما يتلاءم مع احتياجات السوق والمجتمع.
- التواصل شفهياً وكتابةً بمهنية مع المشاركين في العمل.
- التعليم والتطور المستمر واكتساب المهارات الجديدة.
- الوعي بالجوانب الاجتماعية المحيطة والتي تحكمه أحياناً في تصميم وتطوير المنتجات والخدمات الملائمة للمحيط والمجتمع الذي يعمل فيه في إطار مجال هندسة وعلوم الحاسب.
- مواصلة الحصول على درجات علمية أعلى من مرحلة البكالوريوس (للطلبة المتفوقين المتميزين).

المواصفات المميزة لخريج مسار تحليل البيانات الضخمة:

- القدرة على تجهيز البيانات الضخمة واستخراج المعلومات والمعارف منها
- القدرة على توزيع معالجة البيانات الضخمة على خوادم متعددة وتجميع النتائج منها

المواصفات المميزة لخريج مسار الرؤية بالحاسب:

- القدرة على معالجة الصور الرقمية ورفع جودتها
- القدرة على إنشاء برمجيات وانظمة الرؤية بالحاسب وصيانتها
- القدرة على إنشاء نظم التعرف على الأنماط ودمجها بأنظمة الوسائط المتعددة التفاعلية والواقع المختلط والمعزز.

Specific attributes of Software Engineering track:

- The ability to build information systems and decision support systems.
- The ability to design programs used in different kinds of systems.
- The ability to deal with different designs of databases and how to store them.

المواصفات المميزة لخريج مسار هندسة البرمجيات:

- القدرة على بناء نظم المعلومات ونظم دعم اتخاذ القرار
- القدرة على تصميم البرامج المستخدمة في النظم المختلفة
- القدرة على التعامل مع التصميمات المختلفة لقواعد البيانات وطرق تخزينها

Job Opportunities

- International companies working in Information Technology – Programming companies – Information technology departments in environmental and institutional agencies – digital transformation company's startups.

مجالات عمل الخريج

- الشركات الدولية العاملة في تكنولوجيا المعلومات – شركات البرمجة – إدارات تكنولوجيا المعلومات بالجهات الحكومية والخاصة – شركات التحول الرقمي الناشئة.

Benchmark

- The program has adopted the general knowledge and skills issued from the National authority for quality assurance and accreditation of education (NARS) for computer science graduate which can be found through the following link. https://admin.naqaae.eg/api/v1/archive/download/4782
- The bylaws of computer science faculties have been studied and referenced in the following universities: Stanford University. https://www.cs.cmu.edu/academics/undergraduate/programs
- Computer Reference Standard https://www.abet.org/accreditation/accreditation-criteria/criteria-for-accrediting-computing-programs-2018-2019

المرجعية

- وقد تبنى البرنامج المعارف والمهارات الصادرة من الهيئة القومية لضمان جودة التعليم والاعتماد (NARS) لخريجى الحاسبات والمعلومات والتى يمكن الاطلاع عليها من خلال الرابط التالى:
https://admin.naqaae.eg/api/v1/archive/download/4782
- تم الاطلاع والرجوع للوائح كليات الحاسبات فى جامعة ستانفورد
- https://www.cs.cmu.edu/academics/undergraduate/programs
- المعايير المرجعية للحوسبة
https://www.abet.org/accreditation/accreditation-criteria/criteria-for-accrediting-computing-programs-2018-2019

Computer Science Program — برنامج علوم الحاسب

I. Big Data Analytics — I. تحليل البيانات الضخمة

Study Plan — الخطة الدراسية

Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
1	MAT111	Mathematics I	3	0	1	0	4	3	150	6
	PHY211	Physics II	2	2	0	0	4	3	150	6
	CSE014	Structured Programming	2	3	0	0	5	3	180	6
	UC1	University Requirement (1)	2	0	0	0	2	2	90	4
	UC2	University Requirement (2)	2	0	0	0	2	2	90	4
	UE1	Elective University (1)	2	0	0	0	2	2	90	4
							19	**15**	**750**	**30**
2	MAT112	Mathematics II	3	0	1	0	4	3	150	6
	MAT131	Probability & Statistics I	2	1	2	0	5	3	195	6
	CSE015	Object Oriented Programming	2	3	0	0	5	3	180	6
	CSE315	Discrete Mathematics	2	0	2	0	4	3	150	6
	UC3	University Requirement (3)	2	0	0	0	2	2	90	4
	UE2	Elective University (2)	2	0	0	0	2	2	90	4
							22	**16**	**855**	**32**
	MAT212	Linear Algebra	2	2	0	0	4	3	150	6

Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
3	MAT231	Probability & Statistics II	2	2	1	0	5	3	195	6
	CSE111	Data Structures	2	3	0	0	5	3	180	6
	CSE131	Logic Design	2	2	2	0	6	3	195	6
	AIE111	Artificial Intelligence	2	2	1	0	5	3	180	6
	UC4	University Requirement (4)	2	0	0	0	2	2	90	4
							27	**17**	**990**	**34**
4	MAT312	Differential Equations	2	0	2	0	4	3	150	6
	CSE132	Computer Architecture & Organization	2	2	1	0	5	3	165	6
	CSE211	Web Programming	2	2	1	0	5	3	165	6
	CSE221	Database Systems	2	2	1	0	5	3	180	6
	CSE191	Field Training 1 in Computer Science	2	0	0	0	2	2	60	4
	CSE251	Software Engineering	2	2	1	0	5	3	180	6
							26	**17**	**900**	**34**
5	CSE233	Operating Systems	2	2	1	0	5	3	165	6
	E1	Elective Course (1)	2	2	1	0	5	3	180	6
	CSE261	Computer Networks	2	2	1	0	5	3	165	6
	AIE121	Machine Learning	2	2	1	0	5	3	180	6
	CSE352	Systems Analysis & Design	2	2	1	0	5	3	165	6
	UC5	University Requirement (5)	2	0	0	0	2	2	90	4
							27	**17**	**945**	**34**
6	CSE112	Design & Analysis of Algorithms	3	0	3	0	6	4	195	8
	E2	Elective Course (2)	2	2	1	0	5	3	180	6
	CSE363	Cloud Computing	2	2	1	0	5	3	180	6

Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
	CSE373	Parallel Programming	2	2	1	0	5	3	180	6
	CSE292	Field Training 2 in Computer Science	2	0	0	0	2	2	60	4
	UC6	University Requirement (6)	2	0	0	0	2	2	90	4
							25	17	885	34
7	CSE271	Introduction to Parallel Computing	2	2	1	0	5	3	180	6
	CSE322	Big Data Analytics 1	2	2	1	0	5	3	180	6
	AIE323	Data Mining	2	2	1	0	5	3	180	6
	CSE493	Graduation Project 1	2	3	0	0	5	3	180	6
	E3	Elective Course (3)	2	2	1	0	5	3	180	6
	UC7	University Requirement (7)	2	0	0	0	2	2	90	4
							27	17	990	34
8	CSE411	Theory of Computation & Compilers	2	2	1	0	5	3	180	6
	CSE427	Selected Topics in Big Data	2	2	1	0	5	3	180	6
	CSE494	Graduation Project 2	2	3	0	0	5	3	180	6
	E4	Elective Course (4)	2	2	1	0	5	3	180	6
	E5	Elective Course (5)	2	2	1	0	5	3	180	6
	UE3	Elective University (3)	2	0	0	0	2	2	90	4
							27	17	990	34
							200	133	7305	266

Elective (E1)

CSE234 Embedded Systems
CSE281 Image Processing

CSE312	Advanced Web Programming
CSE313	Mobile Programming
CSE323	Advanced Database Systems

Elective (E2)

CSE446	Information & Computer Networks Security
CSE454	Advanced Software Engineering
CSE464	Internet of things
AIE231	Neural Networks

Elective (E3)

CSE383	Computer Vision
AIE241	Natural Language Processing
AIE322	Advanced Machine Learning
BMD311	Introduction to Bioinformatics
BMD312	Clinical Informatics

Elective (E4)

CSE424	Data Warehousing
AIE343	Machine Learning for Text Mining
AIE424	Intelligent Decision Support Systems
AIE425	Intelligent Recommender Systems
AIE314	AI-Based Programming

Elective (E5)

CSE425	Big Data Analytics 2
CSE426	Selected Topics in Data Science
CSE474	Distributed Information Systems
CSE475	High Performance Computing

AIE342	Advanced Methods for Data Analysis
AIE332	Deep Learning

Computer Science Program — برنامج علوم الحاسب

II. Computer Vision — II. الرؤية بالحاسب

Study Plan — الخطة الدراسية

Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
1	MAT111	Mathematics I	3	0	1	0	4	3	150	6
	PHY211	Physics II	2	2	0	0	4	3	150	6
	CSE014	Structured Programming	2	3	0	0	5	3	180	6
	UC1	University Requirement (1)	2	0	0	0	2	2	90	4
	UC2	University Requirement (2)	2	0	0	0	2	2	90	4
	UE1	Elective University (1)	2	0	0	0	2	2	90	4
							19	**15**	**750**	**30**
2	MAT112	Mathematics II	3	0	1	0	4	3	150	6
	MAT131	Probability & Statistics I	2	1	2	0	5	3	195	6
	CSE015	Object Oriented Programming	2	3	0	0	5	3	180	6
	CSE315	Discrete Mathematics	2	0	2	0	4	3	150	6
	UC3	University Requirement (3)	2	0	0	0	2	2	90	4
	UE2	Elective University (2)	2	0	0	0	2	2	90	4
							22	**16**	**855**	**32**

Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
3	MAT212	Linear Algebra	2	2	0	0	4	3	150	6
	MAT231	Probability & Statistics II	2	2	1	0	5	3	195	6
	CSE111	Data Structures	2	3	0	0	5	3	180	6
	CSE131	Logic Design	2	2	2	0	6	3	195	6
	AIE111	Artificial Intelligence	2	2	1	0	5	3	180	6
	UC4	University Requirement (4)	2	0	0	0	2	2	90	4
							27	**17**	**990**	**34**
4	MAT312	Differential Equations	2	0	2	0	4	3	150	6
	CSE132	Computer Architecture & Organization	2	2	1	0	5	3	165	6
	CSE211	Web Programming	2	2	1	0	5	3	165	6
	CSE221	Database Systems	2	2	1	0	5	3	180	6
	CSE191	Field Training 1 in Computer Science	2	0	0	0	2	2	60	4
	CSE251	Software Engineering	2	2	1	0	5	3	180	6
							26	**17**	**900**	**34**
5	CSE233	Operating Systems	2	2	1	0	5	3	165	6
	E1	Elective Course (1)	2	2	1	0	5	3	180	6
	CSE261	Computer Networks	2	2	1	0	5	3	165	6
	CSE281	Image Processing	2	2	1	0	5	3	180	6
	AIE121	Machine Learning	2	2	1	0	5	3	180	6
	UC5	University Requirement (5)	2	0	0	0	2	2	90	4
							27	**17**	**960**	**34**
6	CSE112	Design & Analysis of Algorithms	3	0	3	0	6	4	195	8
	CSE352	Systems Analysis & Design	2	2	1	0	5	3	165	6

Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
	CSE382	Computer Graphics	2	2	1	0	5	3	180	6
	E2	Elective Course (2)	2	2	1	0	5	3	180	6
	CSE292	Field Training 2 in Computer Science	2	0	0	0	2	2	60	4
	UC6	University Requirement (6)	2	0	0	0	2	2	90	4
							25	**17**	**870**	**34**
7	CSE363	Cloud Computing	2	2	1	0	5	3	180	6
	CSE383	Computer Vision	2	2	1	0	5	3	180	6
	AIE323	Data Mining	2	2	1	0	5	3	180	6
	CSE493	Graduation Project 1	2	3	0	0	5	3	180	6
	UC7	University Requirement (7)	2	0	0	0	2	2	90	4
	E3	Elective Course (3)	2	2	1	0	5	3	180	6
							27	**17**	**990**	**34**
8	CSE411	Theory of Computation & Compilers	2	2	1	0	5	3	180	6
	CSE484	Interactive Multimedia	2	2	1	0	5	3	180	6
	CSE494	Graduation Project 2	2	3	0	0	5	3	180	6
	E4	Elective Course (4)	2	2	1	0	5	3	180	6
	E5	Elective Course (5)	2	2	1	0	5	3	180	6
	UE3	Elective University (3)	2	0	0	0	2	2	90	4
							27	**17**	**990**	**34**
							200	**133**	**7305**	**266**

Elective (E1)

CSE234	Embedded Systems
CSE271	Introduction to Parallel Computing
CSE313	Mobile Programming
CSE373	Parallel Programming

Elective (E2)

CSE446	Information & Computer Networks Security
CSE464	Internet of things
AIE212	Knowledge-Based Systems
AIE314	AI-Based Programming

Elective (E3)

AIE231	Neural Networks
AIE241	Natural Language Processing
AIE322	Advanced Machine Learning
BMD431	Medical Image Informatics

Elective (E4)

CSE322	Big Data Analytics 1
CSE474	Distributed Information Systems
CSE475	High Performance Computing
CSE485	Computer Games Theory
AIE332	Deep Learning

Elective (E5)

CSE486	Selected Topics in Computer Vision
CSE272	Parallel & Distributed Systems
CSE487	Mixed & Augmented Reality

AIE342	Advanced Methods for Data Analysis
AIE343	Machine Learning for Text Mining

Computer Science Program — برنامج علوم الحاسب

III. Software Engineering — هندسة البرمجيات .III

Study Plan — الخطة الدراسية

Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
1	MAT111	Mathematics I	3	0	1	0	4	3	150	6
	PHY211	Physics II	2	2	0	0	4	3	150	6
	CSE014	Structured Programming	2	3	0	0	5	3	180	6
	UC1	University Requirement (1)	2	0	0	0	2	2	90	4
	UC2	University Requirement (2)	2	0	0	0	2	2	90	4
	UE1	Elective University (1)	2	0	0	0	2	2	90	4
							19	**15**	**750**	**30**
2	MAT112	Mathematics II	3	0	1	0	4	3	150	6
	MAT131	Probability & Statistics I	2	1	2	0	5	3	195	6
	CSE015	Object Oriented Programming	2	3	0	0	5	3	180	6
	CSE315	Discrete Mathematics	2	0	2	0	4	3	150	6
	UC3	University Requirement (3)	2	0	0	0	2	2	90	4
	UE2	Elective University (2)	2	0	0	0	2	2	90	4
							22	**16**	**855**	**32**
	MAT212	Linear Algebra	2	2	0	0	4	3	150	6

Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
3	MAT231	Probability & Statistics II	2	2	1	0	5	3	195	6
	CSE111	Data Structures	2	3	0	0	5	3	180	6
	CSE131	Logic Design	2	2	2	0	6	3	195	6
	AIE111	Artificial Intelligence	2	2	1	0	5	3	180	6
	UC4	University Requirement (4)	2	0	0	0	2	2	90	4
							27	**17**	**990**	**34**
4	MAT312	Differential Equations	2	0	2	0	4	3	150	6
	CSE132	Computer Architecture & Organization	2	2	1	0	5	3	165	6
	CSE211	Web Programming	2	2	1	0	5	3	165	6
	CSE221	Database Systems	2	2	1	0	5	3	180	6
	CSE191	Field Training 1 in Computer Science	2	0	0	0	2	2	60	4
	CSE251	Software Engineering	2	2	1	0	5	3	180	6
							26	**17**	**900**	**34**
5	CSE233	Operating Systems	2	2	1	0	5	3	165	6
	E1	Elective Course (1)	2	2	1	0	5	3	180	6
	CSE261	Computer Networks	2	2	1	0	5	3	165	6
	AIE121	Machine Learning	2	2	1	0	5	3	180	6
	CSE352	Systems Analysis & Design	2	2	1	0	5	3	165	6
	UC5	University Requirement (5)	2	0	0	0	2	2	90	4
							27	**17**	**945**	**34**
6	CSE112	Design & Analysis of Algorithms	3	0	3	0	6	4	195	8
	CSE241	Security of Information Systems	2	2	1	0	5	3	150	6
	CSE323	Advanced Database Systems	2	2	1	0	5	3	180	6

Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
	E2	Elective Course (2)	2	2	1	0	5	3	180	6
	CSE292	Field Training 2 in Computer Science	2	0	0	0	2	2	60	4
	UC6	University Requirement (6)	2	0	0	0	2	2	90	4
							25	17	855	34
7	CSE363	Cloud Computing	2	2	1	0	5	3	180	6
	CSE454	Advanced Software Engineering	2	2	1	0	5	3	180	6
	AIE323	Data Mining	2	2	1	0	5	3	180	6
	CSE493	Graduation Project 1	2	3	0	0	5	3	180	6
	UC7	University Requirement (7)	2	0	0	0	2	2	90	4
	E3	Elective Course (3)	2	2	1	0	5	3	180	6
							27	17	990	34
8	CSE411	Theory of Computation & Compilers	2	2	1	0	5	3	180	6
	CSE474	Distributed Information Systems	2	2	1	0	5	3	180	6
	CSE494	Graduation Project 2	2	3	0	0	5	3	180	6
	E4	Elective Course (4)	2	2	1	0	5	3	180	6
	E5	Elective Course (5)	2	2	1	0	5	3	180	6
	UE3	Elective University (3)	2	0	0	0	2	2	90	4
							27	17	990	34
							200	133	7305	266

Elective (E1)

CSE234 Embedded Systems

CSE271	Introduction to Parallel Computing
CSE281	Image Processing
CSE313	Mobile Programming

Elective (E2)

CSE322	Big Data Analytics 1
AIE212	Knowledge-Based Systems
AIE231	Neural Networks
AIE241	Natural Language Processing

Elective (E3)

CSE312	Advanced Web Programming
CSE464	Internet of things
AIE343	Machine Learning for Text Mining
AIE424	Intelligent Decision Support Systems
AIE425	Intelligent Recommender Systems

Elective (E4)

CSE426	Selected Topics in Data Science
CSE424	Data Warehousing
CSE453	Building Information Systems
CSE455	Selected Topics in Software Engineering

Elective (E5)

AIE332	Deep Learning
AIE342	Advanced Methods for Data Analysis
AIE314	AI-Based Programming
AIE322	Advanced Machine Learning

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برنامج علوم الذكاء الاصطناعي

Artificial Intelligence Science Program

Program Description

- The program grants the deep knowledge and skills needed to transform big amounts of data into executable decisions. The program focuses on how to use complex inputs like vision, languages, and big databases in decision-making and enhance human power.

وصف البرنامج

- يمنح البرنامج المعرفة والمهارات العميقة المطلوبة لتحويل كميات كبيرة من البيانات إلى قرارات قابلة للتنفيذ. يركز البرنامج ومناهجه على كيفية استخدام المدخلات المعقدة مثل الرؤية واللغة وقواعد البيانات الضخمة في اتخاذ القرارات أو تعزيز القدرات البشرية.

Graduate attributes

In addition, the graduate of the Artificial Intelligence Science program has all the modules of graduates of computer science programs. The graduates have the following characteristics intensively.

- The ability to build software and intelligent agent systems.

مواصفات الخريج

بالإضافة الى تمتع خريج برنامج علوم الذكاء الاصطناعي بكافة مقومات خريجي برامج علوم الحاسب، يتسم الخريجين بالسمات التالية بشكل مكثف.

- القدرة على إنشاء برمجيات وأنظمة العميل الذكي.

- The ability to recognize natural language systems and their implementation.
- The ability to build machine-learning systems, implement them, and measure their performance.

- القدرة على أنظمة التعرف على اللغات الطبيعية وبرمجتهما.
- القدرة على إنشاء أنظمة تعلم الآلة وبرمجتها وقياس أدائها.

Job Opportunities

مجالات عمل الخريج

- International companies working in natural languages processing – Smart programming companies – Smart home components manufacturers – digital transformation companies' startups

- الشركات الدولية العاملة في معالجة اللغات الطبيعية – شركات البرمجة الذكية – شركات تصنيع منتجات المنازل الذكية – شركات التحول الرقمي الناشئة.

Benchmark

المرجعية

- The program has adopted the general knowledge and skills issued from the National authority for quality assurance and accreditation of education (NARS) for computer science graduate which can be found through the following link. http://naqaae.eg/wp-content/uploads/2014/PDF/26.pdf
- The bylaws of computer science faculties have been studied and referenced in the following universities: Stanford University, Carnegie Mellon University, Nile University. https://www.cs.cmu.edu/bs-in-artificial intelligence/curriculum
- Computer Reference Standard https://www.abet.org/accreditation/accreditation-criteria/criteria-for-accrediting-computing-programs-2018-2019

- وقد تبنى البرنامج المعارف والمهارات الصادرة من الهيئة القومية لضمان جودة التعليم والاعتماد (NARS) لخريجى الحاسبات والمعلومات والتى يمكن الاطلاع عليها من خلال الرابط التالى:
http://naqaae.eg/wp-content/uploads/2014/PDF/26.pdf
- تم الاطلاع والرجوع للوائح كليات الحاسبات فى جامعة ستانفورد - كارنيجى ميلون – جامعة النيل.
- المعايير المرجعية للحوسبة
- https://www.abet.org/accreditation/accreditation--criteria/criteria

Artificial Intelligence Science Program

برنامج علوم الذكاء الاصطناعي

Study Plan

الخطة الدراسية

Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
1	MAT111	Mathematics I	3	0	1	0	4	3	150	6
	PHY211	Physics II	2	2	0	0	4	3	150	6
	CSE014	Structured Programming	2	3	0	0	5	3	180	6
	UC1	University Requirement (1)	2	0	0	0	2	2	90	4
	UC2	University Requirement (2)	2	0	0	0	2	2	90	4
	UE1	Elective University (1)	2	0	0	0	2	2	90	4
							19	**15**	**750**	**30**
2	MAT112	Mathematics II	3	0	1	0	4	3	150	6
	MAT131	Probability & Statistics I	2	1	2	0	5	3	195	6
	CSE015	Object Oriented Programming	2	3	0	0	5	3	180	6
	CSE315	Discrete Mathematics	2	0	2	0	4	3	150	6
	UC3	University Requirement (3)	2	0	0	0	2	2	90	4
	UE2	Elective University (2)	2	0	0	0	2	2	90	4
							22	**16**	**855**	**32**
3	MAT212	Linear Algebra	2	2	0	0	4	3	150	6
	MAT231	Probability & Statistics II	2	2	1	0	5	3	195	6

Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
	CSE111	Data Structures	2	3	0	0	5	3	180	6
	CSE131	Logic Design	2	2	2	0	6	3	195	6
	AIE111	Artificial Intelligence	2	2	1	0	5	3	180	6
	UC4	University Requirement (4)	2	0	0	0	2	2	90	4
							27	**17**	**990**	**34**
4	MAT312	Differential Equations	2	0	2	0	4	3	150	6
	CSE132	Computer Architecture & Organization	2	2	1	0	5	3	165	6
	CSE221	Database Systems	2	2	1	0	5	3	180	6
	CSE281	Image Processing	2	2	1	0	5	3	180	6
	AIE121	Machine Learning	2	2	1	0	5	3	180	6
	AIE191	Field Training 1 in AI Science	2	0	0	0	2	2	60	4
							26	**17**	**915**	**34**
5	CSE233	Operating Systems	2	2	1	0	5	3	165	6
	CSE251	Software Engineering	2	2	1	0	5	3	180	6
	CSE261	Computer Networks	2	2	1	0	5	3	165	6
	AIE231	Neural Networks	2	2	1	0	5	3	180	6
	AIE323	Data Mining	2	2	1	0	5	3	180	6
	UC5	University Requirement (5)	2	0	0	0	2	2	90	4
							27	**17**	**960**	**34**
6	CSE112	Design & Analysis of Algorithms	3	0	3	0	6	4	195	8
	AIE212	Knowledge-Based Systems	2	2	1	0	5	3	180	6
	AIE213	Optimization Techniques	3	0	3	0	6	3	180	6
	AIE241	Natural Language Processing	2	2	1	0	5	3	180	6

Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
	AIE292	Field Training 2 in AI Science	2	0	0	0	2	2	60	4
	E1	Elective Course (1)	2	2	1	0	5	3	180	6
							29	18	975	36
7	CSE363	Cloud Computing	2	2	1	0	5	3	180	6
	AIE322	Advanced Machine Learning	2	2	1	0	5	3	180	6
	AIE332	Deep Learning	2	2	1	0	5	3	180	6
	AIE493	Graduation Project 1	2	3	0	0	5	3	180	6
	E2	Elective Course (2)	2	2	1	0	5	3	180	6
	UC6	University Requirement (6)	2	0	0	0	2	2	90	4
							27	17	990	34
8	AIE425	Intelligent Recommender Systems	2	2	1	0	5	3	180	6
	AIE314	AI-Based Programming	2	2	1	0	5	3	180	6
	AIE494	Graduation Project 2	2	3	0	0	5	3	180	6
	E3	Elective Course (3)	2	2	1	0	5	3	180	6
	UC7	University Requirement (7)	2	0	0	0	2	2	90	4
	UE3	Elective University (3)	2	0	0	0	2	2	90	4
							24	16	900	32
							201	133	7335	266

Elective (E1)

CSE211 Web Programming
CSE234 Embedded Systems
CSE271 Introduction to Parallel Computing

CSE322	Big Data Analytics 1
CSE344	Introduction to Cyber Security
CSE382	Computer Graphics
CSE383	Computer Vision
CSE411	Theory of Computation & Compilers

Elective (E2)

CSE464	Internet of things
CSE475	High Performance Computing
CSE485	Computer Games Theory
AIE417	Selected Topics in Artificial Intelligence 1
AIE315	Computational Logic
AIE316	Evolutionary Algorithms
AIE342	Advanced Methods for Data Analysis
AIE343	Machine Learning for Text Mining
AIE351	Robotics Design

Elective (E3)

CSE487	Mixed & Augmented Reality
CSE488	Visualization & Animation
AIE418	Selected Topics in Artificial Intelligence 2
AIE424	Intelligent Decision Support Systems
AIE426	Decision Making Under Uncertainty
AIE427	Statistical Pattern Recognition
AIE444	Question Answering Systems
AIE457	Mobile Robot Development

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برنامج المعلوماتية الطبية الحيوية

Biomedical Informatics Program

Program Description

وصف البرنامج

– The program focuses on developing Informatics systems that depend on medical data and measurements, Radiology images, and magnetic resonance. Then, recognize and analyze these data.

– يهتم البرنامج بتطوير النظم المعلوماتية المعتمدة على بيانات وقياسات طبية وصور أشعة ورنين مغناطيسي، وتحليل تلك البيانات والتعرف عليها.

Graduate attributes

مواصفات الخريج

In addition, the graduate of the Biomedical Informatics program has all the modules of graduates of computer Science programs. The graduates have the following characteristics intensively

– The ability to develop informatics systems that depend on medical data and measurements.

بالإضافة الى تمتع خريج برنامج المعلوماتية الطبية الحيوية بكافة مقومات خريجي برامج علوم الحاسب، يتسم الخريجين بالسمات التالية بشكل مكثف

– القدرة على تطوير النظم المعلوماتية المعتمدة على بيانات وقياسات طبية

– The ability to build software that embed radiology photos, magnetic resonance, and medical measurements; to analyze the data outcome from them

– القدرة على انشاء برمجيات تدمج صور الأشعة الطبية والرنين المغناطيسي، والقياسات الطبية وتقوم بتحليل البيانات الصادرة منها

Job Opportunities

مجالات عمل الخريج

– Companies working in medical systems – Information technology department in health care companies.

– الشركات العاملة في النظم الطبية – إدارات تكنولوجيا المعلومات في شركات التامين الصحي.

Benchmark

المرجعية

– The bylaws of computer science faculties that offer Biomedical Informatics program in Stanford University - Nile University.

– تم الاطلاع والرجوع للوائح كليات الحاسبات التي تقدم برنامج المعلوماتية الطبية الحيوية في جامعة ستانفورد - جامعة النيل

Biomedical Informatics Program

برنامج المعلوماتية الطبية الحيوية

Study Plan

الخطة الدراسية

Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
1	MAT111	Mathematics I	3	0	1	0	4	3	150	6
	PHY281	Biophysics	2	2	0	0	4	3	150	6
	CSE014	Structured Programming	2	3	0	0	5	3	180	6
	UC1	University Requirement (1)	2	0	0	0	2	2	90	4
	UC2	University Requirement (2)	2	0	0	0	2	2	90	4
	UE1	Elective University (1)	2	0	0	0	2	2	90	4
							19	**15**	**750**	**30**
2	MAT112	Mathematics II	3	0	1	0	4	3	150	6
	MAT131	Probability & Statistics I	2	1	2	0	5	3	195	6
	BIO241	Biology II	2	2	0	0	4	3	150	6
	CSE015	Object Oriented Programming	2	3	0	0	5	3	180	6
	UC3	University Requirement (3)	2	0	0	0	2	2	90	4
	UE2	Elective University (2)	2	0	0	0	2	2	90	4
							22	**16**	**855**	**32**
3	MAT212	Linear Algebra	2	2	0	0	4	3	150	6
	CSE111	Data Structures	2	3	0	0	5	3	180	6
	CSE131	Logic Design	2	2	2	0	6	3	195	6

Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
	AIE111	Artificial Intelligence	2	2	1	0	5	3	180	6
	BIO221	Molecular Biology I	2	2	0	0	4	3	150	6
	BMD191	Field Training 1 in Biomedical Informatics	2	0	0	0	2	2	60	4
							26	**17**	**915**	**34**
4	MAT312	Differential Equations	2	0	2	0	4	3	150	6
	CSE221	Database Systems	2	2	1	0	5	3	180	6
	CSE315	Discrete Mathematics	2	0	2	0	4	3	150	6
	BMD241	Human Physiology	2	2	1	0	5	3	180	6
	BMD311	Introduction to Bioinformatics	2	2	1	0	5	3	180	6
	UC4	University Requirement (4)	2	0	0	0	2	2	90	4
							25	**17**	**930**	**34**
5	CSE251	Software Engineering	2	2	1	0	5	3	180	6
	E1	Elective Course (1)	2	2	1	0	5	3	180	6
	CSE281	Image Processing	2	2	1	0	5	3	180	6
	BMD312	Clinical Informatics	2	2	1	0	5	3	180	6
	BMD351	Biomedical Data Acquisition	2	2	1	0	5	3	180	6
	UC5	University Requirement (5)	2	0	0	0	2	2	90	4
							27	**17**	**990**	**34**
6	CSE112	Design & Analysis of Algorithms	3	0	3	0	6	4	195	8
	CSE352	Systems Analysis & Design	2	2	1	0	5	3	165	6
	BMD361	Biomedical Statistics	2	2	1	0	5	3	180	6
	BMD292	Field Training 2 in Biomedical Informatics	2	0	0	0	2	2	60	4

Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
	CSE261	Computer Networks	2	2	1	0	5	3	165	6
	UC6	University Requirement (6)	2	0	0	0	2	2	90	4
							25	17	855	34
7	BMD421	Biomedical Information Systems	2	2	1	0	5	3	180	6
	BMD431	Medical Image Informatics	2	2	1	0	5	3	180	6
	BMD493	Graduation Project 1	2	3	0	0	5	3	180	6
	E2	Elective Course (2)	2	2	1	0	5	3	180	6
	E3	Elective Course (3)	2	2	1	0	5	3	180	6
	UE3	Elective University (3)	2	0	0	0	2	2	90	4
							27	17	990	34
8	BMD413	Structural Bioinformatics	2	2	1	0	5	3	180	6
	BMD494	Graduation Project 2	2	3	0	0	5	3	180	6
	E4	Elective Course (4)	2	2	1	0	5	3	180	6
	E5	Elective Course (5)	2	2	1	0	5	3	180	6
	E6	Elective Course (6)	2	2	1	0	5	3	180	6
	UC7	University Requirement (7)	2	0	0	0	2	2	90	4
							27	17	990	34
							198	133	7275	266

Elective (E1)

BIO131 Biology I

MAT231	Probability & Statistics II
MAT315	Numerical Analysis
ELE113	Electrical Drawing
ELE432	Digital Signal Processing
CSE132	Computer Architecture & Organization
CSE211	Web Programming
CSE233	Operating Systems
CSE241	Security of Information Systems

Elective (E2)

CSE234	Embedded Systems
CSE272	Parallel & Distributed Systems
CSE322	Big Data Analytics 1
CSE363	Cloud Computing
CSE344	Introduction to Cyber Security
CSE383	Computer Vision
AIE121	Machine Learning

Elective (E3)

CSE446	Information & Computer Networks Security
AIE231	Neural Networks
AIE241	Natural Language Processing
AIE317	Artificial intelligence in medicine
AIE323	Data Mining

Elective (E4)

AIE342	Advanced Methods for Data Analysis
AIE343	Machine Learning for Text Mining
AIE424	Intelligent Decision Support Systems
AIE425	Intelligent Recommender Systems

Elective (E5)

CSE453	Building Information Systems
CSE464	Internet of Things
CSE475	High Performance Computing
CSE484	Interactive Multimedia
BMD414	Selected Topics in Biomedical Informatics 1: Data Analysis & Visualization

Elective (E6)

BMD415	Selected Topics in Biomedical Informatics 2: Data Mining & Machine Learning
BMD422	Systems Biology
BMD452	Biomedical Text Processing
BMD462	Bio-Inspired Computing

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برنامج تكنولوجيا المعلومات

Information Technology Program

Program Description

– The program focuses on qualifying graduates to meet the user's technology needs. Scientific programs in information technology began to appear in the 1990s and developed to keep pace with the needs of the current and future labor market.

وصف البرنامج

– يهتم البرنامج بالتركيز على تأهيل خريجين لتلبية احتياجات المستخدم من التكنولوجيا. وقد بدأت البرامج العلمية في تكنولوجيا المعلومات في الظهور في التسعينيات. وتطورت لتواكب احتياجات سوق العمل الحالية والمستقبلية.

Graduate attributes

مواصفات الخريج

In addition, the graduate of the Information Technology program has all the modules of graduates of computer Science programs. The graduates have the following characteristics intensively

- The ability to apply critical thinking methods to solve and diagnose various IT problems including information security problems.
- Ability to manage and monitor the performance of operating systems, networks and databases.

بالإضافة الى تمتع خريج برنامج تكنولوجيا المعلومات بكافة مقومات خريجي برامج علوم الحاسب، يتسم الخريجين بالسمات التالية بشكل مكثف

- القدرة على تطبيق أساليب التفكير النقدي لحل وتشخيص مشاكل تكنولوجيا المعلومات المختلفة بما في ذلك مشاكل أمن المعلومات.
- القدرة على إدارة ومراقبة أداء أنظمة التشغيل والشبكات وقواعد البيانات.

Job Opportunities

مجالات عمل الخريج

- Companies working in the field of information technology - Information technology departments in different companies. Develop and support the requirements of modern institutions of computing technology

- الشركات العاملة في مجال تكنولوجيا المعلومات – إدارات تكنولوجيا المعلومات في الشركات المختلفة. تطوير ودعم متطلبات المؤسسات الحديثة من تكنولوجيا الحوسبة

Benchmark

المرجعية

– The bylaws of computer science faculties that offer Biomedical Informatics program in James Madison University - Information Technology Program at Arizona State University.

– تم الاطلاع والرجوع للوائح كليات الحاسبات التي تقدم برنامج تكنولوجيا المعلومات في جامعة جيمس ماديسون - برنامج تكنولوجيا المعلومات بجامعة ولاية أريزونا.

Information Technology Program

برنامج تكنولوجيا المعلومات

Study Plan

الخطة الدراسية

Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
1	MAT111	Mathematics I	3	0	1	0	4	3	150	6
	PHY211	Physics II	2	2	0	0	4	3	150	6
	BIS251	Business Information Systems	3	1	0	0	4	3	150	6
	CSE014	Structured Programming	2	3	0	0	5	3	180	6
	UC1	University Requirement (1)	2	0	0	0	2	2	90	4
	UC2	University Requirement (2)	2	0	0	0	2	2	90	4

Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
							21	**16**	**810**	**32**
2	MAT112	Mathematics II	3	0	1	0	4	3	150	6
	MAT131	Probability & Statistics I	2	1	2	0	5	3	195	6
	CSE015	Object Oriented Programming	2	3	0	0	5	3	180	6
	CSE315	Discrete Mathematics	2	0	2	0	4	3	150	6
	UC3	University Requirement (3)	2	0	0	0	2	2	90	4
	UE1	Elective University (1)	2	0	0	0	2	2	90	4
							22	**16**	**855**	**32**
3	MAT212	Linear Algebra	2	2	0	0	4	3	150	6
	MAT231	Probability & Statistics II	2	2	1	0	5	3	195	6
	CSE111	Data Structures	2	3	0	0	5	3	180	6
	CSE131	Logic Design	2	2	2	0	6	3	195	6
	UC4	University Requirement (4)	2	0	0	0	2	2	90	4
	UE2	Elective University (2)	2	0	0	0	2	2	90	4
							24	**16**	**900**	**32**
4	MAT315	Numerical Analysis	2	0	2	0	4	3	150	6
	CSE132	Computer Architecture & Organization	2	2	1	0	5	3	165	6
	CSE211	Web Programming	2	2	1	0	5	3	165	6
	CSE221	Database Systems	2	2	1	0	5	3	180	6
	CSE241	Security of Information Systems	2	2	1	0	5	3	150	6
	CSE192	Field Training 1 in Information Technology	2	0	0	0	2	2	60	4
							26	**17**	**870**	**34**
	CSE232	Operating Systems Administration	2	2	1	0	5	3	180	6

Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
5	CSE251	Software Engineering	2	2	1	0	5	3	180	6
	BIS356	Information Technology Governance	3	0	0	0	3	3	135	6
	CSE262	Introduction to Telecommunications, Networking and Security	2	2	1	0	5	3	180	6
	AIE122	Basics of Machine Learning and Data Science	2	2	1	0	5	3	180	6
	UC5	University Requirement (5)	2	0	0	0	2	2	90	4
							25	**17**	**945**	**34**
6	CSE321	Societal and Ethical Issues in Information Technology	2	2	1	0	5	3	180	6
	CSE324	IT Management and Operations	2	2	1	0	5	3	180	6
	CSE344	Introduction to Cyber Security	2	2	1	0	5	3	180	6
	E1	Elective Course (1)	2	2	1	0	5	3	180	6
	CSE293	Field Training 2 in Information Technology	2	0	0	0	2	2	60	4
	UC6	University Requirement (6)	2	0	0	0	2	2	90	4
							24	**16**	**870**	**32**
7	CSE325	Database Systems Administration	2	2	1	0	5	3	180	6
	CSE363	Cloud Computing	2	2	1	0	5	3	180	6
	CSE361	Networks Virtualization and Administration	2	2	1	0	5	3	180	6
	CSE493	Graduation Project 1	2	3	0	0	5	3	180	6
	UC6	University Requirement (6)	2	0	0	0	2	2	90	4
	E2	Elective Course (2)	2	2	1	0	5	3	180	6
							27	**17**	**990**	**34**
8	CSE353	Interaction Design	2	2	1	0	5	3	180	6
	CSE463	Advanced Networking	2	2	1	0	5	3	180	6
	CSE494	Graduation Project 2	2	3	0	0	5	3	180	6

Semester	Code	Course Name	LCT	LAB	TUT	OTH	CNTCT	CH	SWL	ECTS
	E3	Elective Course (3)	2	2	1	0	5	3	180	6
	E4	Elective Course (4)	2	2	1	0	5	3	180	6
	UE3	Elective University (3)	2	0	0	0	2	2	90	4
							27	**17**	**990**	**34**
							196	**132**	**7230**	**264**

Elective (E1)

CSE112	Design & Analysis of Algorithms
CSE312	Advanced Web Programming
CSE313	Mobile Programming
CSE314	Android and iOS Development
AIE111	Artificial Intelligence

Elective (E2)

CSE351	Software Project Management
CSE362	Industrial Networks
CSE426	Selected Topics in Data Science
CSE427	Selected Topics in Big Data
AIE121	Machine Learning

Elective (E3)

CSE322	Big Data Analytics 1
CSE446	Information & Computer Networks Security
CSE447	Selected Topics in Computer Security
CSE448	Cyber Forensics
CSE464	Internet of things
CSE467	Client/Server Technologies & Applications
CSE468	Web Server Administration

Elective (E4)

AIE212	Knowledge-Based Systems
ELE232	Fundamentals of Communication Systems
BIS452	Integrated Information Systems
CSE465	Selected Topics in Cloud Computing
CSE466	Selected Topics in IoT

Alamein- CE PROGRAM/Embedded Systems Track- COURSE MAP 2020 - 157 CR HRS

Level	Term	Code	Course	CR	CNTC	STWL	LEC	TUT	LAB
LEVEL 0	FALL (17 CREDIT HOURS, 24 CONTACT HOURS, 855)	MAT 111	Mathematics I	3	4	150	3	1	0
		MEC 011	Engineering Drawing (1)	3	5	135	2	3	0
		PHY 212	Introduction to Engineering Physics	3	4	150	2	0	2
		CSE 014	Structured Programming	3	5	180	2	0	3
		UC1 1xx	Univ Req1	2	2	90	2	0	0
		MAT 123	Mechanics	3	4	150	2	0	2
	SPRING (16 CREDIT HOURS, 22 CONTACT HOURS, 855)	MAT 112	Mathematics II	3	4	150	3	1	0
		MAT 131	Probability & Statistics I	3	5	195	2	2	1
		CSE 315	Discrete Mathematics	3	4	150	2	2	0
		CSE 015	Object Oriented Programming	3	5	180	2	0	3
		UC2 1xx	Univ Req2	2	2	90	2	0	0
		UC3 2xx	Univ Req3	2	2	90	2	0	0
LEVEL 1	FALL (16 CREDIT HOURS, 25 CONTACT HOURS, 885)	MAT 212	Linear Algebra	3	4	150	2	0	2
		CSE 111	Data Structures	3	5	180	2	0	3
		CSE 113	Electric & Electronic Circuits	3	6	180	1	2	3
		CSE 131	Logic Design	3	6	195	2	2	2
		UC4 2xx	Univ Req4	2	2	90	2	0	0
		UC5 2xx	Univ Req5	2	2	90	2	0	0
	SPRING (15 CREDIT HOURS, 24 CONTACT HOURS, 870)	MAT 231	Probability & Statistics II	3	5	195	2	1	2
		MAT 312	Differential Equations	3	4	150	2	2	0
		CSE 132	Computer Architecture & Organization	3	5	165	2	1	2
		AIE 111	Artificial Intelligence	3	5	180	2	1	2
		E1 2xx	ENG Elect 1	3	5	180	2	1	2
LEVEL 2	FALL (16 CREDIT HOURS, 24 CONTACT HOURS, 855)	E2 2xx	ENG Elect 2	3	5	180	2	2	1
		CSE 221	Database Systems	3	5	180	2	1	2
		UE1 1xx	Univ Elect1	2	2	90	2	0	0
		CSE 233	Operating Systems	3	5	165	2	1	2
		AIE 121	Machine Learning	3	5	180	2	1	2
		CSE 291	Field Training 1 in Computer Engineering	2	2	60	2	0	0
	SPRING (16 CREDIT HOURS, 26 CONTACT HOURS, 885)	CSE 211	Web Programming	3	5	165	2	1	2
		CSE 251	Software Engineering	3	5	180	2	1	2
		CSE 261	Computer Networks	3	5	165	2	1	2
		CSE 112	Design and Analysis of Algorithms	4	6	195	3	3	0
		CSE 234	Embedded Systems	3	5	180	2	1	2
LEVEL 3	FALL (17 CREDIT HOURS, 26 CONTACT HOURS, 960)	ELE 432	Digital Signals Processing	3	4	150	2	1	1
		CSE 281	Image Processing	3	5	180	2	1	2
		CSE 242	Cryptography	3	5	180	2	1	2
		CSE 344	Introduction to Cyber Security	3	5	180	2	1	2
		CSE 363	Cloud Computing	3	5	180	2	1	2
		UE2 1xx	Univ Elect2	2	2	90	2	0	0
	SPRING (16 CREDIT HOURS, 24 CONTACT HOURS, 840)	CSE 272	Parallel & Distributed Systems	3	5	150	2	1	2
		CSE 446	Information & Computer Networks Security	3	5	180	2	1	2
		CSE 464	Internet of Things	3	5	180	2	1	2
		CSE 335	Real Time & Embedded Systems Design	3	5	180	2	1	2
		CSE 392	Field Training 2 in Computer Engineering	2	2	60	2	0	0
		UC6 3xx	Univ Req6	2	2	90	2	0	0
LEVEL 4	FALL (14 CREDIT HOURS, 22 CONTACT HOURS, 810)	E3 3xx	Tech ELECT 3	3	5	180	2	1	2
		E4 3xx	Tech ELECT 4	3	5	180	2	1	2
		CSE 436	Selected Topics in Embedded Systems	3	5	180	2	1	2
		CSE 493	Graduation Project 1	3	5	180	2	0	3
		UC7 3xx	Univ Req7	2	2	90	2	0	0
	SPRING (14 CREDIT HOURS, 22 CONTACT HOURS, 810)	E5 3xx	Tech ELECT 5	3	5	180	2	1	2
		E6 4xx	Tech ELECT 6	3	5	180	2	1	2
		CSE 466	Selected Topics in IoT	3	5	180	2	1	2
		CSE 494	Graduation Project 2	3	5	180	2	0	3
		UE3 3xx	Univ Elect3	2	2	90	2	0	0

COURSE MAP LEGEND

- AI Courses
- University Req
- Basic Science and Eng
- MAJOR ELECTIVE
- Major Compulsory
- Comp Science and Eng
- Graduation Project

COURSE CODE
COURSE NAME
CR | CNTC | STWL
LEC | TUT | LAB

Alamein- CE PROGRAM/cloud Track- COURSE MAP 2020 - 157 CR HRS

LEVEL 0

FALL — CREDIT HOURS 17 — CONTACT HOURS 24 — 855

Course	Title	CR	CNTC	STWL	LEC	TUT	LAB
MAT 111	Mathematics I	3	4	150	3	1	0
MEC 011	Engineering Drawing (1)	3	5	135	2	3	0
PHY 212	Introduction to Engineering Physics	3	4	150	2	0	2
CSE 014	Structured Programming	3	5	180	2	0	3
UC1 1xx	Univ Req1	2	2	90	2	0	0
MAT 123	Mechanics	3	4	150	2	0	2

SPRING — CREDIT HOURS 16 — CONTACT HOURS 22 — 855

Course	Title	CR	CNTC	STWL	LEC	TUT	LAB
MAT 112	Mathematics II	3	4	150	3	1	0
MAT 131	Probability & Statistics I	3	5	195	2	2	1
CSE 315	Discrete Mathematics	3	4	150	2	2	0
CSE 015	Object Oriented Programming	3	5	180	2	0	3
UC2 1xx	Univ Req2	2	2	90	2	0	0
UC3 2xx	Univ Req3	2	2	90	2	0	0

LEVEL 1

FALL — CREDIT HOURS 16 — CONTACT HOURS 25 — 885

Course	Title	CR	CNTC	STWL	LEC	TUT	LAB
MAT 212	Linear Algebra	3	4	150	2	0	2
CSE 111	Data Structures	3	5	180	2	0	3
CSE 113	Electric & Electronic Circuits	3	6	180	1	2	3
CSE 131	Logic Design	3	6	195	2	2	2
UC4 2xx	Univ Req4	2	2	90	2	0	0
UC5 2xx	Univ Req5	2	2	90	2	0	0

SPRING — CREDIT HOURS 15 — CONTACT HOURS 24 — 870

Course	Title	CR	CNTC	STWL	LEC	TUT	LAB
MAT 231	Probability & Statistics II	3	5	195	2	1	2
MAT 312	Differential Equations	3	4	150	2	2	0
CSE 132	Computer Architecture & Organization	3	5	165	2	1	2
AIE 111	Artificial Intelligence	3	5	180	2	1	2
E1 2xx	ENG Elect 1	3	5	180	2	1	2

LEVEL 2

FALL — CREDIT HOURS 16 — CONTACT HOURS 24 — 855

Course	Title	CR	CNTC	STWL	LEC	TUT	LAB
E2 2xx	ENG Elect 2	3	5	180	2	2	1
CSE 221	Database Systems	3	5	180	2	1	2
UE1 1xx	Univ Elect1	2	2	90	2	0	0
CSE 233	Operating Systems	3	5	165	2	1	2
AIE 121	Machine Learning	3	5	180	2	1	2
CSE 291	Field Training 1 in Computer Engineering	2	2	60	2	0	0

SPRING — CREDIT HOURS 16 — CONTACT HOURS 26 — 885

Course	Title	CR	CNTC	STWL	LEC	TUT	LAB
CSE 211	Web Programming	3	5	165	2	1	2
CSE 251	Software Engineering	3	5	180	2	1	2
CSE 261	Computer Networks	3	5	165	2	1	2
CSE 112	Design and Analysis of Algorithms	4	6	195	3	3	0
CSE 234	Embedded Systems	3	5	180	2	1	2

LEVEL 3

FALL — CREDIT HOURS 17 — CONTACT HOURS 26 — 960

Course	Title	CR	CNTC	STWL	LEC	TUT	LAB
ELE 432	Digital Signals Processing	3	4	150	2	1	1
CSE 242	Cryptography	3	5	180	2	1	2
CSE 271	Introduction to Parallel Computing	3	5	180	2	1	2
CSE 344	Introduction to Cyber Security	3	5	180	2	1	2
CSE 363	Cloud Computing	3	5	180	2	1	2
UE2 1xx	Univ Elect2	2	2	90	2	0	0

SPRING — CREDIT HOURS 16 — CONTACT HOURS 24 — 840

Course	Title	CR	CNTC	STWL	LEC	TUT	LAB
CSE 272	Parallel & Distributed Systems	3	5	150	2	1	2
CSE 446	Information & Computer Networks Security	3	5	180	2	1	2
CSE 464	Internet of Things	3	5	180	2	1	2
CSE 322	Big Data Analytics 1	3	5	180	2	1	2
CSE 392	Field Training 2 in Computer Engineering	2	2	60	2	0	0
UC6 3xx	Univ Req6	2	2	90	2	0	0

LEVEL 4

FALL — CREDIT HOURS 14 — CONTACT HOURS 22 — 810

Course	Title	CR	CNTC	STWL	LEC	TUT	LAB
E3 3xx	Tech ELECT 3	3	5	180	2	1	2
E4 3xx	Tech ELECT 4	3	5	180	2	1	2
CSE 465	Selected Topics in Cloud Computing	3	5	180	2	1	2
CSE 493	Graduation Project 1	3	5	180	2	0	3
UC7 3xx	Univ Req7	2	2	90	2	0	0

SPRING — CREDIT HOURS 14 — CONTACT HOURS 22 — 810

Course	Title	CR	CNTC	STWL	LEC	TUT	LAB
E5 3xx	Tech ELECT 5	3	5	180	2	1	2
E6 4xx	Tech ELECT 6	3	5	180	2	1	2
CSE 427	Selected Topics in Big Data	3	5	180	2	1	2
CSE 494	Graduation Project 2	3	5	180	2	0	3
UE3 3xx	Univ Elect3	2	2	90	2	0	0

COURSE MAP LEGEND

- AI Courses
- University Req
- Basic Science and Eng
- MAJOR ELECTIVE
- Major Compulsory
- Comp Science and Eng
- Graduation Project

COURSE CODE
COURSE NAME
CR | CNTC | STWL
LEC | TUT | LAB

Alamein- CE PROGRAM/HPC Track- COURSE MAP 2020 - 157 CR HRS

Level	Term	Course	Name	CR	CNTC	STWL	LEC	TUT	LAB
LEVEL 0	FALL (17 CREDIT HOURS, 24 CONTACT HOURS, 855)	MAT 111	Mathematics I	3	4	150	3	1	0
		MEC 011	Engineering Drawing (1)	3	5	135	2	3	0
		PHY 212	Introduction to Engineering Physics	3	4	150	2	0	2
		CSE 014	Structured Programming	3	5	180	2	0	3
		UC1 1xx	Univ Req1	2	2	90	2	0	0
		MAT 123	Mechanics	3	4	150	2	0	2
	SPRING (16 CREDIT HOURS, 22 CONTACT HOURS, 855)	MAT 112	Mathematics II	3	4	150	3	1	0
		MAT 131	Probability & Statistics I	3	5	195	2	2	1
		CSE 315	Discrete Mathematics	3	4	150	2	2	0
		CSE 015	Object Oriented Programming	3	5	180	2	0	3
		UC2 1xx	Univ Req2	2	2	90	2	0	0
		UC3 2xx	Univ Req3	2	2	90	2	0	0
LEVEL 1	FALL (16 CREDIT HOURS, 25 CONTACT HOURS, 885)	MAT 212	Linear Algebra	3	4	150	2	0	2
		CSE 111	Data Structures	3	5	180	2	0	3
		CSE 113	Electric & Electronic Circuits	3	6	180	1	2	3
		CSE 131	Logic Design	3	6	195	2	2	2
		UC4 2xx	Univ Req4	2	2	90	2	0	0
		UC5 2xx	Univ Req5	2	2	90	2	0	0
	SPRING (15 CREDIT HOURS, 24 CONTACT HOURS, 870)	MAT 231	Probability & Statistics II	3	5	195	2	1	2
		MAT 312	Differential Equations	3	4	150	2	2	0
		CSE 132	Computer Architecture & Organization	3	5	165	2	1	2
		AIE 111	Artificial Intelligence	3	5	180	2	1	2
		E1 2xx	ENG Elect 1	3	5	180	2	1	2
LEVEL 2	FALL (16 CREDIT HOURS, 24 CONTACT HOURS, 855)	E2 2xx	ENG Elect 2	3	5	180	2	2	1
		CSE 221	Database Systems	3	5	180	2	1	2
		UE1 1xx	Univ Elect1	2	2	90	2	0	0
		CSE 233	Operating Systems	3	5	165	2	1	2
		AIE 121	Machine Learning	3	5	180	2	1	2
		CSE 291	Field Training 1 in Computer Engineering	2	2	60	2	0	0
	SPRING (16 CREDIT HOURS, 26 CONTACT HOURS, 885)	CSE 211	Web Programming	3	5	165	2	1	2
		CSE 251	Software Engineering	3	5	180	2	1	2
		CSE 261	Computer Networks	3	5	165	2	1	2
		CSE 112	Design and Analysis of Algorithms	4	6	195	3	3	0
		CSE 234	Embedded Systems	3	5	180	2	1	2
LEVEL 3	FALL (17 CREDIT HOURS, 26 CONTACT HOURS, 930)	ELE 432	Digital Signals Processing	3	4	150	2	1	1
		CSE 271	Introduction to Parallel Computing	3	5	180	2	1	2
		CSE 272	Parallel & Distributed Systems	3	5	150	2	1	2
		CSE 344	Introduction to Cyber Security	3	5	180	2	1	2
		CSE 363	Cloud Computing	3	5	180	2	1	2
		UE2 1xx	Univ Elect2	2	2	90	2	0	0
	SPRING (16 CREDIT HOURS, 24 CONTACT HOURS, 870)	CSE 373	Parallel Programming	3	5	180	2	1	2
		CSE 446	Information & Computer Networks Security	3	5	180	2	1	2
		CSE 464	Internet of Things	3	5	180	2	1	2
		CSE 475	High Performance Computing	3	5	180	2	1	2
		CSE 392	Field Training 2 in Computer Engineering	2	2	60	2	0	0
		UC6 3xx	Univ Req6	2	2	90	2	0	0
LEVEL 4	FALL (14 CREDIT HOURS, 22 CONTACT HOURS, 810)	E3 3xx	Tech ELECT 3	3	5	180	2	1	2
		E4 3xx	Tech ELECT 4	3	5	180	2	1	2
		CSE 411	Theory of Computation & Compilers	3	5	180	2	1	2
		CSE 493	Graduation Project 1	3	5	180	2	0	3
		UC7 3xx	Univ Req7	2	2	90	2	0	0
	SPRING (14 CREDIT HOURS, 22 CONTACT HOURS, 810)	E5 3xx	Tech ELECT 5	3	5	180	2	1	2
		E6 4xx	Tech ELECT 6	3	5	180	2	1	2
		CSE 476	Selected Topics in High Performance Computing	3	5	180	2	1	2
		CSE 494	Graduation Project 2	3	5	180	2	0	3
		UE3 3xx	Univ Elect3	2	2	90	2	0	0

COURSE MAP LEGEND

- AI Courses
- University Req
- Basic Science and Eng
- MAJOR ELECTIVE
- Major Compulsory
- Comp Science and Eng
- Graduation Project

COURSE CODE
COURSE NAME
CR | CNTC | STWL
LEC | TUT | LAB

Alamein- CE PROGRAM/Security Track- COURSE MAP 2020 - 157 CR HRS

LEVEL 0

FALL — CREDIT HOURS 17 — CONTACT HOURS 24 — 855

Course	Name	CR	CNTC	STWL	LEC	TUT	LAB
MAT 111	Mathematics I	3	4	150	3	1	0
MEC 011	Engineering Drawing (1)	3	5	135	2	3	0
PHY 212	Introduction to Engineering Physics	3	4	150	2	0	2
CSE 014	Structured Programming	3	5	180	2	0	3
UC1 1xx	Univ Req1	2	2	90	2	0	0
MAT 123	Mechanics	3	4	150	2	0	2

SPRING — CREDIT HOURS 16 — CONTACT HOURS 22 — 855

Course	Name	CR	CNTC	STWL	LEC	TUT	LAB
MAT 112	Mathematics II	3	4	150	3	1	0
MAT 131	Probability & Statistics I	3	5	195	2	2	1
CSE 315	Discrete Mathematics	3	4	150	2	2	0
CSE 015	Object Oriented Programming	3	5	180	2	0	3
UC2 1xx	Univ Req2	2	2	90	2	0	0
UC3 2xx	Univ Req3	2	2	90	2	0	0

LEVEL 1

FALL — CREDIT HOURS 16 — CONTACT HOURS 25 — 885

Course	Name	CR	CNTC	STWL	LEC	TUT	LAB
MAT 212	Linear Algebra	3	4	150	2	0	2
CSE 111	Data Structures	3	5	180	2	0	3
CSE 113	Electric & Electronic Circuits	3	6	180	1	2	3
CSE 131	Logic Design	3	6	195	2	2	2
UC4 2xx	Univ Req4	2	2	90	2	0	0
UC5 2xx	Univ Req5	2	2	90	2	0	0

SPRING — CREDIT HOURS 15 — CONTACT HOURS 24 — 870

Course	Name	CR	CNTC	STWL	LEC	TUT	LAB
MAT 231	Probability & Statistics II	3	5	195	2	1	2
MAT 312	Differential Equations	3	4	150	2	2	0
CSE 132	Computer Architecture & Organization	3	5	165	2	1	2
AIE 111	Artificial Intelligence	3	5	180	2	1	2
E1 2xx	ENG Elect 1	3	5	180	2	1	2

LEVEL 2

FALL — CREDIT HOURS 16 — CONTACT HOURS 24 — 855

Course	Name	CR	CNTC	STWL	LEC	TUT	LAB
E2 2xx	ENG Elect 2	3	5	180	2	2	1
CSE 221	Database Systems	3	5	180	2	1	2
UE1 1xx	Univ Elect1	2	2	90	2	0	0
CSE 233	Operating Systems	3	5	165	2	1	2
AIE 121	Machine Learning	3	5	180	2	1	2
CSE 291	Field Training 1 in Computer Engineering	2	2	60	2	0	0

SPRING — CREDIT HOURS 16 — CONTACT HOURS 26 — 885

Course	Name	CR	CNTC	STWL	LEC	TUT	LAB
CSE 211	Web Programming	3	5	165	2	1	2
CSE 251	Software Engineering	3	5	180	2	1	2
CSE 261	Computer Networks	3	5	165	2	1	2
CSE 112	Design and Analysis of Algorithms	4	6	195	3	3	0
CSE 234	Embedded Systems	3	5	180	2	1	2

LEVEL 3

FALL — CREDIT HOURS 17 — CONTACT HOURS 26 — 930

Course	Name	CR	CNTC	STWL	LEC	TUT	LAB
ELE 432	Digital Signals Processing	3	4	150	2	1	1
CSE 241	Security of Information Systems	3	5	150	2	1	2
CSE 242	Cryptography	3	5	180	2	1	2
CSE 344	Introduction to Cyber Security	3	5	180	2	1	2
CSE 363	Cloud Computing	3	5	180	2	1	2
UE2 1xx	Univ Elect2	2	2	90	2	0	0

SPRING — CREDIT HOURS 16 — CONTACT HOURS 24 — 870

Course	Name	CR	CNTC	STWL	LEC	TUT	LAB
CSE 243	Secure Programming	3	5	180	2	0	3
CSE 446	Information & Computer Networks Security	3	5	180	2	1	2
CSE 464	Internet of Things	3	5	180	2	1	2
CSE 373	Parallel Programming	3	5	180	2	1	2
CSE 392	Field Training 2 in Computer Engineering	2	2	60	2	0	0
UC6 3xx	Univ Req6	2	2	90	2	0	0

LEVEL 4

FALL — CREDIT HOURS 14 — CONTACT HOURS 22 — 810

Course	Name	CR	CNTC	STWL	LEC	TUT	LAB
E3 3xx	Tech ELECT 3	3	5	180	2	1	2
E4 3xx	Tech ELECT 4	3	5	180	2	1	2
CSE 445	Selected Topics in Information Security	3	5	180	2	1	2
CSE 493	Graduation Project 1	3	5	180	2	0	3
UC7 3xx	Univ Req7	2	2	90	2	0	0

SPRING — CREDIT HOURS 14 — CONTACT HOURS 22 — 810

Course	Name	CR	CNTC	STWL	LEC	TUT	LAB
E5 3xx	Tech ELECT 5	3	5	180	2	1	2
E6 4xx	Tech ELECT 6	3	5	180	2	1	2
CSE 447	Selected Topics in Computer Security	3	5	180	2	1	2
CSE 494	Graduation Project 2	3	5	180	2	0	3
UE3 3xx	Univ Elect3	2	2	90	2	0	0

COURSE MAP LEGEND

AI Courses	
University Req	
Basic Science and Eng	
MAJOR ELECTIVE	
Major Compulsory	
Comp Science and Eng	
Graduation Project	

COURSE CODE
COURSE NAME
CR | CNTC | STWL
LEC | TUT | LAB

Alamein- AIE PROGRAM- COURSE MAP 2020 - 157 CR HRS

Level	Term	Course	Name	CR	CNTC	STWL	LEC	TUT	LAB
LEVEL 0	FALL (17 CREDIT HOURS, 24 CONTACT HOURS, 855)	MAT 111	Mathematics I	3	4	150	3	1	0
		MEC 011	Engineering Drawing (1)	3	5	135	2	3	0
		PHY 212	Introduction to Engineering Physics	3	4	150	2	0	2
		CSE 014	Structured Programming	3	5	180	2	0	3
		UC1 1xx	Univ Req1	2	2	90	2	0	0
		MAT 123	Mechanics	3	4	150	2	0	2
	SPRING (16 CREDIT HOURS, 22 CONTACT HOURS, 855)	MAT 112	Mathematics II	3	4	150	3	1	0
		MAT 131	Probability & Statistics I	3	5	195	2	2	1
		CSE 315	Discrete Mathematics	3	4	150	2	2	0
		CSE 015	Object Oriented Programming	3	5	180	2	0	3
		UC2 1xx	Univ Req2	2	2	90	2	0	0
		UC3 2xx	Univ Req3	2	2	90	2	0	0
LEVEL 1	FALL (16 CREDIT HOURS, 25 CONTACT HOURS, 885)	MAT 212	Linear Algebra	3	4	150	2	0	2
		CSE 111	Data Structures	3	5	180	2	0	3
		CSE 113	Electric & Electronic Circuits	3	6	180	1	2	3
		CSE 131	Logic Design	3	6	195	2	2	2
		UC4 2xx	Univ Req4	2	2	90	2	0	0
		UC5 2xx	Univ Req5	2	2	90	2	0	0
	SPRING (15 CREDIT HOURS, 24 CONTACT HOURS, 870)	MAT 231	Probability & Statistics II	3	5	195	2	1	2
		MAT 312	Differential Equations	3	4	150	2	2	0
		CSE 132	Computer Architecture & Organization	3	5	165	2	1	2
		AIE 111	Artificial Intelligence	3	5	180	2	1	2
		E1 2xx	ENG Elect 1	3	5	180	2	2	1
LEVEL 2	FALL (16 CREDIT HOURS, 24 CONTACT HOURS, 855)	E2 2xx	ENG Elect 2	3	5	180	2	2	1
		CSE 221	Database Systems	3	5	180	2	1	2
		UE1 1xx	Univ Elect1	2	2	90	2	0	0
		CSE 233	Operating Systems	3	5	165	2	1	2
		AIE 121	Machine Learning	3	5	180	2	1	2
		AIE 291	Field Training 1 in Artificial Intelligence Engineering	2	2	60	2	0	0
	SPRING (16 CREDIT HOURS, 26 CONTACT HOURS, 900)	CSE 281	Image Processing	3	5	180	2	1	2
		CSE 251	Software Engineering	3	5	180	2	1	2
		CSE 261	Computer Networks	3	5	165	2	1	2
		CSE 112	Design and Analysis of Algorithms	4	6	195	3	3	0
		CSE 234	Embedded Systems	3	5	180	2	1	2
LEVEL 3	FALL (17 CREDIT HOURS, 26 CONTACT HOURS, 960)	ELE 432	Digital Signals Processing	3	4	150	2	1	1
		AIE 323	Data Mining	3	5	180	2	1	2
		AIE 241	Natural Language Processing	3	5	180	2	1	2
		CSE 344	Introduction to Cyber Security	3	5	180	2	1	2
		CSE 363	Cloud Computing	3	5	180	2	1	2
		UE2 1xx	Univ Elect2	2	2	90	2	0	0
	SPRING (16 CREDIT HOURS, 25 CONTACT HOURS, 870)	CSE 383	Computer Vision	3	5	180	2	1	2
		AIE 212	Knowledge-Based Systems	3	5	180	2	1	2
		AIE 213	Optimization Techniques	3	6	180	3	3	0
		AIE 231	Neural Networks	3	5	180	2	1	2
		AIE 392	Field Training 2 in Artificial Intelligence Engineering	2	2	60	2	0	0
		UC6 3xx	Univ Req6	2	2	90	2	0	0
LEVEL 4	FALL (14 CREDIT HOURS, 22 CONTACT HOURS, 810)	E3 3xx	Tech ELECT 3	3	5	180	2	1	2
		E4 3xx	Tech ELECT 4	3	5	180	2	1	2
		AIE 351	Robotics Design	3	5	180	2	1	2
		AIE 493	Graduation Project 1	3	5	180	2	0	3
		UC7 3xx	Univ Req7	2	2	90	2	0	0
	SPRING (14 CREDIT HOURS, 22 CONTACT HOURS, 810)	E5 3xx	Tech ELECT 5	3	5	180	2	1	2
		E6 4xx	Tech ELECT 6	3	5	180	2	1	2
		AIE 332	Deep Learning	3	5	180	2	1	2
		AIE 494	Graduation Project 2	3	5	180	3	0	3
		UE3 3xx	Univ Elect3	2	2	90	2	0	0

COURSE MAP LEGEND

- AI Courses
- University Req
- Basic Science and Eng
- MAJOR ELECTIVE
- Major Compulsory
- Comp Science and Eng
- Graduation Project

COURSE CODE
COURSE NAME
CR | CNTC | STWL
LEC | TUT | LAB

Alamein- CS PROGRAM/Big data Track- COURSE MAP 2020 - 127 CR HRS

Level	Term	Course	Name	CR	CNTC	STWL	LEC	TUT	LAB
LEVEL 1	FALL (15 CREDIT HOURS, 19 CONTACT HOURS, 750)	CSE 014	Structured Programming	3	5	180	2	0	3
		PHY 211	Physics II	3	4	150	3	0	2
		MAT 111	Mathematics I	3	4	150	3	1	0
		UC1 1xx	Univ Req1	2	2	90	2	0	0
		UC2 1xx	Univ Req2	2	2	90	2	0	0
		UE1 1xx	Univ Elect1	2	2	90	2	0	0
	SPRING (16 CREDIT HOURS, 22 CONTACT HOURS, 855)	CSE 015	Object Oriented Programming	3	5	180	2	0	3
		MAT 131	Probability & Statistics I	3	5	195	2	2	1
		CSE 315	Discrete Mathematics	3	4	150	2	2	0
		MAT 112	Mathematics II	3	4	150	3	1	0
		UC3 2xx	Univ Req3	2	2	90	2	0	0
		UE2 2xx	Univ Elect2	2	2	90	2	0	0
LEVEL 2	FALL (17 CREDIT HOURS, 27 CONTACT HOURS, 990)	CSE 111	Data Structures	3	5	180	2	0	3
		CSE 131	Logic Design	3	6	195	2	2	2
		MAT 231	Probability & Statistics II	3	5	195	2	1	2
		AIE 111	Artificial Intelligence	3	5	180	2	1	2
		MAT 212	Linear Algebra	3	4	150	2	0	2
		UC4 2xx	Univ Req4	2	2	90	2	0	0
	SPRING (17 CREDIT HOURS, 26 CONTACT HOURS, 900)	CSE 211	Web Programming	3	5	165	2	1	2
		CSE 221	Database Systems	3	5	180	2	1	2
		CSE 132	Computer Architecture and Organization	3	5	165	2	1	2
		MAT 312	Differential Equations	3	4	150	2	2	0
		CSE 191	Field Training 1 In Computer Science	2	2	60	2	0	0
		CSE 251	Software Engineering	3	5	180	2	1	2
LEVEL 3	FALL (17 CREDIT HOURS, 27 CONTACT HOURS, 945)	CSE 261	Computer Networks	3	5	165	2	1	2
		E1 3xx	Tech ELECT 1	3	5	180	2	1	2
		AIE 121	Machine Learning	3	5	180	2	1	2
		CSE 352	Systems Analysis & Design	3	5	165	2	1	2
		CSE 233	Operating Systems	3	5	165	2	1	2
		UC5 2xx	Univ Req5	2	2	90	2	0	0
	SPRING (17 CREDIT HOURS, 25 CONTACT HOURS, 885)	CSE 112	Design and Analysis of Algorithms	4	6	195	3	3	0
		CSE 292	Field Training 2 Computer Science	2	2	60	2	0	0
		CSE 373	Parallel Programming	3	5	180	2	1	2
		E2 4xx	Tech ELECT 2	3	5	180	2	1	2
		CSE 363	Cloud Computing	3	5	180	2	1	2
		UC6 3xx	Univ Req6	2	2	90	2	0	0
LEVEL 4	FALL (17 CREDIT HOURS, 27 CONTACT HOURS, 990)	CSE 322	Big Data Analytics 1	3	5	180	2	1	2
		CSE 271	Introduction to Parallel Computing	3	5	180	2	1	2
		CSE 493	Graduation Project 1	3	5	180	2	0	3
		AIE 323	Data Mining	3	5	180	2	1	2
		UC7 3xx	Univ Req7	2	2	90	2	0	0
		E3 4xx	Tech ELECT 3	3	5	180	2	1	2
	SPRING (17 CREDIT HOURS, 27 CONTACT HOURS, 990)	CSE 427	Selected Topics in Big Data	3	5	180	2	1	2
		CSE 494	Graduation Project 2	3	5	180	2	0	3
		CSE 411	Theory of Computation & Compilers	3	5	180	2	1	2
		E4 4xx	Tech ELECT 4	3	5	180	2	1	2
		E5 4xx	Tech ELECT 5	3	5	180	2	1	2
		UE3 2xx	Univ Elect3	2	2	90	2	0	0

COURSE MAP LEGEND

COURSE CODE
COURSE NAME
CR | CNTC | STWL
LEC | TUT | LAB

University Req
Basic Science and Eng
MAJOR ELECTIVE
Major Compulsory
Comp Science and Eng
Graduation Project

Alamein- CS PROGRAM/Vision Track- COURSE MAP 2020 - 127 CR HRS

Level	Term	Course	Course Name	CR	CNTC	STWL	LEC	TUT	LAB
LEVEL 1	FALL (15 CREDIT HOURS, 19 CONTACT HOURS, 750)	CSE 014	Structured Programming	3	5	180	2	0	3
		PHY 211	Physics II	3	4	150	3	0	2
		MAT 111	Mathematics I	3	4	150	3	1	0
		UC1 1xx	Univ Req1	2	2	90	2	0	0
		UC2 1xx	Univ Req2	2	2	90	2	0	0
		UE1 1xx	Univ Elect1	2	2	90	2	0	0
	SPRING (16 CREDIT HOURS, 22 CONTACT HOURS, 855)	CSE 015	Object Oriented Programming	3	5	180	2	0	3
		MAT 131	Probability & Statistics I	3	5	195	2	2	1
		CSE 315	Discrete Mathematics	3	4	150	2	2	0
		MAT 112	Mathematics II	3	4	150	3	1	0
		UC3 2xx	Univ Req3	2	2	90	2	0	0
		UE2 2xx	Univ Elect2	2	2	90	2	0	0
LEVEL 2	FALL (17 CREDIT HOURS, 27 CONTACT HOURS, 990)	CSE 111	Data Structures	3	5	180	2	0	3
		CSE 131	Logic Design	3	6	195	2	2	2
		MAT 231	Probability & Statistics II	3	5	195	2	1	2
		AIE 111	Artificial Intelligence	3	5	180	2	1	2
		MAT 212	Linear Algebra	3	4	150	2	0	2
		UC4 2xx	Univ Req4	2	2	90	2	0	0
	SPRING (17 CREDIT HOURS, 26 CONTACT HOURS, 900)	CSE 211	Web Programming	3	5	165	2	1	2
		CSE 221	Database Systems	3	5	180	2	1	2
		CSE 132	Computer Architecture and Organization	3	5	165	2	1	2
		MAT 312	Differential Equations	3	4	150	2	2	0
		CSE 191	Field Training 1 In Computer Science	2	2	60	2	0	0
		CSE 251	Software Engineering	3	5	180	2	1	2
LEVEL 3	FALL (17 CREDIT HOURS, 27 CONTACT HOURS, 960)	CSE 261	Computer Networks	3	5	165	2	1	2
		CSE 281	Image Processing	3	5	180	2	1	2
		AIE 121	Machine Learning	3	5	180	2	1	2
		E1 3xx	Tech ELECT 1	3	5	180	2	1	2
		CSE 233	Operating Systems	3	5	165	2	1	2
		UC5 2xx	Univ Req5	2	2	90	2	0	0
	SPRING (17 CREDIT HOURS, 25 CONTACT HOURS, 870)	CSE 112	Design and Analysis of Algorithms	4	6	195	3	3	0
		CSE 292	Field Training 2 Computer Science	2	2	60	2	0	0
		CSE 382	Computer Graphics	3	5	180	2	1	2
		CSE 352	Systems Analysis & Design	3	5	165	2	1	2
		E2 3xx	Tech ELECT 2	3	5	180	2	1	2
		UC6 3xx	Univ Req6	2	2	90	2	0	0
LEVEL 4	FALL (17 CREDIT HOURS, 27 CONTACT HOURS, 990)	CSE 363	Cloud Computing	3	5	180	2	1	2
		CSE 383	Computer Vision	3	5	180	2	1	2
		CSE 493	Graduation Project 1	3	5	180	2	0	3
		AIE 323	Data Mining	3	5	180	2	1	2
		UC7 3xx	Univ Req7	2	2	90	2	0	0
		E3 4xx	Tech ELECT 3	3	5	180	2	1	2
	SPRING (17 CREDIT HOURS, 27 CONTACT HOURS, 990)	CSE 484	Interactive Multimedia	3	5	180	2	1	2
		CSE 494	Graduation Project 2	3	5	180	2	0	3
		CSE 411	Theory of Computation & Compilers	3	5	180	2	1	2
		E4 4xx	Tech ELECT 4	3	5	180	2	1	2
		E5 4xx	Tech ELECT 5	3	5	180	2	1	2
		UE3 3xx	Univ Elect3	2	2	90	2	0	0

COURSE MAP LEGEND

COURSE CODE
COURSE NAME
CR | CNTC | STWL
LEC | TUT | LAB

University Req
Basic Science and Eng
MAJOR ELECTIVE
Major Compulsory
Comp Science and Eng
Graduation Project

Alamein- CS PROGRAM/Big data Track- COURSE MAP 2020 - 127 CR HRS

Level	Term	Course	Name	CR	CNTC	STWL	LEC	TUT	LAB
LEVEL 1	FALL (15 CREDIT HOURS, 19 CONTACT HOURS, 750)	CSE 014	Structured Programming	3	5	180	2	0	3
		PHY 211	Physics II	3	4	150	2	0	2
		MAT 111	Mathematics I	3	4	150	3	1	0
		UC1 1xx	Univ Req1	2	2	90	2	0	0
		UC2 1xx	Univ Req2	2	2	90	2	0	0
		UE1 1xx	Univ Elect1	2	2	90	2	0	0
	SPRING (16 CREDIT HOURS, 22 CONTACT HOURS, 855)	CSE 015	Object Oriented Programming	3	5	180	2	0	3
		MAT 131	Probability & Statistics I	3	5	195	2	2	1
		CSE 315	Discrete Mathematics	3	4	150	2	2	0
		MAT 112	Mathematics II	3	4	150	3	1	0
		UC3 2xx	Univ Req3	2	2	90	2	0	0
		UE2 2xx	Univ Elect2	2	2	90	2	0	0
LEVEL 2	FALL (17 CREDIT HOURS, 27 CONTACT HOURS, 990)	CSE 111	Data Structures	3	5	180	2	0	3
		CSE 131	Logic Design	3	6	195	2	2	2
		MAT 231	Probability & Statistics II	3	5	195	2	1	2
		AIE 111	Artificial Intelligence	3	5	180	2	1	2
		MAT 212	Linear Algebra	3	4	150	2	0	2
		UC4 2xx	Univ Req4	2	2	90	2	0	0
	SPRING (17 CREDIT HOURS, 26 CONTACT HOURS, 900)	CSE 211	Web Programming	3	5	165	2	1	2
		CSE 221	Database Systems	3	5	180	2	1	2
		CSE 132	Computer Architecture and Organization	3	5	165	2	1	2
		MAT 312	Differential Equations	3	4	150	2	2	0
		CSE 191	Field Training 1 In Computer Science	2	2	60	2	0	0
		CSE 251	Software Engineering	3	5	180	2	1	2
LEVEL 3	FALL (17 CREDIT HOURS, 27 CONTACT HOURS, 945)	CSE 261	Computer Networks	3	5	165	2	1	2
		E1 3xx	Tech ELECT 1	3	5	180	2	1	2
		AIE 121	Machine Learning	3	5	180	2	1	2
		CSE 251	Systems Analysis & Design	3	5	165	2	1	2
		CSE 233	Operating Systems	3	5	165	2	1	2
		UC5 2xx	Univ Req5	2	2	90	2	0	0
	SPRING (17 CREDIT HOURS, 25 CONTACT HOURS, 885)	CSE 112	Design and Analysis of Algorithms	4	6	195	3	3	0
		CSE 292	Field Training 2 Computer Science	2	2	60	2	0	0
		CSE 241	Security of Information Systems	3	5	180	2	1	2
		E2 4xx	Tech ELECT 2	3	5	180	2	1	2
		CSE 323	Advanced Database Systems	3	5	180	2	1	2
		UC6 3xx	Univ Req6	2	2	90	2	0	0
LEVEL 4	FALL (17 CREDIT HOURS, 27 CONTACT HOURS, 990)	CSE 363	Cloud Computing	3	5	180	2	1	2
		CSE 454	Advanced Software Engineering	3	5	180	2	1	2
		CSE 493	Graduation Project 1	3	5	180	2	0	3
		AIE 323	Data Mining	3	5	180	2	1	2
		UC7 3xx	Univ Req7	2	2	90	2	0	0
		E3 4xx	Tech ELECT 3	3	5	180	2	1	2
	SPRING (17 CREDIT HOURS, 27 CONTACT HOURS, 990)	CSE 474	Distributed Information Systems	3	5	180	2	1	2
		CSE 494	Graduation Project 2	3	5	180	2	0	3
		CSE 411	Theory of Computation & Compilers	3	5	180	2	1	2
		E4 4xx	Tech ELECT 4	3	5	180	2	1	2
		E5 4xx	Tech ELECT 5	3	5	180	2	1	2
		UE3 3xx	Univ Elect3	2	2	90	2	0	0

COURSE MAP LEGEND

COURSE CODE
COURSE NAME
CR | CNTC | STWL
LEC | TUT | LAB

University Req
Basic Science and Eng
MAJOR ELECTIVE
Major Compulsory
Comp Science and Eng
Graduation Project

Alamein- AIS PROGRAM- COURSE MAP 2020 - 133 CR HRS

Level	Term	Course	Name	CR	CNTC	STWL	LEC	TUT	LAB
LEVEL 1	FALL (15 CREDIT HOURS, 19 CONTACT HOURS, 750)	CSE 014	Structured Programming	3	5	180	2	0	3
		PHY 211	Physics II	3	4	150	3	0	2
		MAT 111	Mathematics I	3	4	150	3	1	0
		UC1 1xx	Univ Req1	2	2	90	2	0	0
		UC2 1xx	Univ Req2	2	2	90	2	0	0
		UE1 1xx	Univ Elect1	2	2	90	2	0	0
	SPRING (16 CREDIT HOURS, 22 CONTACT HOURS, 855)	CSE 015	Object Oriented Programming	3	5	180	2	0	3
		MAT 131	Probability & Statistics I	3	5	195	2	2	1
		CSE 315	Discrete Mathematics	3	4	150	2	2	0
		MAT 112	Mathematics II	3	4	150	3	1	0
		UC3 2xx	Univ Req3	2	2	90	2	0	0
		UE2 2xx	Univ Elect2	2	2	90	2	0	0
LEVEL 2	FALL (17 CREDIT HOURS, 27 CONTACT HOURS, 990)	CSE 111	Data Structures	3	5	180	2	0	3
		CSE 131	Logic Design	3	6	195	2	2	2
		MAT 231	Probability & Statistics II	3	5	195	2	1	2
		AIE 111	Artificial Intelligence	3	5	180	2	1	2
		MAT 212	Linear Algebra	3	4	150	2	0	2
		UC4 2xx	Univ Req4	2	2	90	2	0	0
	SPRING (17 CREDIT HOURS, 26 CONTACT HOURS, 915)	CSE 132	Computer Architecture & Organization	3	5	165	2	1	2
		CSE 221	Database Systems	3	5	180	2	1	2
		CSE 281	Image Processing	3	5	180	2	1	2
		MAT 312	Differential Equations	3	4	150	2	2	0
		AIE 191	Field Training 1 in AI Science	2	2	60	2	0	0
		AIE 121	Machine Learning	3	5	180	2	1	2
LEVEL 3	FALL (17 CREDIT HOURS, 27 CONTACT HOURS, 960)	CSE 251	Software Engineering	3	5	180	2	1	2
		AIE 231	Neural Networks	3	5	180	2	1	2
		AIE 323	Data Mining	3	5	180	2	1	2
		CSE 261	Computer Networks	3	5	165	2	1	2
		CSE 233	Operating Systems	3	5	165	2	1	2
		UC5 2xx	Univ Req5	2	2	90	2	0	0
	SPRING (18 CREDIT HOURS, 29 CONTACT HOURS, 975)	CSE 112	Design and Analysis of Algorithms	4	6	195	3	3	0
		AIE 292	Field Training 1 in AI Science	2	2	60	2	0	0
		AIE 213	Optimization Techniques	3	6	180	3	3	0
		AIE 212	Knowledge-Based Systems	3	5	180	2	1	2
		AIE 241	Natural Language Processing	3	5	180	2	1	2
		E1 4xx	Tech ELECT 1	3	5	180	2	1	2
LEVEL 4	FALL (17 CREDIT HOURS, 27 CONTACT HOURS, 990)	AIE 322	Advanced Machine Learning	3	5	180	2	1	2
		CSE 363	Cloud Computing	3	5	180	2	1	2
		AIE 493	Graduation Project 1	3	5	180	2	0	3
		AIE 332	Deep Learning	3	5	180	2	1	2
		UC6 3xx	Univ Req6	2	2	90	2	0	0
		E2 4xx	Tech ELECT 2	3	5	180	2	1	2
	SPRING (16 CREDIT HOURS, 24 CONTACT HOURS, 900)	AIE 425	Intelligent Recommender Systems	3	5	180	2	1	2
		AIE 494	Graduation Project 2	3	5	180	2	0	3
		AIE 314	AI-Based Programming	3	5	180	2	1	2
		E3 4xx	Tech ELECT 3	3	5	180	2	1	2
		UC7 3xx	Univ Req7	2	2	90	2	0	0
		UE3 1xx	Univ Elect3	2	2	90	2	0	0

COURSE MAP LEGEND

COURSE CODE
COURSE NAME
CR | CNTC | STWL
LEC | TUT | LAB

- University Req
- Basic Science and Eng
- MAJOR ELECTIVE
- Major Compulsory
- Comp Science and Eng
- Graduation Project

Alamein- Biomedical Info PROGRAM- COURSE MAP 2020- 133 CR HRS

Level	Term	Course	Name	CR	CNTC	STWL	LEC	TUT	LAB
LEVEL 1	FALL (15 CREDIT HOURS, 19 CONTACT HOURS, 750)	UE1 1xx	Univ Elect1	2	2	90	2	0	0
		UC1 1xx	Univ Req1	2	2	90	2	0	0
		MAT 111	Mathematics I	3	4	150	3	1	0
		PHY 281	Biophysics	3	4	150	2	0	2
		CSE 014	Structured Programming	3	5	180	2	0	3
		UC2 1xx	Univ Req2	2	2	90	2	0	0
	SPRING (16 CREDIT HOURS, 22 CONTACT HOURS, 855)	UC3 2xx	Univ Req3	2	2	90	2	0	0
		MAT 112	Mathematics II	3	4	150	3	1	0
		MAT 131	Probability & Statistics I	3	5	195	2	2	1
		BIO 241	Biology II	3	4	150	2	0	2
		CSE 015	Object Oriented Programming	3	5	180	2	0	3
		UE2 2xx	Univ Elect2	2	2	90	2	0	0
LEVEL 2	FALL (17 CREDIT HOURS, 26 CONTACT HOURS, 915)	MAT 212	Linear Algebra	3	4	150	2	0	2
		CSE 131	Logic Design	3	6	195	2	2	2
		BMD 191	Field Training 1 in Biomedical Informatics	2	2	60	2	0	0
		BIO 221	Molecular Biology I	3	4	150	2	0	2
		CSE 111	Data Structures	3	5	180	2	0	3
		AIE 111	Artificial Intelligence	3	5	180	2	1	2
	SPRING (17 CREDIT HOURS, 25 CONTACT HOURS, 930)	UC4 2xx	Univ Req4	2	2	90	2	0	0
		CSE 315	Discrete Mathematics	3	4	150	2	2	0
		MAT 312	Differential Equations	3	4	150	2	2	0
		CSE 221	Database Systems	3	5	180	2	1	2
		BMD 311	Introduction to Bioinformatics	3	5	180	2	1	2
		BMD 241	Human physiology	3	5	180	2	1	2
LEVEL 3	FALL (17 CREDIT HOURS, 27 CONTACT HOURS, 975)	UC5 2xx	Univ Req5	2	2	90	2	0	0
		CSE 281	Image Processing	3	5	180	2	1	2
		CSE 261	Computer Networks	3	5	165	2	1	2
		BMD 312	Clinical Informatics	3	5	180	2	1	2
		BMD 351	Biomedical Data Acquisition	3	5	180	2	1	2
		CSE 251	Software Engineering	3	5	180	2	1	2
	SPRING (17 CREDIT HOURS, 25 CONTACT HOURS, 870)	BMD 292	Field Training 2 in Biomedical Informatics	2	2	60	2	0	0
		UC6 3xx	Univ Req6	2	2	90	2	0	0
		CSE 352	Systems Analysis and Design	3	5	165	2	1	2
		CSE 112	Design & Analysis of Algorithms	4	6	195	3	0	3
		BMD 361	Biomedical Statistics	3	5	180	2	1	2
		E1 3xx	Tech ELECT 1	3	5	180	2	1	2
LEVEL 4	FALL (17 CREDIT HOURS, 27 CONTACT HOURS, 990)	E2 3xx	Tech ELECT 2	3	5	180	2	1	2
		BMD 421	Biomedical Information systems	3	5	180	2	1	2
		BMD 431	Medical Image Informatics	3	5	180	2	1	2
		E3 3xx	Tech ELECT 3	3	5	180	2	1	2
		BMD 493	Graduation Project 1	3	5	180	2	0	3
		UE3 2xx	Univ Elect3	2	2	90	2	0	0
	SPRING (17 CREDIT HOURS, 27 CONTACT HOURS, 990)	BMD 413	Structural Bioinformatics	3	5	180	2	1	2
		E4 4xx	Tech ELECT 4	3	5	180	2	1	2
		E5 4xx	Tech ELECT 5	3	5	180	2	1	2
		E6 4xx	Tech ELECT 6	3	5	180	2	1	2
		BMD 494	Graduation Project 2	3	5	180	2	0	3
		UC7 3xx	Univ Req7	2	2	90	2	0	0

COURSE MAP LEGEND

AI Courses

COURSE CODE
COURSE NAME
CR | CNTC | STWL
LEC | TUT | LAB

University Req | Basic Science and Eng | MAJOR ELECTIVE | Major Compulsory | Comp Science and Eng | Graduation Project

Alamein- Information Technology PROGRAM- COURSE MAP 2020- 132 CR HRS

Level	Term	Course	Code	CR	CNTC	STWL	LEC	TUT	LAB
LEVEL 1	FALL (16 CREDIT HOURS, 21 CONTACT HOURS, 810)	Business Information Systems	BIS 251	3	4	150	3	0	1
		Univ Req1	UC1 1xx	2	2	90	2	0	0
		Mathematics I	MAT 111	3	4	150	3	1	0
		Physics II	PHY 211	3	4	150	3	0	2
		Structured Programming	CSE 014	3	5	180	2	0	3
		Univ Req2	UC2 1xx	2	2	90	2	0	0
	SPRING (16 CREDIT HOURS, 22 CONTACT HOURS, 855)	Univ Req3	UC3 2xx	2	2	90	2	0	0
		Mathematics II	MAT 112	3	4	150	3	1	0
		Probability & Statistics I	MAT 131	3	5	195	2	2	1
		Discrete Mathematics	CSE 315	3	4	150	2	2	0
		Object Oriented Programming	CSE 015	3	5	180	2	0	3
		Univ Elect1	UE1 2xx	2	2	90	2	0	0
LEVEL 2	FALL (16 CREDIT HOURS, 24 CONTACT HOURS, 900)	Linear Algebra	MAT 212	3	4	150	2	0	2
		Logic Design	CSE 131	3	6	195	2	2	2
		Univ Req4	UC4 2xx	2	2	90	2	0	0
		Probability & Statistics II	MAT 231	3	5	195	2	1	2
		Data Structures	CSE 111	3	5	180	2	0	3
		Univ Elect2	UE2 2xx	2	2	90	2	0	0
	SPRING (17 CREDIT HOURS, 26 CONTACT HOURS, 870)	Computer Architecture & Organization	CSE 132	3	5	165	2	1	2
		Field Training 1 in Information Technology	CSE 192	2	2	60	2	0	0
		Numerical Analysis	MAT 315	3	4	150	2	2	0
		Database Systems	CSE 221	3	5	180	2	1	2
		Web Programming	CSE 211	3	5	165	2	1	2
		Security of Information Systems	CSE 241	3	5	150	2	1	2
LEVEL 3	FALL (17 CREDIT HOURS, 25 CONTACT HOURS, 945)	Univ Req5	UC5 2xx	2	2	90	2	0	0
		Operating Systems Administration	CSE 232	3	5	180	2	1	2
		Introduction to Telecommunications, Networking and Security	CSE 262	3	5	180	2	1	2
		Information Technology Governance	BIS 356	3	3	135	3	0	0
		Basics of Machine Learning and Data Science	AIE 122	3	5	180	2	1	2
		Software Engineering	CSE 251	3	5	180	2	1	2
	SPRING (16 CREDIT HOURS, 24 CONTACT HOURS, 870)	Field Training 2 in Information Technology	CSE 293	2	2	60	2	0	0
		Univ Req6	UC6 3xx	2	2	90	2	0	0
		Introduction to Cyber Security	CSE 344	3	5	180	2	1	2
		Societal and Ethical Issues in Information Technology	CSE 321	3	5	180	2	1	2
		IT Management and Operations	CSE 324	3	5	180	2	1	2
		Tech ELECT 1	E1 3xx	3	5	180	2	1	2
LEVEL 4	FALL (17 CREDIT HOURS, 27 CONTACT HOURS, 990)	Tech ELECT 2	E2 3xx	3	5	180	2	1	2
		Database Systems Administration	CSE 325	3	5	180	2	1	2
		Cloud Computing	CSE 363	3	5	180	2	1	2
		Networks Virtualization and Administration	CSE 361	3	5	180	2	1	2
		Graduation Project 1	CSE 493	3	5	180	2	0	3
		Univ Req7	UC7 3xx	2	2	90	2	0	0
	SPRING (17 CREDIT HOURS, 27 CONTACT HOURS, 990)	Interaction Design	CSE 353	3	5	180	2	1	2
		Tech ELECT 3	E3 4xx	3	5	180	2	1	2
		Tech ELECT 4	E4 4xx	3	5	180	2	1	2
		Advanced Networking	CSE 463	3	5	180	2	1	2
		Graduation Project 2	CSE 494	3	5	180	2	0	3
		Univ Elect3	UE3 2xx	2	2	90	2	0	0

COURSE MAP LEGEND

AI Courses

COURSE CODE
COURSE NAME
CR | CNTC | STWL
LEC | TUT | LAB

University Req | Basic Science and Eng | MAJOR ELECTIVE | Major Compulsory | Comp Science and Eng | Graduation Project